

Modern Practical Control Systems

Section 0

Mathematical Reference

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Draft Version
January 19, 2026

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Chapter 1

Complex Numbers and Imaginary Arithmetic

Complex numbers are fundamental to control systems analysis. They appear naturally in the solution of differential equations, in the Laplace transform, in frequency response analysis, and in stability analysis.

1.1 Definition and Representation

Definition 0.1: Complex Number

A **complex number** z is defined as:

$$z = a + jb \quad (1.1)$$

where $a, b \in \mathbb{R}$, and $j = \sqrt{-1}$ is the **imaginary unit**. The real part is $\text{Re}(z) = a$ and the imaginary part is $\text{Im}(z) = b$.

Note 1.1. In electrical engineering, j is often used instead of i for the imaginary unit to avoid confusion with current. We adopt this convention throughout.

Complex numbers can be represented in multiple forms:

Rectangular Form

$$z = a + jb \quad (1.2)$$

Polar Form

$$z = r(\cos \theta + j \sin \theta) = r \angle \theta \quad (1.3)$$

where $r = |z| = \sqrt{a^2 + b^2}$ is the **magnitude** (or modulus) and $\theta = \arg(z) = \arctan(b/a)$ is the **argument** (or phase angle).

Exponential Form (Euler's Formula)

$$z = re^{j\theta} \quad (1.4)$$

Euler's formula, one of the most elegant results in mathematics, states:

$$\boxed{e^{j\theta} = \cos \theta + j \sin \theta} \quad (1.5)$$

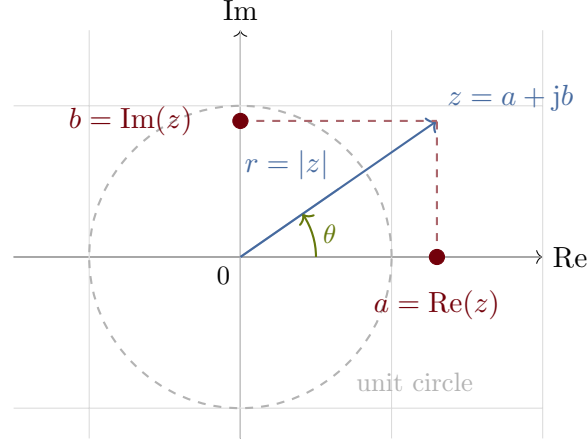


Figure 1.1: Geometric representation of a complex number in the complex plane (Argand diagram). The blue vector shows the complex number z , with dashed red projections onto the real and imaginary axes. The green angle θ represents the argument, and the unit circle (dashed gray) provides a reference. Colors: Atlantic (blue) for magnitude, Garnet (red) for projections, Horseshoe (green) for angle.

1.2 Arithmetic Operations

Addition and Subtraction

$$(a + jb) \pm (c + jd) = (a \pm c) + j(b \pm d) \quad (1.6)$$

Multiplication

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc) \quad (1.7)$$

In polar form: $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$

Division

$$\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2} \quad (1.8)$$

In polar form: $\frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$

Complex Conjugate

$$z^* = \bar{z} = a - jb \quad (1.9)$$

Important property: $z \cdot z^* = |z|^2 = a^2 + b^2$

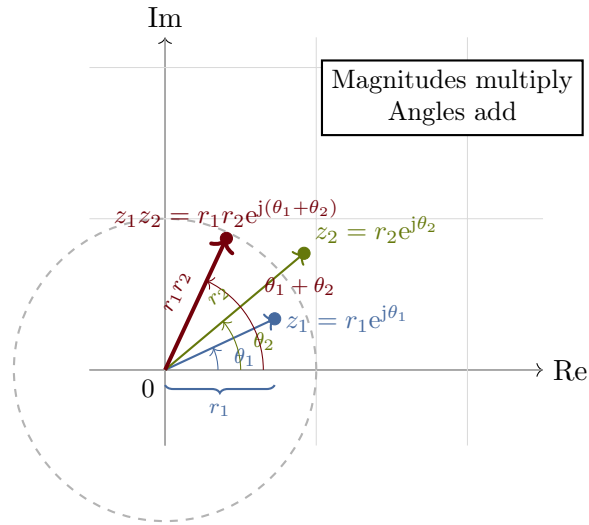


Figure 1.2: Complex multiplication in polar form. When multiplying two complex numbers $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$, the magnitudes multiply ($r_1 r_2$) and the angles add ($\theta_1 + \theta_2$). This geometric interpretation makes polar form ideal for multiplication and division operations.

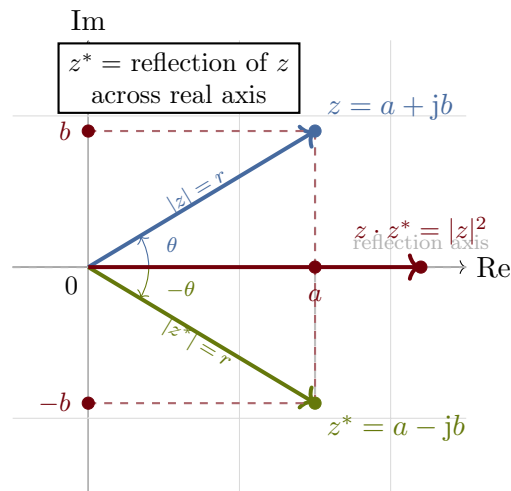


Figure 1.3: Complex conjugate visualization. The conjugate $z^* = a - jb$ is the reflection of $z = a + jb$ across the real axis. Both have the same magnitude $|z| = |z^*| = r$, but opposite angles $\pm\theta$. The product $z \cdot z^* = |z|^2 = a^2 + b^2$ is always a real, non-negative number lying on the real axis.

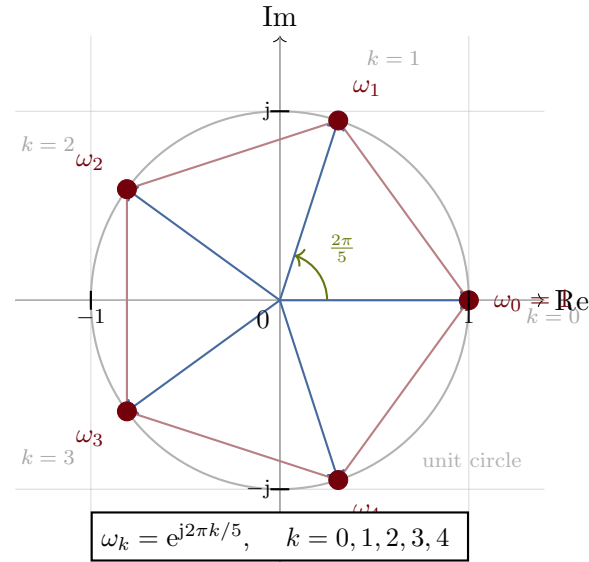


Figure 1.4: The fifth roots of unity. The n th roots of unity are n complex numbers evenly distributed on the unit circle, satisfying $z^n = 1$. For $n = 5$, these roots $\omega_k = e^{j2\pi k/5}$ are separated by angles of $2\pi/5 = 72^\circ$ and form a regular pentagon. The root $\omega_0 = 1$ always lies on the positive real axis.

Powers (De Moivre's Theorem)

$$z^n = r^n e^{jn\theta} = r^n (\cos n\theta + j \sin n\theta) \quad (1.10)$$

Roots

The n th roots of $z = re^{j\theta}$ are:

$$z^{1/n} = r^{1/n} e^{j(\theta+2\pi k)/n}, \quad k = 0, 1, \dots, n-1 \quad (1.11)$$

Example 0.1: Complex Number Operations

Find the polar form, magnitude, and phase of $z = 3 + 4j$.

Solution:

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \quad (1.12)$$

$$\theta = \arctan\left(\frac{4}{3}\right) \approx 53.13^\circ = 0.927 \text{ rad} \quad (1.13)$$

$$z = 5e^{j(0.927)} = 5 \angle 53.13^\circ \quad (1.14)$$

Example 0.2: Multiplication and Division in Polar Form

Given $z_1 = 2e^{j\pi/6}$ and $z_2 = 3e^{j\pi/4}$, compute $z_1 \cdot z_2$ and z_1/z_2 .

Solution:

For multiplication, we multiply magnitudes and add angles:

$$z_1 \cdot z_2 = (2)(3)e^{j(\pi/6+\pi/4)} = 6e^{j(5\pi/12)} \quad (1.15)$$

Converting to rectangular form: $z_1 \cdot z_2 = 6 \cos(5\pi/12) + 6j \sin(5\pi/12) \approx 1.55 + 5.80j$.

For division, we divide magnitudes and subtract angles:

$$\frac{z_1}{z_2} = \frac{2}{3} e^{j(\pi/6 - \pi/4)} = \frac{2}{3} e^{-j\pi/12} \quad (1.16)$$

Converting to rectangular form: $z_1/z_2 = \frac{2}{3} \cos(-\pi/12) + \frac{2}{3} j \sin(-\pi/12) \approx 0.644 - 0.173j$.

Key insight: Polar form makes multiplication and division much simpler than rectangular form, which is why it is preferred in frequency response analysis.

Example 0.3: Finding All Cube Roots of a Complex Number

Find all cube roots of $z = 8$.

Solution:

First, write $z = 8$ in polar form: $z = 8e^{j \cdot 0} = 8e^{j \cdot 2\pi k}$ for any integer k .

The n th roots of $z = re^{j\theta}$ are given by:

$$z^{1/n} = r^{1/n} e^{j(\theta + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1$$

For the cube roots ($n = 3$) of $8e^{j \cdot 0}$:

$$z_0 = 8^{1/3} e^{j(0 + 2\pi \cdot 0)/3} = 2e^{j \cdot 0} = 2 \quad (1.17)$$

$$z_1 = 8^{1/3} e^{j(0 + 2\pi \cdot 1)/3} = 2e^{j2\pi/3} = 2 \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = -1 + j\sqrt{3} \quad (1.18)$$

$$z_2 = 8^{1/3} e^{j(0 + 2\pi \cdot 2)/3} = 2e^{j4\pi/3} = 2 \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) = -1 - j\sqrt{3} \quad (1.19)$$

Verification: We can verify $(2)^3 = 8$, $(-1 + j\sqrt{3})^3 = 8$, and $(-1 - j\sqrt{3})^3 = 8$.

Geometric interpretation: The three roots are equally spaced on a circle of radius 2 in the complex plane, separated by 120° (i.e., $2\pi/3$ radians).

Example 0.4: Deriving Trigonometric Identities Using Euler's Formula

Use Euler's formula to derive the double-angle formulas for $\cos(2\theta)$ and $\sin(2\theta)$.

Solution:

From Euler's formula: $e^{j\theta} = \cos \theta + j \sin \theta$.

Consider $e^{j(2\theta)}$. We can write this in two ways:

Method 1: Direct application of Euler's formula:

$$e^{j(2\theta)} = \cos(2\theta) + j \sin(2\theta)$$

Method 2: Using properties of exponents:

$$\begin{aligned} e^{j(2\theta)} &= \left(e^{j\theta} \right)^2 = (\cos \theta + j \sin \theta)^2 \\ &= \cos^2 \theta + 2j \cos \theta \sin \theta + j^2 \sin^2 \theta \\ &= (\cos^2 \theta - \sin^2 \theta) + j(2 \cos \theta \sin \theta) \end{aligned}$$

Equating real and imaginary parts from both methods:

$$\text{Re : } \cos(2\theta) = \cos^2 \theta - \sin^2 \theta \quad (1.20)$$

$$\text{Im : } \sin(2\theta) = 2 \cos \theta \sin \theta \quad (1.21)$$

Extension: This technique can be used to derive any multiple-angle formula. For example, expanding $(e^{j\theta})^3$ yields the triple-angle formulas.

1.3 Properties of Complex Operations

This section provides formal proofs of fundamental properties of complex number operations. These properties are essential for understanding frequency response analysis and stability theory.

Key Property: Magnitude of a Product

For any two complex numbers z_1 and z_2 :

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2| \quad (1.22)$$

The magnitude of a product equals the product of the magnitudes.

Proof. Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$ be two complex numbers in polar form, where $r_1 = |z_1|$ and $r_2 = |z_2|$.

The product is:

$$z_1 \cdot z_2 = r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

Taking the magnitude:

$$|z_1 \cdot z_2| = |r_1 r_2 e^{j(\theta_1 + \theta_2)}| = r_1 r_2 = |z_1| \cdot |z_2|$$

where we used the fact that $|e^{j\phi}| = 1$ for any real ϕ . □

Key Property: Argument of a Product

For any two complex numbers z_1 and z_2 (with $z_1, z_2 \neq 0$):

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2) \pmod{2\pi} \quad (1.23)$$

The argument of a product equals the sum of the arguments (modulo 2π).

Proof. Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$, where $\theta_1 = \arg(z_1)$ and $\theta_2 = \arg(z_2)$.

The product is:

$$z_1 \cdot z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

By definition of the argument (the angle in polar form):

$$\arg(z_1 \cdot z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2) \pmod{2\pi}$$

The modulo 2π accounts for the fact that angles differing by 2π represent the same direction. □

Key Property: Magnitude Squared via Conjugate

For any complex number z :

$$|z|^2 = z \cdot z^* \quad (1.24)$$

The squared magnitude equals the product of a number and its complex conjugate.

Proof. Let $z = a + jb$ where $a, b \in \mathbb{R}$.

The complex conjugate is $z^* = a - jb$.

Computing the product:

$$\begin{aligned} z \cdot z^* &= (a + jb)(a - jb) \\ &= a^2 - jab + jab - j^2 b^2 \\ &= a^2 - (-1)b^2 \\ &= a^2 + b^2 \end{aligned}$$

By definition of magnitude: $|z|^2 = (\sqrt{a^2 + b^2})^2 = a^2 + b^2$.

Therefore: $|z|^2 = z \cdot z^*$.

Alternative proof using polar form: Let $z = re^{j\theta}$. Then $z^* = re^{-j\theta}$, and:

$$z \cdot z^* = re^{j\theta} \cdot re^{-j\theta} = r^2 e^{j \cdot 0} = r^2 = |z|^2$$

□

Note 1.2. The property $|z|^2 = z \cdot z^*$ is particularly useful in control systems when computing power spectral densities and when rationalizing complex-valued transfer functions for frequency response analysis.

1.4 Complex Numbers in Control Systems

Complex numbers appear throughout control theory:

- **Poles and Zeros:** The roots of transfer function polynomials are generally complex.
- **Eigenvalues:** System matrices often have complex eigenvalues that determine stability and oscillatory behavior.
- **Frequency Response:** Evaluating transfer functions at $s = j\omega$ yields complex numbers representing magnitude and phase.
- **Sinusoidal Steady-State:** Complex exponentials simplify analysis of sinusoidal inputs.

Chapter 2

Calculus Review

2.1 Differential Calculus

2.1.1 Derivatives and Rules

The derivative of a function $f(t)$ represents its instantaneous rate of change:

$$\frac{df}{dt} = f'(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (2.1)$$

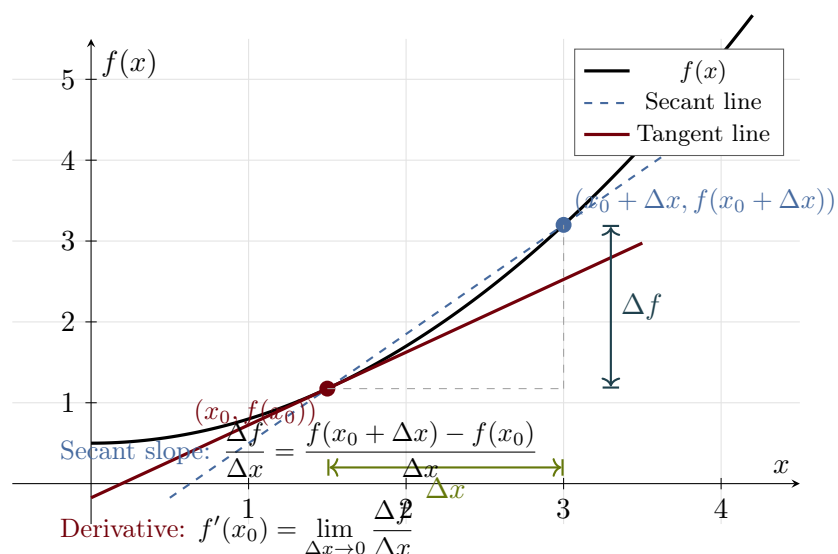


Figure 2.1: Visualization of the derivative as the limit of secant line slopes. The curve $f(x)$ is shown in black. The secant line (dashed Atlantic blue) connects two points separated by Δx , with slope $\Delta f/\Delta x$. As $\Delta x \rightarrow 0$, the secant line approaches the tangent line (Garnet red) at x_0 , whose slope equals the derivative $f'(x_0)$. The intervals Δx (Horseshoe green) and Δf (Congaree teal) are marked.

Rule	Formula
Power Rule	$\frac{d}{dt}(t^n) = nt^{n-1}$
Exponential	$\frac{d}{dt}(e^{at}) = ae^{at}$
Logarithm	$\frac{d}{dt}(\ln t) = \frac{1}{t}$
Sine	$\frac{d}{dt}(\sin \omega t) = \omega \cos \omega t$
Cosine	$\frac{d}{dt}(\cos \omega t) = -\omega \sin \omega t$
Product Rule	$\frac{d}{dt}(fg) = f'g + fg'$
Quotient Rule	$\frac{d}{dt}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$
Chain Rule	$\frac{d}{dt}f(g(t)) = f'(g(t)) \cdot g'(t)$

Table 2.1: Essential differentiation rules for control systems.

Key Differentiation Rules

Product Rule Application

Differentiate $f(t) = t^2 e^{-at}$ where $a > 0$ is a constant.

Solution: Using the product rule with $u = t^2$ and $v = e^{-at}$:

$$\frac{df}{dt} = \frac{d}{dt}(t^2) \cdot e^{-at} + t^2 \cdot \frac{d}{dt}(e^{-at}) \quad (2.2)$$

$$= 2t \cdot e^{-at} + t^2 \cdot (-ae^{-at}) \quad (2.3)$$

$$= e^{-at}(2t - at^2) \quad (2.4)$$

$$= te^{-at}(2 - at) \quad (2.5)$$

This type of function appears frequently in control systems as a damped response with a time-varying amplitude.

2.1.2 Partial Derivatives

For functions of multiple variables $f(x_1, x_2, \dots, x_n)$, partial derivatives hold all variables constant except one:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta x_i, \dots, x_n) - f(x_1, \dots, x_n)}{\Delta x_i} \quad (2.6)$$

The **gradient** is the vector of all partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \dots \quad \frac{\partial f}{\partial x_n} \right]^\top \quad (2.7)$$

The **Hessian** is the matrix of second partial derivatives:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (2.8)$$

2.1.3 Taylor Series Expansion

Taylor series are essential for linearization of nonlinear systems:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots \quad (2.9)$$

For multivariate functions about point $\mathbf{x}_0$:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) \quad (2.10)$$

Linearization in Control

For nonlinear systems $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, we linearize about an equilibrium point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\Delta \dot{\mathbf{x}} \approx \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u} \quad (2.11)$$

where $\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$ and $\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$

Taylor Expansion of Sine Function

Find the Taylor expansion of $\sin(x)$ about $x = 0$ (Maclaurin series) to fourth order.

Solution: We need the derivatives of $f(x) = \sin(x)$ evaluated at $x = 0$:

$$f(x) = \sin(x) \quad f(0) = 0 \quad (2.12)$$

$$f'(x) = \cos(x) \quad f'(0) = 1 \quad (2.13)$$

$$f''(x) = -\sin(x) \quad f''(0) = 0 \quad (2.14)$$

$$f'''(x) = -\cos(x) \quad f'''(0) = -1 \quad (2.15)$$

$$f^{(4)}(x) = \sin(x) \quad f^{(4)}(0) = 0 \quad (2.16)$$

Substituting into the Taylor series formula:

$$\sin(x) = 0 + 1 \cdot x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{0}{4!}x^4 + \dots = x - \frac{x^3}{6} + \mathcal{O}(x^5) \quad (2.17)$$

For small angles ($|x| \ll 1$), we often use the approximation $\sin(x) \approx x$, which is the first-order Taylor expansion. This is the “small angle approximation” commonly used in linearizing pendulum and rotational dynamics.

2.2 Integral Calculus

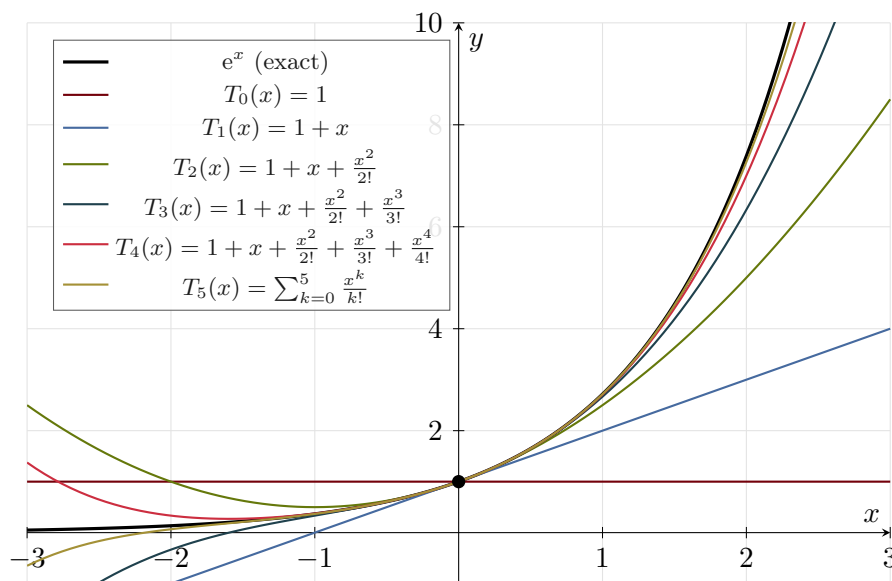
2.2.1 Definite and Indefinite Integrals

The indefinite integral (antiderivative):

$$\int f(t) dt = F(t) + C \quad \text{where} \quad F'(t) = f(t) \quad (2.18)$$

The definite integral:

$$\int_a^b f(t) dt = F(b) - F(a) \quad (2.19)$$



Taylor Series:
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Figure 2.2: Convergence of Taylor series approximations to e^x centered at $x = 0$. The exact function e^x is shown in black. Successive Taylor polynomial approximations $T_n(x) = \sum_{k=0}^n x^k/k!$ are shown for orders $n = 0$ (Garnet), $n = 1$ (Atlantic), $n = 2$ (Horseshoe), $n = 3$ (Congaree), $n = 4$ (Rose), and $n = 5$ (Honeycomb). Higher-order approximations provide better accuracy over a wider range of x values. All approximations are exact at the expansion point $x = 0$.

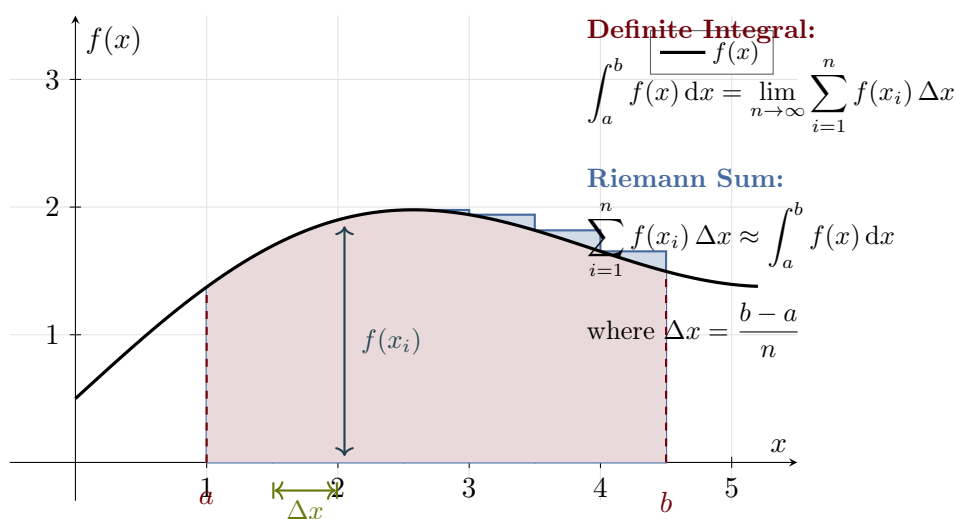


Figure 2.3: Visualization of the definite integral as area under a curve. The function $f(x)$ is shown in black. The shaded Garnet region represents the exact area $\int_a^b f(x) dx$. The Atlantic blue rectangles show a Riemann sum approximation with $n = 7$ rectangles of width $\Delta x = (b - a)/n$. As $n \rightarrow \infty$, the Riemann sum converges to the definite integral.

Key Integration Formulas

Function	Integral
t^n ($n \neq -1$)	$\frac{t^{n+1}}{n+1} + C$
$\frac{1}{t}$	$\ln t + C$
e^{at}	$\frac{1}{a}e^{at} + C$
$\sin \omega t$	$-\frac{1}{\omega} \cos \omega t + C$
$\cos \omega t$	$\frac{1}{\omega} \sin \omega t + C$
$\frac{1}{a^2+t^2}$	$\frac{1}{a} \arctan \frac{t}{a} + C$

Table 2.2: Essential integration formulas.

2.2.2 Integration by Parts

$$\int u \, dv = uv - \int v \, du \quad (2.20)$$

This technique is essential for deriving Laplace transform properties.

Integration by Parts

Evaluate the integral $\int te^{-t} \, dt$.

Solution: Choose $u = t$ and $dv = e^{-t} \, dt$. Then:

$$u = t \quad du = dt \quad (2.21)$$

$$dv = e^{-t} \, dt \quad v = -e^{-t} \quad (2.22)$$

Applying integration by parts:

$$\int te^{-t} \, dt = uv - \int v \, du \quad (2.23)$$

$$= t(-e^{-t}) - \int (-e^{-t}) \, dt \quad (2.24)$$

$$= -te^{-t} + \int e^{-t} \, dt \quad (2.25)$$

$$= -te^{-t} - e^{-t} + C \quad (2.26)$$

$$= -e^{-t}(t+1) + C \quad (2.27)$$

This integral appears when computing Laplace transforms of polynomial-exponential products.

2.2.3 Integration Techniques

Substitution (u-Substitution)

When an integrand contains a composite function, substitution can simplify the integral:

$$\int f(g(t))g'(t) dt = \int f(u) du \quad \text{where } u = g(t) \quad (2.28)$$

Choosing a Substitution

Look for an “inner function” $u = g(t)$ whose derivative $g'(t)$ also appears in the integrand. Common patterns include:

- $\int e^{at} \cdot a dt$: let $u = at$
- $\int \frac{f'(t)}{f(t)} dt$: let $u = f(t)$, giving $\ln |f(t)| + C$
- $\int t^n f(t^{n+1}) dt$: let $u = t^{n+1}$

Partial Fraction Decomposition

For rational functions $\frac{P(s)}{Q(s)}$ where $\deg(P) < \deg(Q)$, partial fractions decompose the integrand into simpler terms. This is essential for inverse Laplace transforms.

Case 1: Distinct real roots. If $Q(s) = (s - p_1)(s - p_2) \cdots (s - p_n)$:

$$\frac{P(s)}{Q(s)} = \frac{A_1}{s - p_1} + \frac{A_2}{s - p_2} + \cdots + \frac{A_n}{s - p_n} \quad (2.29)$$

Case 2: Repeated real roots. If $Q(s) = (s - p)^n$:

$$\frac{P(s)}{(s - p)^n} = \frac{A_1}{s - p} + \frac{A_2}{(s - p)^2} + \cdots + \frac{A_n}{(s - p)^n} \quad (2.30)$$

Case 3: Complex conjugate roots. If $Q(s)$ contains $(s^2 + as + b)$ with $a^2 - 4b < 0$:

$$\frac{P(s)}{s^2 + as + b} = \frac{As + B}{s^2 + as + b} \quad (2.31)$$

Cover-Up Method for Residues

For distinct real poles, the residue A_k can be found by:

$$A_k = \lim_{s \rightarrow p_k} (s - p_k) \frac{P(s)}{Q(s)} = \frac{P(p_k)}{Q'(p_k)} \quad (2.32)$$

This “cover-up” method is faster than solving systems of equations.

2.2.4 Convolution Integral

The convolution of two functions $f(t)$ and $g(t)$ is:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \quad (2.33)$$

For causal systems (signals zero for $t < 0$):

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad (2.34)$$

Convolution in Control Systems

The output $y(t)$ of a linear time-invariant (LTI) system with impulse response $h(t)$ to input $u(t)$ is:

$$y(t) = (h * u)(t) = \int_0^t h(\tau)u(t - \tau) d\tau \quad (2.35)$$

Convolution of Two Exponentials

Compute the convolution $(f * g)(t)$ for $t \geq 0$ where $f(t) = e^{-at}$ and $g(t) = e^{-bt}$ with $a \neq b$ and both signals are zero for $t < 0$.

Solution: Using the causal convolution formula:

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad (2.36)$$

$$= \int_0^t e^{-a\tau} e^{-b(t-\tau)} d\tau \quad (2.37)$$

$$= e^{-bt} \int_0^t e^{-a\tau} e^{b\tau} d\tau \quad (2.38)$$

$$= e^{-bt} \int_0^t e^{(b-a)\tau} d\tau \quad (2.39)$$

Evaluating the integral:

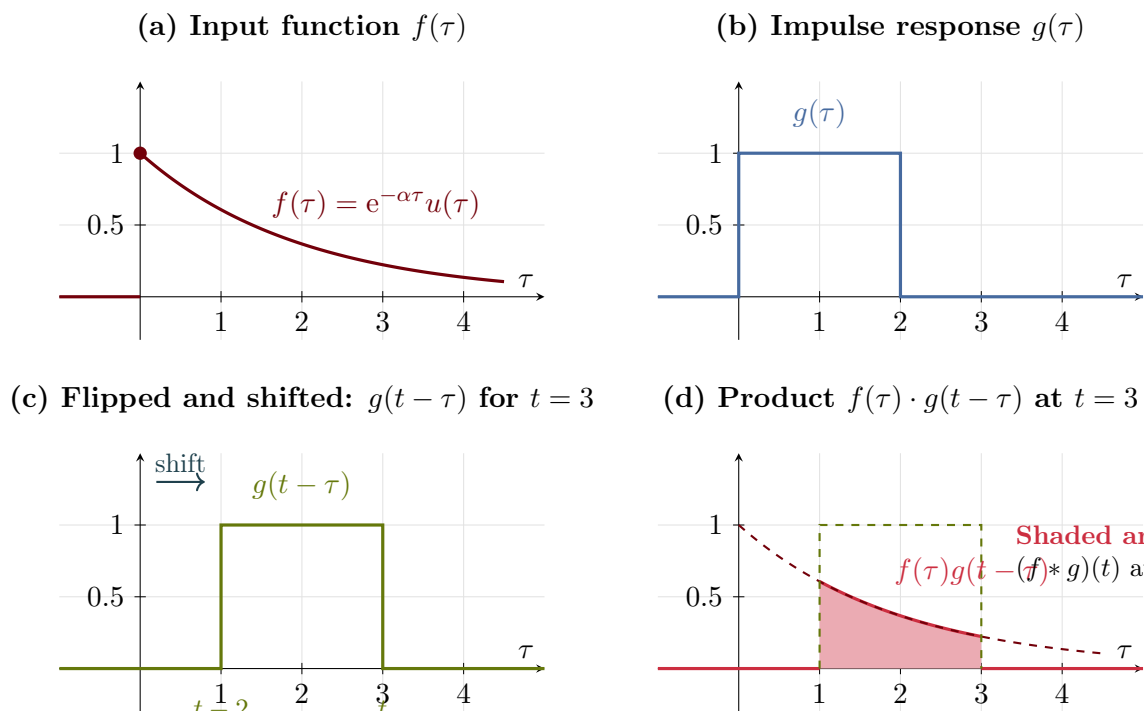
$$(f * g)(t) = e^{-bt} \left[\frac{e^{(b-a)\tau}}{b-a} \right]_0^t \quad (2.40)$$

$$= e^{-bt} \cdot \frac{e^{(b-a)t} - 1}{b-a} \quad (2.41)$$

$$= \frac{e^{-bt} e^{(b-a)t} - e^{-bt}}{b-a} \quad (2.42)$$

$$= \frac{e^{-at} - e^{-bt}}{b-a} \quad (2.43)$$

This result is symmetric in the sense that swapping a and b gives the same answer (note the sign change in both numerator and denominator). This convolution appears when analyzing cascaded first-order systems.



Convolution Integral: $(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$

Figure 2.4: Visualization of the convolution operation $(f * g)(t)$. (a) The input function $f(\tau) = e^{-\alpha\tau}u(\tau)$ in Garnet. (b) The impulse response $g(\tau)$, a rectangular pulse, in Atlantic blue. (c) The flipped and shifted function $g(t - \tau)$ for $t = 3$ in Horseshoe green—convolution involves sliding this function across f . (d) The product $f(\tau) \cdot g(t - \tau)$ in Rose; the shaded area equals the convolution value $(f * g)(3)$. As t varies, this area traces out the complete convolution.

Chapter 3

Differential Equations

Differential equations describe how systems evolve over time and are the mathematical language of dynamics. In control systems, we encounter differential equations when modeling physical systems, analyzing stability, and designing controllers.

Definition: Ordinary Differential Equation (ODE)

An **ordinary differential equation** is an equation involving a function $y(t)$ and its derivatives with respect to a single independent variable t :

$$F\left(t, y, \frac{dy}{dt}, \frac{d^2y}{dt^2}, \dots, \frac{d^ny}{dt^n}\right) = 0 \quad (3.1)$$

The **order** of the ODE is the highest derivative that appears. An ODE is **linear** if the function y and all its derivatives appear linearly (to the first power and not multiplied together).

3.1 Ordinary Differential Equations (ODEs)

3.1.1 First-Order Linear ODEs

The general first-order linear ODE:

$$\frac{dy}{dt} + p(t)y = q(t) \quad (3.2)$$

Solution using Integrating Factor

Multiply by $\mu(t) = e^{\int p(t) dt}$:

$$y(t) = \frac{1}{\mu(t)} \left[\int \mu(t)q(t) dt + C \right] \quad (3.3)$$

For constant coefficients $\frac{dy}{dt} + ay = b$:

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a} \right) e^{-at} \quad (3.4)$$

Time Constant

For a first-order system $\tau \frac{dy}{dt} + y = Ku$, the **time constant** τ determines:

- The speed of response: smaller τ means faster response
- At $t = \tau$, the response reaches 63.2% of its final value
- At $t = 4\tau$, the response is within 2% of steady state
- The bandwidth is $\omega_{bw} = 1/\tau$ rad/s

Example: RC Circuit

An RC circuit with voltage source V_s satisfies:

$$RC \frac{dv_c}{dt} + v_c = V_s \quad (3.5)$$

With $v_c(0) = 0$, the capacitor voltage is:

$$v_c(t) = V_s (1 - e^{-t/RC}) \quad (3.6)$$

The time constant $\tau = RC$ determines how quickly the system responds.

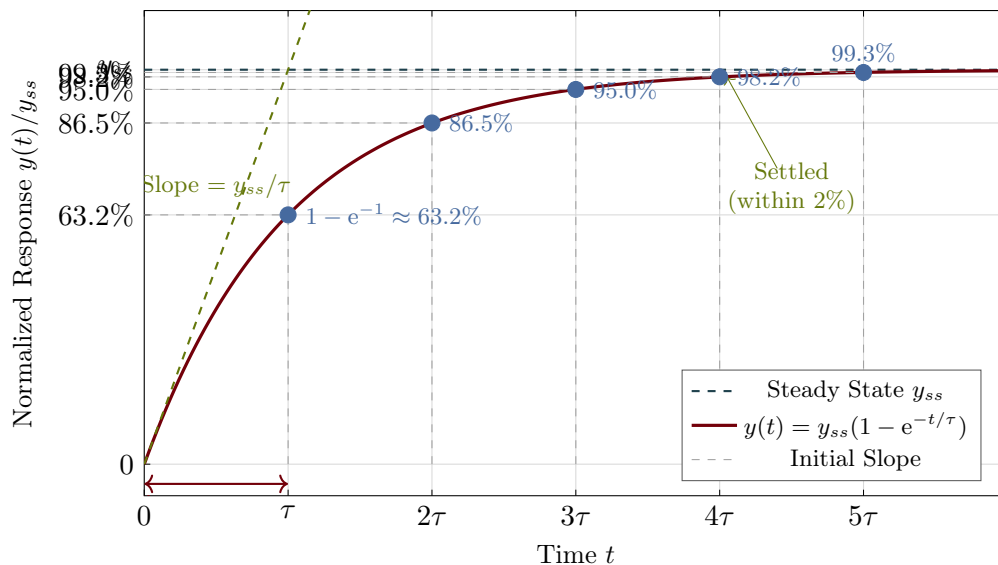


Figure 3.1: Step response of a first-order system $y(t) = y_{ss}(1 - e^{-t/\tau})$ showing the characteristic exponential approach to steady state. The time constant τ uniquely determines the response speed: at $t = \tau$, the response reaches 63.2% of the final value. Key milestones are marked at each multiple of τ : 63.2% at τ , 86.5% at 2τ , 95.0% at 3τ , 98.2% at 4τ , and 99.3% at 5τ . The initial tangent line (Horseshoe green) has slope y_{ss}/τ and would reach y_{ss} at $t = \tau$ if the rate were maintained. The system is typically considered settled (within 2% of y_{ss}) after 4τ .

3.1.2 Second-Order Linear ODEs with Constant Coefficients

The standard form:

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = \omega_n^2 u(t) \quad (3.7)$$

where:

- ω_n = natural frequency (rad/s)
- ζ = damping ratio (dimensionless)

The **characteristic equation** is:

$$\lambda^2 + 2\zeta\omega_n \lambda + \omega_n^2 = 0 \quad (3.8)$$

with roots:

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (3.9)$$

Cases based on damping ratio

1. **Overdamped** ($\zeta > 1$): Two distinct real roots. Response: $y(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$
2. **Critically Damped** ($\zeta = 1$): Repeated real root $\lambda = -\omega_n$. Response: $y(t) = (C_1 + C_2 t) e^{-\omega_n t}$
3. **Underdamped** ($0 < \zeta < 1$): Complex conjugate roots $\lambda = -\zeta\omega_n \pm j\omega_d$ where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Response:

$$y(t) = e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t) \quad (3.10)$$
4. **Undamped** ($\zeta = 0$): Pure imaginary roots. Response: $y(t) = C_1 \cos \omega_n t + C_2 \sin \omega_n t$

Example: Mass-Spring-Damper System

Consider a mass-spring-damper system with $m = 2$ kg, $c = 4$ N·s/m, and $k = 10$ N/m. The equation of motion is:

$$2\ddot{x} + 4\dot{x} + 10x = 0 \quad (3.11)$$

Step 1: Write in standard form by dividing by m :

$$\ddot{x} + 2\dot{x} + 5x = 0 \quad (3.12)$$

Step 2: Identify parameters. Comparing with $\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$:

- $\omega_n^2 = 5 \Rightarrow \omega_n = \sqrt{5} \approx 2.24$ rad/s
- $2\zeta\omega_n = 2 \Rightarrow \zeta = 1/\omega_n = 1/\sqrt{5} \approx 0.447$

Step 3: Since $\zeta < 1$, the system is **underdamped**. The damped frequency is:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = \sqrt{5} \sqrt{1 - 0.2} = \sqrt{4} = 2 \text{ rad/s} \quad (3.13)$$

Step 4: The general solution is:

$$x(t) = e^{-t} (C_1 \cos 2t + C_2 \sin 2t) \quad (3.14)$$

With initial conditions $x(0) = 1$ m, $\dot{x}(0) = 0$:

- From $x(0) = 1$: $C_1 = 1$
- From $\dot{x}(0) = 0$: $-C_1 + 2C_2 = 0 \Rightarrow C_2 = 0.5$

Final answer:

$$x(t) = e^{-t} (\cos 2t + 0.5 \sin 2t) \text{ m} \quad (3.15)$$

Example: Overdamped Second-Order System

Solve $\ddot{y} + 5\dot{y} + 4y = 0$ with $y(0) = 1$, $\dot{y}(0) = 0$.

Step 1: Find the characteristic equation:

$$\lambda^2 + 5\lambda + 4 = 0 \quad (3.16)$$

Step 2: Factor to find roots:

$$(\lambda + 1)(\lambda + 4) = 0 \Rightarrow \lambda_1 = -1, \quad \lambda_2 = -4 \quad (3.17)$$

Since both roots are real and distinct (and negative), the system is **overdamped and stable**.

Step 3: Write the general solution:

$$y(t) = C_1 e^{-t} + C_2 e^{-4t} \quad (3.18)$$

Step 4: Apply initial conditions:

$$y(0) = C_1 + C_2 = 1 \quad (3.19)$$

$$\dot{y}(0) = -C_1 - 4C_2 = 0 \quad (3.20)$$

Solving: $C_1 = 4/3$, $C_2 = -1/3$.

Final answer:

$$y(t) = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t} \quad (3.21)$$

Note: The response is dominated by the slower mode e^{-t} as t increases.

Example: Critically Damped System

Solve $\ddot{y} + 6\dot{y} + 9y = 0$ with $y(0) = 2$, $\dot{y}(0) = 3$.

Step 1: Characteristic equation:

$$\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0 \quad (3.22)$$

Step 2: Repeated root $\lambda = -3$ (critically damped, $\zeta = 1$, $\omega_n = 3$).

Step 3: General solution for repeated roots:

$$y(t) = (C_1 + C_2 t)e^{-3t} \quad (3.23)$$

Step 4: Apply initial conditions:

$$y(0) = C_1 = 2 \quad (3.24)$$

$$\dot{y}(t) = C_2 e^{-3t} - 3(C_1 + C_2 t)e^{-3t} \quad (3.25)$$

$$\dot{y}(0) = C_2 - 3C_1 = 3 \Rightarrow C_2 = 9 \quad (3.26)$$

Final answer:

$$y(t) = (2 + 9t)e^{-3t} \quad (3.27)$$

This represents the fastest decay without oscillation.

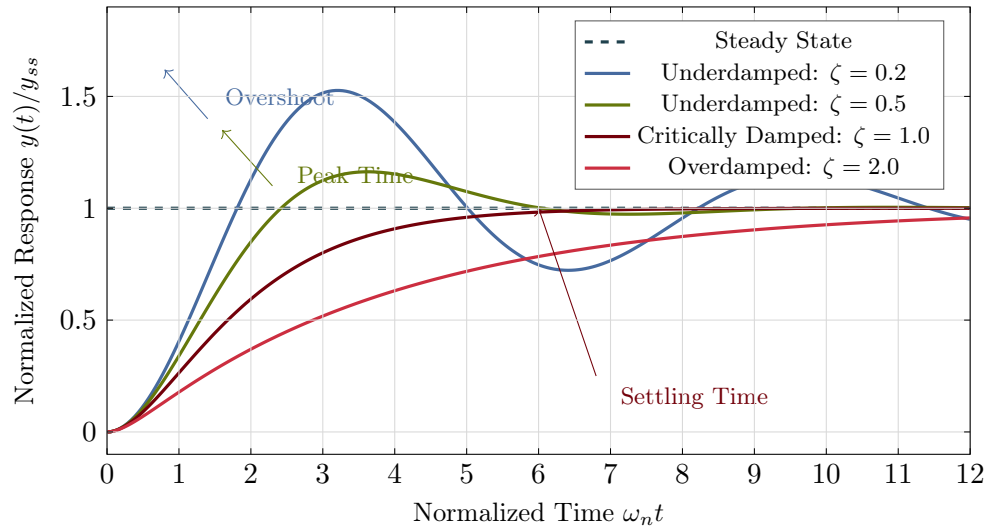


Figure 3.2: Step response of second-order systems for different damping ratios. The response shows how damping affects the transient behavior: underdamped ($\zeta < 1$) exhibits overshoot and oscillation, critically damped ($\zeta = 1$) provides fast response without overshoot, and overdamped ($\zeta > 1$) is slow but smooth. Key features include overshoot (Atlantic blue), peak time (Horseshoe green), and settling time (Garnet red) annotations. Dashed line shows steady-state value.

Common Pitfall: Sign Errors in Characteristic Equation

When solving the characteristic equation, be careful with signs:

- The ODE $\ddot{y} + a\dot{y} + by = 0$ gives characteristic equation $\lambda^2 + a\lambda + b = 0$
- Stable systems require $a > 0$ and $b > 0$ (all coefficients positive)
- A negative coefficient indicates instability

Example: $\ddot{y} - 2\dot{y} + y = 0$ has $\lambda = 1$ (repeated), which is **unstable** despite the positive coefficient of y .

3.1.3 Higher-Order Linear ODEs

The general n th-order linear ODE with constant coefficients:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = f(t) \quad (3.28)$$

The solution consists of:

- **Homogeneous solution** $y_h(t)$: From the characteristic equation roots

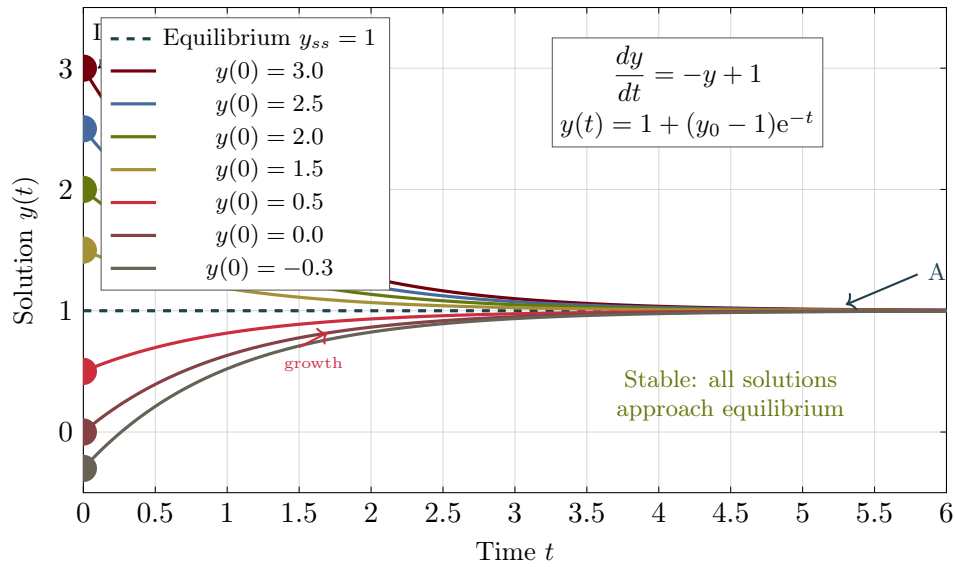


Figure 3.3: Family of solutions to the first-order ODE $\frac{dy}{dt} = -y + 1$ for different initial conditions. The general solution $y(t) = 1 + (y_0 - 1)e^{-t}$ demonstrates that all trajectories converge exponentially to the stable equilibrium $y_{ss} = 1$ (dashed Congaree line). Solutions starting above the equilibrium decay downward, while solutions starting below the equilibrium grow upward. This convergence behavior, regardless of initial condition, characterizes asymptotic stability: the equilibrium is globally attractive. The rate of convergence is determined by the eigenvalue $\lambda = -1$ (time constant $\tau = 1$).

- **Particular solution** $y_p(t)$: Depends on the forcing function $f(t)$

The complete solution is $y(t) = y_h(t) + y_p(t)$.

Example: Converting 3rd Order ODE to State-Space Form

Convert the following third-order ODE to state-space form:

$$\ddot{y} + 6\ddot{y} + 11\dot{y} + 6y = u(t) \quad (3.29)$$

Step 1: Define state variables. Choose:

$$x_1 = y \quad (3.30)$$

$$x_2 = \dot{y} = \dot{x}_1 \quad (3.31)$$

$$x_3 = \ddot{y} = \dot{x}_2 \quad (3.32)$$

Step 2: Write the first-order equations. From the original ODE:

$$\dot{x}_3 = \ddot{y} = -6\ddot{y} - 11\dot{y} - 6y + u = -6x_3 - 11x_2 - 6x_1 + u \quad (3.33)$$

Step 3: Assemble into matrix form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad (3.34)$$

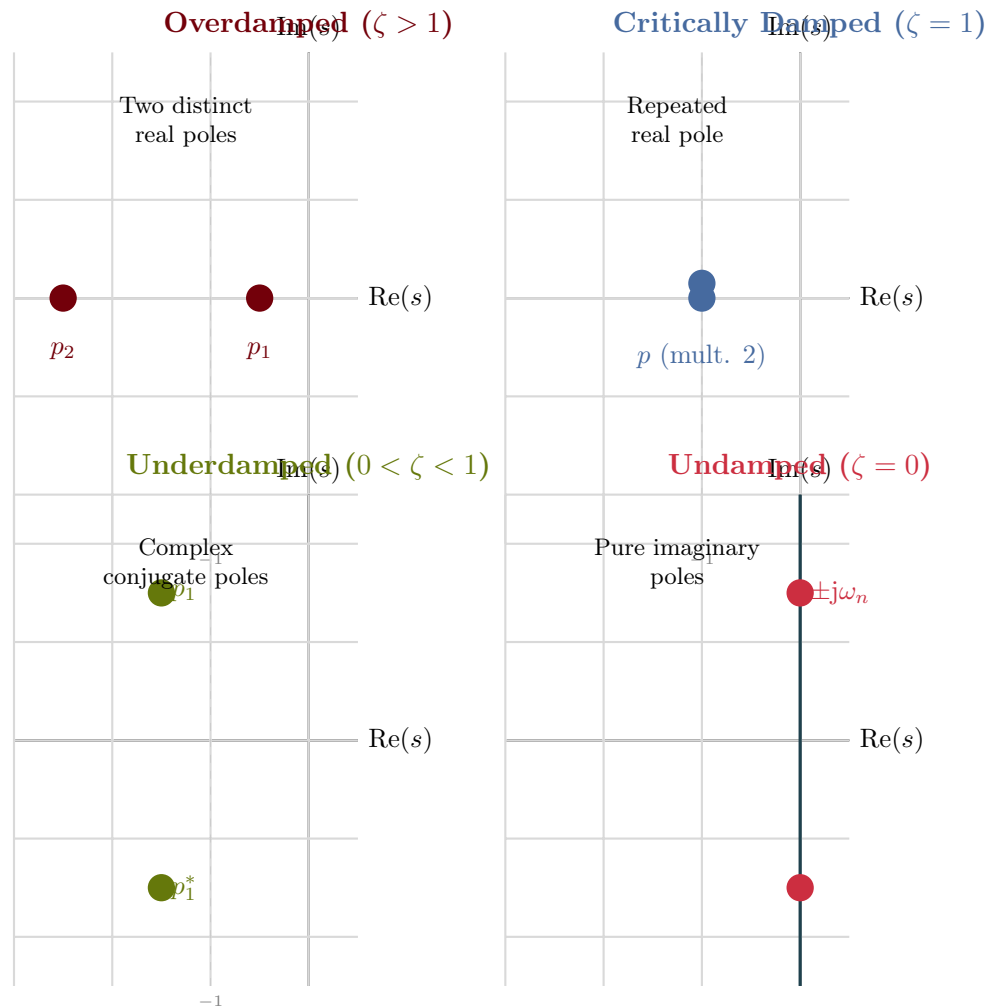


Figure 3.4: Pole locations in the s -plane for second-order systems under different damping conditions. Each subplot shows the characteristic pole configuration: overdamped (Garnet, two real poles on negative axis), critically damped (Atlantic, repeated real pole), underdamped (Horseshoe, complex conjugate pair), and undamped (UofSCRose, pure imaginary pair on the imaginary axis). The stability region (left half-plane) ensures all poles are in the left half-plane for stable responses.

Step 4: Output equation (if y is the output):

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{C}\mathbf{x} \quad (3.35)$$

This is the **controllable canonical form**. The characteristic polynomial of $\mathbf{A}$ matches the original ODE: $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^3 + 6\lambda^2 + 11\lambda + 6 = (\lambda + 1)(\lambda + 2)(\lambda + 3)$.

State-Space Conversion Recipe

For an n th-order ODE: $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1\dot{y} + a_0y = b_0u$

1. Define states: $x_1 = y, x_2 = \dot{y}, \dots, x_n = y^{(n-1)}$
2. The companion matrix form is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} \quad (3.36)$$

3. Output: $\mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}, \mathbf{D} = 0$

3.2 Solution Techniques for Nonhomogeneous ODEs

The complete solution of a nonhomogeneous linear ODE $L[y] = f(t)$ is:

$$y(t) = y_h(t) + y_p(t) \quad (3.37)$$

where $y_h(t)$ is the homogeneous solution (from Section 3) and $y_p(t)$ is a particular solution.

3.2.1 Method of Undetermined Coefficients

This method works when the forcing function $f(t)$ is a polynomial, exponential, sine, cosine, or a combination thereof.

Definition: Method of Undetermined Coefficients

For an ODE with constant coefficients $L[y] = f(t)$:

1. Identify the form of $f(t)$ from the table below
2. Guess a particular solution $y_p(t)$ with undetermined coefficients
3. Substitute $y_p(t)$ into the ODE and solve for the coefficients
4. If any term in y_p is a solution of the homogeneous equation, multiply by t (or t^2 for repeated roots)

Table of Particular Solution Forms

Forcing Function $f(t)$	Particular Solution Form $y_p(t)$
K (constant)	A
Kt^n	$A_nt^n + A_{n-1}t^{n-1} + \cdots + A_1t + A_0$
Ke^{at}	Ae^{at}
$K \cos(\omega t)$ or $K \sin(\omega t)$	$A \cos(\omega t) + B \sin(\omega t)$
$Ke^{at} \cos(\omega t)$ or $Ke^{at} \sin(\omega t)$	$e^{at}(A \cos(\omega t) + B \sin(\omega t))$
$Kt^n e^{at}$	$(A_nt^n + \cdots + A_0)e^{at}$

Modification Rule for Resonance

If any term in the guessed $y_p(t)$ is also a solution of the homogeneous equation:

- Multiply the entire guess by t
- If still a homogeneous solution (repeated roots), multiply by t^2

This accounts for **resonance** conditions.

Example: Undetermined Coefficients with Exponential Forcing

Solve $\ddot{y} + 3\dot{y} + 2y = 10e^{-3t}$ with $y(0) = 0$, $\dot{y}(0) = 0$.

Step 1: Find homogeneous solution. Characteristic equation: $\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0$

$$y_h(t) = C_1 e^{-t} + C_2 e^{-2t} \quad (3.38)$$

Step 2: Guess particular solution. Since $f(t) = 10e^{-3t}$ and e^{-3t} is not a homogeneous solution:

$$y_p(t) = Ae^{-3t} \quad (3.39)$$

Step 3: Substitute into ODE:

$$\dot{y}_p = -3Ae^{-3t}, \quad \ddot{y}_p = 9Ae^{-3t} \quad (3.40)$$

$$9Ae^{-3t} + 3(-3Ae^{-3t}) + 2(Ae^{-3t}) = 10e^{-3t} \quad (3.41)$$

$$(9A - 9A + 2A)e^{-3t} = 10e^{-3t} \quad (3.42)$$

$$2A = 10 \Rightarrow A = 5 \quad (3.43)$$

Step 4: General solution: $y(t) = C_1 e^{-t} + C_2 e^{-2t} + 5e^{-3t}$

Step 5: Apply initial conditions:

$$y(0) = C_1 + C_2 + 5 = 0 \quad (3.44)$$

$$\dot{y}(0) = -C_1 - 2C_2 - 15 = 0 \quad (3.45)$$

Solving: $C_1 = -20$, $C_2 = 15$.

Final answer:

$$y(t) = -20e^{-t} + 15e^{-2t} + 5e^{-3t} \quad (3.46)$$

Example: Undetermined Coefficients with Sinusoidal Forcing

Find the particular solution of $\ddot{y} + 4y = 3 \cos(2t)$.

Step 1: Check homogeneous solutions. Characteristic equation: $\lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2j$

$$y_h(t) = C_1 \cos(2t) + C_2 \sin(2t) \quad (3.47)$$

Step 2: Initial guess would be $A \cos(2t) + B \sin(2t)$, but this IS a homogeneous solution. Apply modification rule:

$$y_p(t) = t(A \cos(2t) + B \sin(2t)) \quad (3.48)$$

Step 3: Compute derivatives:

$$\dot{y}_p = A \cos(2t) + B \sin(2t) + t(-2A \sin(2t) + 2B \cos(2t)) \quad (3.49)$$

$$\ddot{y}_p = -4A \sin(2t) + 4B \cos(2t) - 4t(A \cos(2t) + B \sin(2t)) \quad (3.50)$$

Step 4: Substitute into $\ddot{y}_p + 4y_p = 3 \cos(2t)$:

$$-4A \sin(2t) + 4B \cos(2t) = 3 \cos(2t) \quad (3.51)$$

Comparing coefficients: $B = 3/4$, $A = 0$.

Final answer:

$$y_p(t) = \frac{3}{4}t \sin(2t) \quad (3.52)$$

Note: The factor of t indicates **resonance**—the response grows linearly with time.

3.2.2 Variation of Parameters

The method of variation of parameters provides a general formula that works for *any* forcing function $f(t)$, not just the special forms required for undetermined coefficients.

Variation of Parameters Formula (Second-Order)

For $\ddot{y} + p(t)\dot{y} + q(t)y = f(t)$ with homogeneous solutions $y_1(t)$ and $y_2(t)$:

$$y_p(t) = -y_1(t) \int \frac{y_2(t)f(t)}{W(t)} dt + y_2(t) \int \frac{y_1(t)f(t)}{W(t)} dt \quad (3.53)$$

where $W(t) = y_1\dot{y}_2 - \dot{y}_1y_2$ is the **Wronskian**.

For constant-coefficient equations, the Wronskian is constant (or a simple exponential), making the integrals more tractable.

When to Use Each Method

- **Undetermined coefficients:** Use when forcing is polynomial, exponential, sine/cosine, or products thereof. Faster and simpler.
- **Variation of parameters:** Use when forcing is more complex (e.g., $\tan t$, $\sec t$, arbitrary functions). Always works but integrals may be difficult.

- **Laplace transforms:** Often the easiest approach for control systems (Chapter 4). Handles initial conditions automatically.

3.3 Phase Plane Analysis

The phase plane provides a geometric visualization of second-order system behavior without explicitly solving the differential equation.

Definition: Phase Portrait

For a second-order system written as two first-order equations:

$$\dot{x}_1 = f_1(x_1, x_2) \quad (3.54)$$

$$\dot{x}_2 = f_2(x_1, x_2) \quad (3.55)$$

The **phase portrait** is the collection of all trajectories in the (x_1, x_2) plane, showing how the system state evolves from different initial conditions.

For linear systems $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, the eigenvalues of $\mathbf{A}$ determine the type of equilibrium:

Eigenvalues	Equilibrium Type	Stability
$\lambda_1, \lambda_2 < 0$ (real, distinct)	Stable node	Asymptotically stable
$\lambda_1, \lambda_2 > 0$ (real, distinct)	Unstable node	Unstable
$\lambda_1 < 0 < \lambda_2$ (real, opposite signs)	Saddle point	Unstable
$\lambda = \sigma \pm j\omega, \sigma < 0$	Stable focus (spiral)	Asymptotically stable
$\lambda = \sigma \pm j\omega, \sigma > 0$	Unstable focus (spiral)	Unstable
$\lambda = \pm j\omega$ (pure imaginary)	Center	Marginally stable

3.4 Stability Analysis

Stability is a fundamental property in control systems. A system is stable if its response remains bounded and eventually settles to equilibrium.

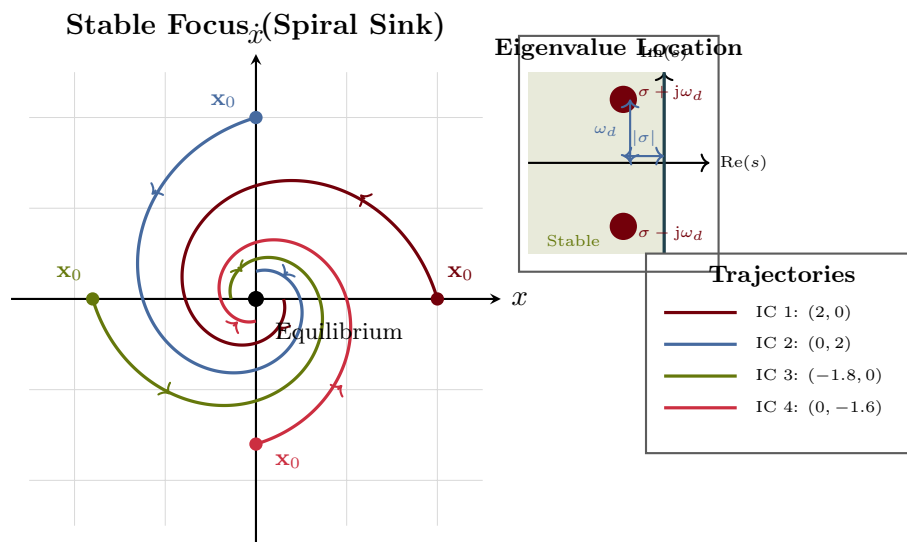
3.4.1 Definitions of Stability

Definition: Lyapunov Stability

An equilibrium point $\mathbf{x}_e$ is **Lyapunov stable** (or stable in the sense of Lyapunov) if, for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\|\mathbf{x}(0) - \mathbf{x}_e\| < \delta \Rightarrow \|\mathbf{x}(t) - \mathbf{x}_e\| < \epsilon \quad \forall t \geq 0 \quad (3.56)$$

In other words, trajectories starting close to equilibrium remain close.



$|\sigma|$ controls **decay rate**, ω_d controls **spiral frequency**

Figure 3.5: Phase portrait for a stable second-order system (stable focus, $\zeta = 0.3$, $\omega_n = 1$). Trajectories in the state space $(x, \dot{x})$ spiral inward toward the equilibrium at the origin, demonstrating asymptotic stability. The spiral pattern results from complex conjugate eigenvalues $\lambda = \sigma \pm j\omega_d$ with negative real part ($\sigma < 0$). The inset shows eigenvalue locations in the s -plane: the real part σ determines the decay rate (how quickly spirals shrink), while the imaginary part ω_d determines the oscillation frequency (how tightly spirals wind). All trajectories from different initial conditions converge to the stable equilibrium.

Definition: Asymptotic Stability

An equilibrium is **asymptotically stable** if it is Lyapunov stable AND:

$$\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}_e \quad (3.57)$$

for all initial conditions in some neighborhood of $\mathbf{x}_e$. The system not only stays close but converges to equilibrium.

Definition: BIBO Stability

A system is **Bounded-Input Bounded-Output (BIBO) stable** if every bounded input produces a bounded output:

$$|u(t)| \leq M_u < \infty \quad \forall t \geq 0 \Rightarrow |y(t)| \leq M_y < \infty \quad \forall t \geq 0 \quad (3.58)$$

This is the most common stability notion for input-output (transfer function) descriptions.

Stability Criterion for LTI Systems

For a linear time-invariant (LTI) system, the following are equivalent:

1. The system is asymptotically stable
2. All eigenvalues of $\mathbf{A}$ have negative real parts: $\text{Re}(\lambda_i) < 0$
3. All poles of the transfer function lie in the open left half-plane
4. The impulse response decays to zero: $\lim_{t \rightarrow \infty} h(t) = 0$

For BIBO stability, the additional requirement is that there are no pole-zero cancellations in the right half-plane.

3.4.2 Pole Locations and System Response

The location of poles (eigenvalues) in the complex plane directly determines system behavior:

- **Real part** $\sigma = \text{Re}(\lambda)$: Determines stability and decay/growth rate
 - $\sigma < 0$: Decaying response (stable)
 - $\sigma = 0$: Sustained oscillation or constant (marginally stable)
 - $\sigma > 0$: Growing response (unstable)
- **Imaginary part** $\omega = \text{Im}(\lambda)$: Determines oscillation frequency
 - $\omega = 0$: No oscillation (real pole)
 - $\omega \neq 0$: Oscillation at frequency $|\omega|$ rad/s
- **Distance from origin** $|\lambda|$: Determines speed of response
 - Larger $|\lambda|$: Faster response
 - Smaller $|\lambda|$: Slower response (dominant pole)

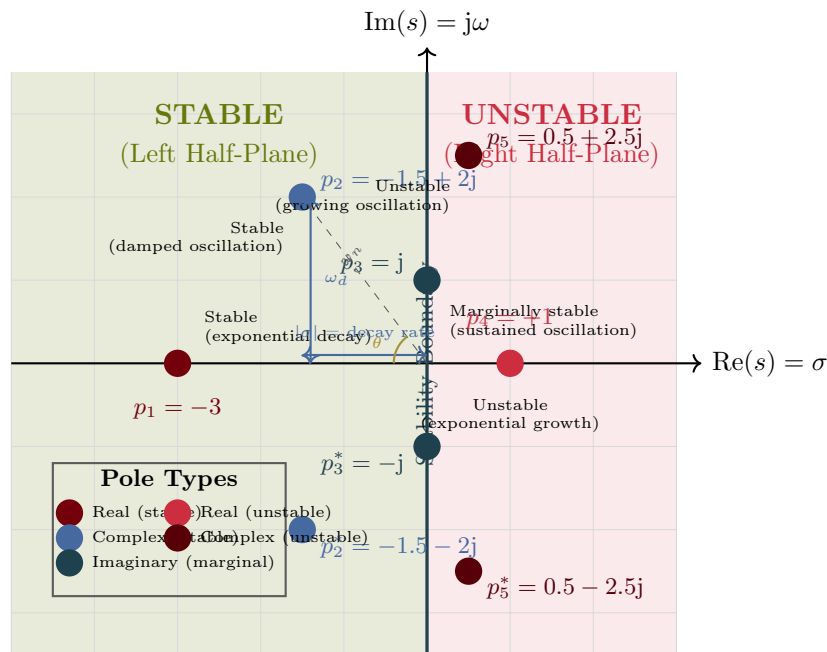


Figure 3.6: Stability regions in the complex s -plane. The left half-plane (LHP, shaded green) contains stable poles where $\text{Re}(s) < 0$, producing decaying responses. The right half-plane (RHP, shaded pink) contains unstable poles where $\text{Re}(s) > 0$, producing unbounded responses. The imaginary axis (Congaree, emphasized) is the stability boundary: poles on this axis yield marginally stable systems with sustained oscillations. Example poles demonstrate different behaviors: real stable pole (p_1 , exponential decay), complex stable pair (p_2, p_2^* , damped oscillation), imaginary pair (p_3, p_3^* , sustained oscillation), real unstable pole (p_4 , exponential growth), and complex unstable pair (p_5, p_5^* , growing oscillation).

3.4.3 The Routh-Hurwitz Criterion (Preview)

For higher-order systems, directly computing eigenvalues can be difficult. The **Routh-Hurwitz criterion** provides an algebraic test for stability without solving the characteristic equation.

Routh-Hurwitz Stability Criterion

For a characteristic polynomial:

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0 \quad (3.59)$$

Necessary conditions (quick check):

1. All coefficients a_i must be present (no missing terms)
2. All coefficients must have the same sign (typically all positive)

If either condition fails, the system is unstable.

The **sufficient condition** requires constructing the Routh array and checking that all entries in the first column have the same sign. The number of sign changes equals the number of right half-plane poles.

Note: The complete Routh-Hurwitz procedure is covered in detail in the classical control chapters.

Example: Quick Stability Check

Determine if the following characteristic polynomials represent stable systems:

(a) $s^3 + 4s^2 + 6s + 4 = 0$

All coefficients present and positive. Passes necessary conditions. (Routh array confirms stability: roots are $-2, -1 \pm j$.)

(b) $s^3 + 2s^2 + s + 8 = 0$

All coefficients present and positive. Passes necessary conditions. However, Routh array reveals two RHP poles—**unstable**.

(c) $s^4 + 2s^2 + 1 = 0$

Missing s^3 and s^1 terms. Fails necessary condition—**unstable** (or marginally stable with imaginary axis poles).

(d) $s^3 - 2s^2 + s + 4 = 0$

Coefficient of s^2 is negative while others are positive. Sign inconsistency—**unstable**.

Marginal Stability Warning

Systems with poles exactly on the imaginary axis ($\text{Re}(\lambda) = 0$) are **marginally stable**:

- Bounded response to bounded inputs only for some input types
- Sustained oscillations (not decaying, not growing)
- Any perturbation can push the system to instability
- In practice, treat marginally stable systems as potentially unstable

3.5 Systems of First-Order ODEs

Any higher-order ODE can be written as a system of first-order equations. This is the **state-space** representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (3.60)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector, and $\mathbf{A}$, $\mathbf{B}$ are system matrices.

The solution is:

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau \quad (3.61)$$

where $e^{\mathbf{A}t}$ is the **matrix exponential** (discussed in Chapter 6).

Connection: Poles, Eigenvalues, and Stability

For a linear system, the following are all equivalent characterizations:

Representation	Stability Determined By	Condition
ODE: $a_n y^{(n)} + \cdots + a_0 y = f(t)$	Roots of characteristic polynomial	$\text{Re}(\lambda_i) < 0$
Transfer function: $G(s) = N(s)/D(s)$	Poles (roots of $D(s)$)	$\text{Re}(p_i) < 0$
State-space: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$	Eigenvalues of $\mathbf{A}$	$\text{Re}(\lambda_i) < 0$

All three approaches yield the same stability conclusion for a given system.

Chapter 4

Laplace Transforms

The Laplace transform converts differential equations into algebraic equations, making system analysis much simpler.

4.1 Definition

Definition 0.2: Laplace Transform

The **Laplace transform** of a function $f(t)$ defined for $t \geq 0$ is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt \quad (4.1)$$

where $s = \sigma + j\omega$ is a complex variable.

The complex variable $s = \sigma + j\omega$ defines a plane called the **s-plane** or **complex frequency plane**. The real part σ determines exponential growth or decay, while the imaginary part ω determines oscillation frequency.

The **inverse Laplace transform** is:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds \quad (4.2)$$

In practice, we use tables and partial fraction expansion rather than direct integration.

Example: Deriving the Laplace Transform of te^{-at} from Definition

Derive the Laplace transform of $f(t) = te^{-at}$ directly from the definition.

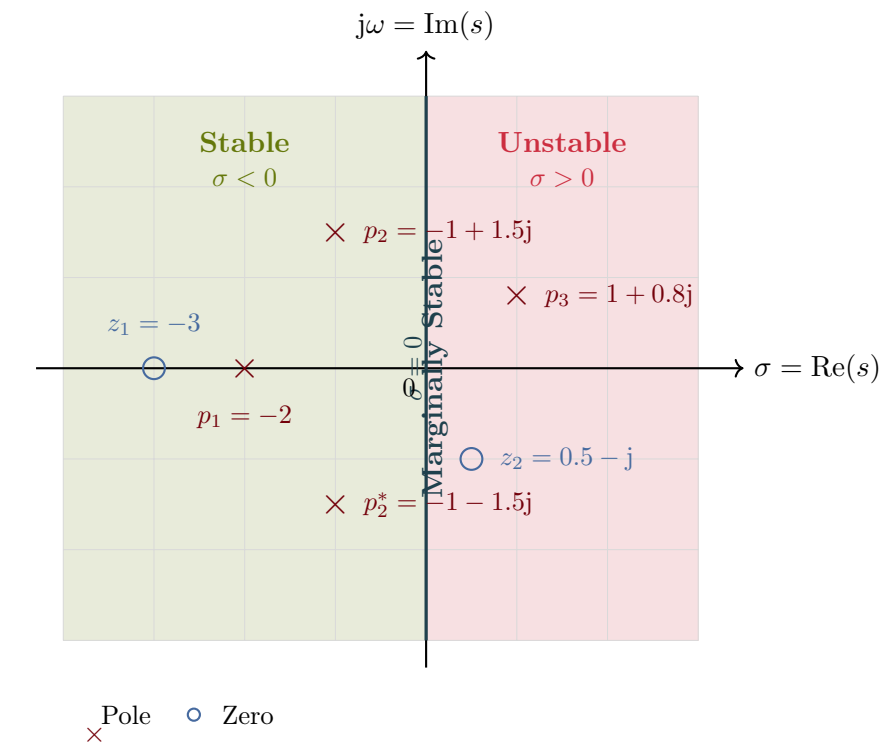
Solution: Applying the definition:

$$F(s) = \int_0^{\infty} te^{-at}e^{-st} dt = \int_0^{\infty} te^{-(s+a)t} dt \quad (4.3)$$

Use integration by parts with $u = t$ and $dv = e^{-(s+a)t}dt$:

$$du = dt \quad (4.4)$$

$$v = \frac{-1}{s+a}e^{-(s+a)t} \quad (4.5)$$



Time Response Characteristics by Pole Location

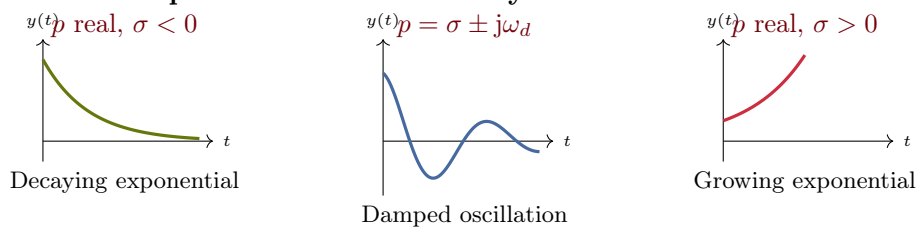


Figure 4.1: The complex s -plane showing stability regions and pole-zero locations. The left half-plane (Horseshoe green shading, $\sigma < 0$) corresponds to stable system responses, while the right half-plane (Rose shading, $\sigma > 0$) produces unstable responses. The imaginary axis (Congaree, $\sigma = 0$) represents marginal stability. Poles are marked with Garnet $\times$ symbols, zeros with Atlantic $\circ$ symbols. Bottom panels show representative time responses: stable real poles produce decaying exponentials, complex conjugate poles in the LHP produce damped oscillations, and RHP poles lead to unbounded growth.

Applying integration by parts:

$$F(s) = \left[\frac{-t}{s+a} e^{-(s+a)t} \right]_0^\infty + \frac{1}{s+a} \int_0^\infty e^{-(s+a)t} dt \quad (4.6)$$

$$= 0 + \frac{1}{s+a} \cdot \frac{1}{s+a} \quad (4.7)$$

$$= \frac{1}{(s+a)^2} \quad (4.8)$$

where we used the fact that $\lim_{t \rightarrow \infty} te^{-(s+a)t} = 0$ for $\text{Re}(s) > -a$.

Therefore:

$$\boxed{\mathcal{L}\{te^{-at}\} = \frac{1}{(s+a)^2}} \quad (4.9)$$

4.2 Region of Convergence

The Laplace transform integral does not converge for all values of s . The set of s values for which the integral converges is called the **Region of Convergence (ROC)**.

Definition: Region of Convergence

The **Region of Convergence (ROC)** of a Laplace transform $F(s)$ is the set of all complex values s for which the integral $\int_0^\infty f(t)e^{-st} dt$ converges (exists and is finite).

For most signals encountered in control systems, the ROC is a half-plane of the form $\text{Re}(s) > \sigma_0$, where σ_0 is called the **abscissa of convergence**.

ROC and Causality

The ROC determines the causality of the corresponding time-domain signal:

- **Right-sided signals** (causal, $f(t) = 0$ for $t < 0$): ROC is a right half-plane $\text{Re}(s) > \sigma_0$
- **Left-sided signals** (anti-causal): ROC is a left half-plane $\text{Re}(s) < \sigma_0$
- **Two-sided signals**: ROC is a vertical strip $\sigma_1 < \text{Re}(s) < \sigma_2$

Important Note

In control systems, we **always assume causal (right-sided) signals**. This means we work with ROCs that are right half-planes extending to $\text{Re}(s) \rightarrow +\infty$. This assumption is implicit in all transform pair tables and partial fraction methods.

For example, the transform pair $e^{-at} \leftrightarrow \frac{1}{s+a}$ assumes causality. The same $F(s) = \frac{1}{s+a}$ with a left half-plane ROC would correspond to $-e^{-at}u(-t)$, an anti-causal signal.

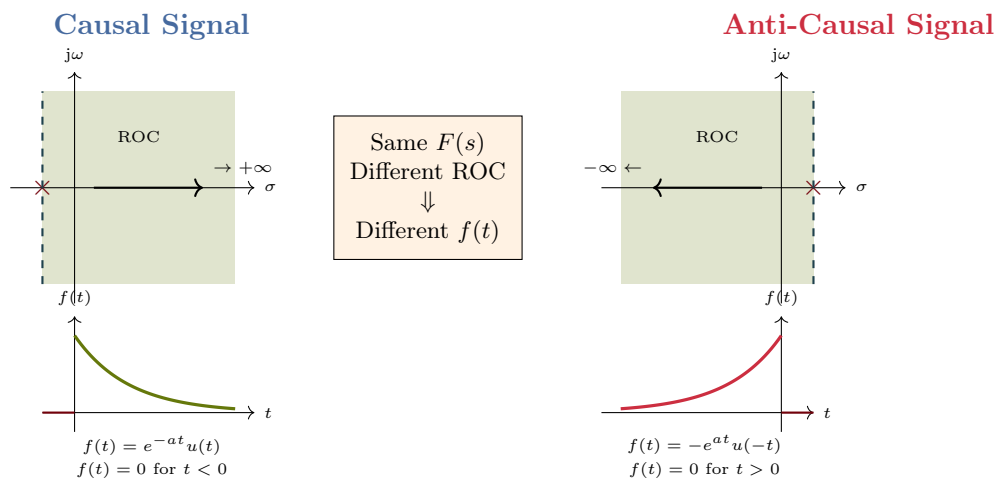
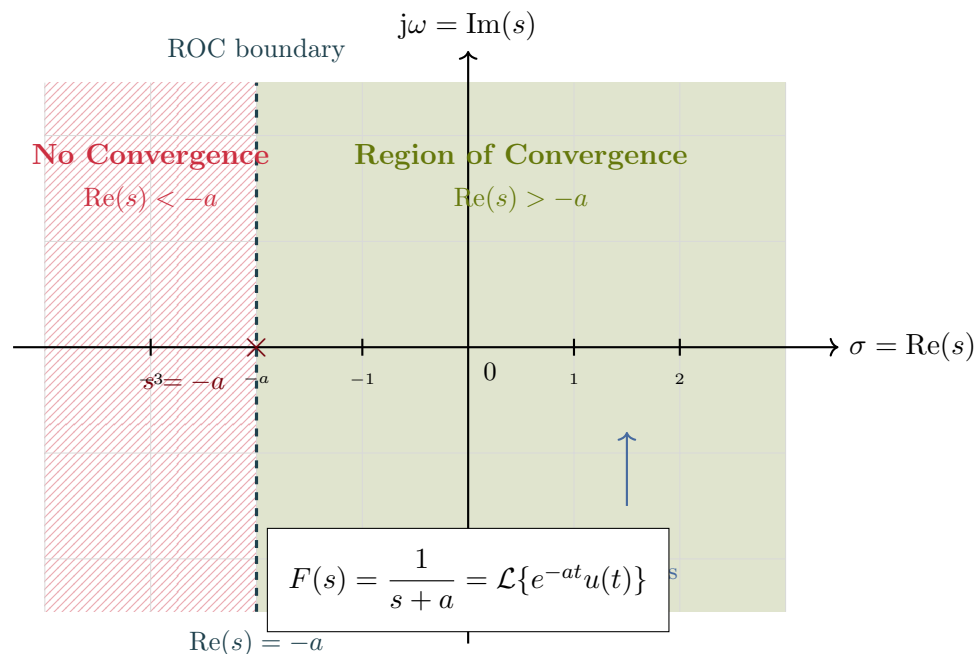


Figure 4.2: Region of Convergence (ROC) for the Laplace transform $F(s) = 1/(s + a)$. The ROC (Horseshoe green shading) is the half-plane $\text{Re}(s) > -a$, to the right of the pole at $s = -a$ (Garnet $\times$). The hatched region shows where the Laplace integral does not converge. The ROC determines whether the inverse transform yields a causal or anti-causal time-domain signal: ROC extending to $+\infty$ corresponds to the causal signal $e^{-at}u(t)$, while ROC extending to $-\infty$ yields the anti-causal signal $-e^{at}u(-t)$. For control systems, we typically assume causal signals, so the ROC extends rightward from the rightmost pole.

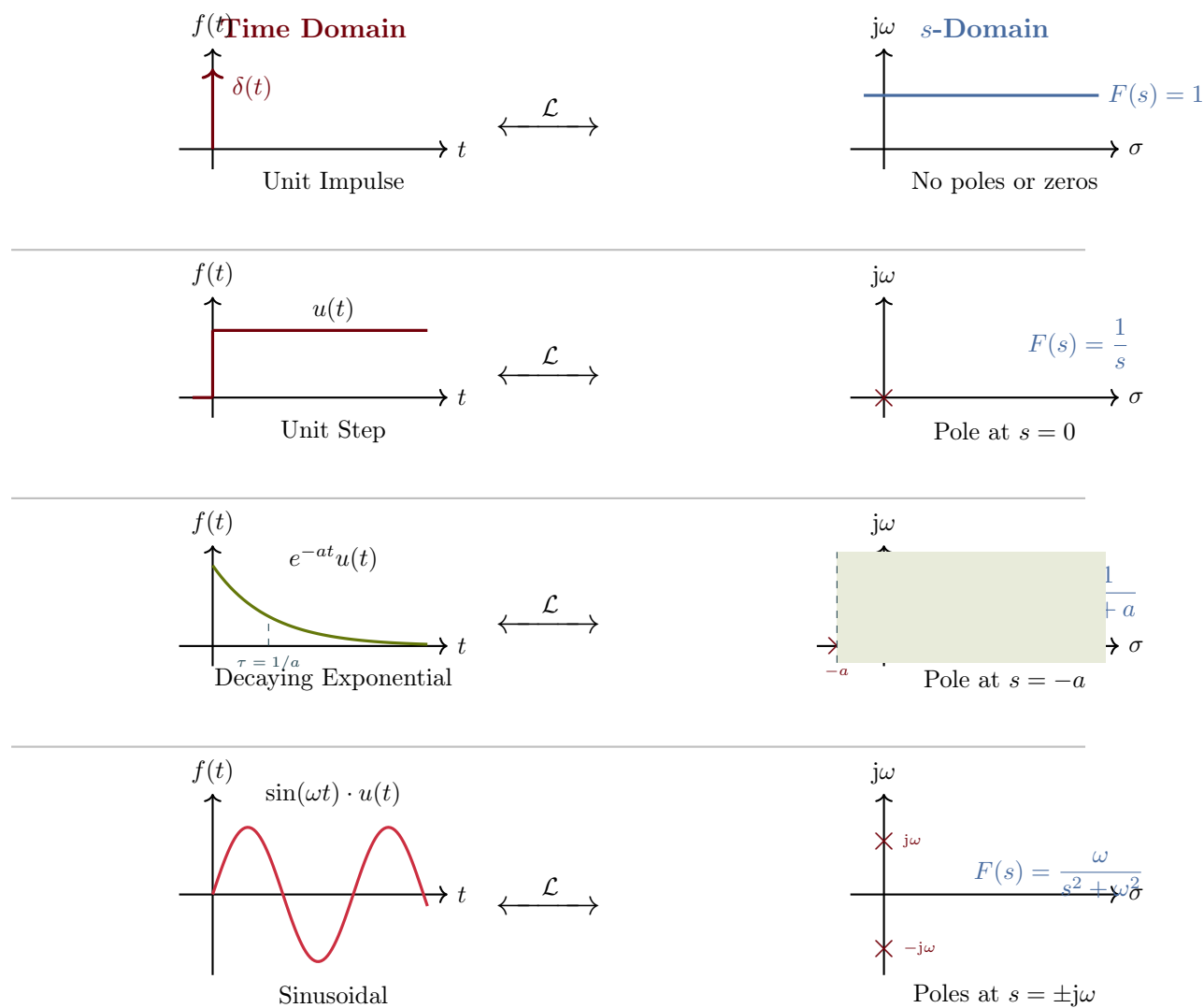


Figure 4.3: Visual comparison of fundamental Laplace transform pairs showing the correspondence between time-domain signals (left, Garnet/Horseshoe/Rose) and their s -domain representations (right, Atlantic). From top to bottom: (1) the unit impulse $\delta(t)$ transforms to constant 1 with no poles; (2) the unit step $u(t)$ transforms to $1/s$ with a pole at the origin; (3) the decaying exponential $e^{-at}u(t)$ transforms to $1/(s+a)$ with a pole at $s = -a$ (the shaded region shows the ROC); (4) the sine wave $\sin(\omega t)u(t)$ transforms to $\omega/(s^2 + \omega^2)$ with complex conjugate poles on the imaginary axis. Pole locations directly determine the time-domain behavior.

Signal Name	$f(t)$ for $t \geq 0$	$F(s)$
Impulse	$\delta(t)$	1
Unit step	$u(t) = 1$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Parabolic	t^2	$\frac{2}{s^3}$
Power	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Decaying ramp	te^{-at}	$\frac{1}{(s+a)^2}$
Decaying power	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
Sine	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
Cosine	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
Damped sine	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
Damped cosine	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Table 4.1: Common Laplace transform pairs.

4.3 Common Laplace Transform Pairs

4.4 Laplace Transform Properties

Property	Time Domain	s -Domain
Linearity	$af(t) + bg(t)$	$aF(s) + bG(s)$
Differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
Second derivative	$\frac{d^2f}{dt^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
n th derivative	$\frac{d^nf}{dt^n}$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$
Integration	$\int_0^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Time delay	$f(t - \tau)u(t - \tau)$	$e^{-\tau s}F(s)$
Frequency shift	$e^{-at}f(t)$	$F(s + a)$
Time scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Multiplication by t	$tf(t)$	$-\frac{dF}{ds}$
Convolution	$f(t) * g(t)$	$F(s)G(s)$
Initial value	$\lim_{t \rightarrow 0^+} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$

Table 4.2: Laplace transform properties.

Example: Using the Differentiation Property to Find $\mathcal{L}\{t^2\}$

Use the differentiation property of Laplace transforms to find $\mathcal{L}\{t^2\}$.

Solution: The “multiplication by t ” property states:

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds} \quad (4.10)$$

We know that $\mathcal{L}\{t\} = \frac{1}{s^2}$.

Apply the property to find $\mathcal{L}\{t \cdot t\} = \mathcal{L}\{t^2\}$:

$$\mathcal{L}\{t^2\} = \mathcal{L}\{t \cdot t\} = -\frac{d}{ds} \left(\frac{1}{s^2} \right) \quad (4.11)$$

$$= - \left(\frac{-2}{s^3} \right) \quad (4.12)$$

$$= \frac{2}{s^3} \quad (4.13)$$

Therefore:

$$\boxed{\mathcal{L}\{t^2\} = \frac{2}{s^3}} \quad (4.14)$$

This pattern generalizes: applying the property n times gives $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$.

Limitations of the Final Value Theorem

The **Final Value Theorem** only applies if all poles of $sF(s)$ are in the left half-plane (LHP). If any poles are on the imaginary axis or in the right half-plane, the theorem gives incorrect results.

Examples of invalid applications:

- $F(s) = \frac{\omega}{s^2 + \omega^2}$ (sine wave): Poles at $s = \pm j\omega$ on imaginary axis. FVT would incorrectly give 0.
- $F(s) = \frac{1}{s-1}$ (exponentially growing): Pole at $s = 1$ in RHP. FVT would incorrectly give -1 .
- $F(s) = \frac{1}{s^2}$ (ramp): Double pole at $s = 0$. Limit does not exist.

Example: Initial and Final Value Theorems

For the transfer function $F(s) = \frac{10(s+2)}{s(s+1)(s+5)}$, find the initial and final values of $f(t)$.

Solution:

Initial Value Theorem:

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} s \cdot \frac{10(s+2)}{s(s+1)(s+5)} \quad (4.15)$$

$$= \lim_{s \rightarrow \infty} \frac{10(s+2)}{(s+1)(s+5)} \quad (4.16)$$

$$= \lim_{s \rightarrow \infty} \frac{10s+20}{s^2+6s+5} \quad (4.17)$$

$$= \lim_{s \rightarrow \infty} \frac{10/s+20/s^2}{1+6/s+5/s^2} = 0 \quad (4.18)$$

Final Value Theorem: First, check validity: the poles of $sF(s)$ are at $s = -1$ and $s = -5$,

both in the LHP. The theorem is valid.

$$f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} s \cdot \frac{10(s+2)}{s(s+1)(s+5)} \quad (4.19)$$

$$= \lim_{s \rightarrow 0} \frac{10(s+2)}{(s+1)(s+5)} \quad (4.20)$$

$$= \frac{10(0+2)}{(0+1)(0+5)} = \frac{20}{5} = 4 \quad (4.21)$$

Therefore:

$$\boxed{f(0^+) = 0, \quad f(\infty) = 4} \quad (4.22)$$

4.5 Partial Fraction Expansion

To find the inverse Laplace transform, we decompose rational functions into simpler terms.

Distinct Real Poles

$$\frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} = \frac{A_1}{s+p_1} + \frac{A_2}{s+p_2} + \cdots + \frac{A_n}{s+p_n} \quad (4.23)$$

The residue A_i is found by:

$$A_i = \lim_{s \rightarrow -p_i} (s+p_i)F(s) \quad (4.24)$$

Repeated Poles

For a pole of multiplicity m at $s = -p$:

$$\frac{N(s)}{(s+p)^m} = \frac{A_1}{s+p} + \frac{A_2}{(s+p)^2} + \cdots + \frac{A_m}{(s+p)^m} \quad (4.25)$$

The coefficients are found by:

$$A_m = \lim_{s \rightarrow -p} (s+p)^m F(s) \quad (4.26)$$

$$A_{m-1} = \lim_{s \rightarrow -p} \frac{d}{ds} [(s+p)^m F(s)] \quad (4.27)$$

$$A_{m-k} = \lim_{s \rightarrow -p} \frac{1}{k!} \frac{d^k}{ds^k} [(s+p)^m F(s)] \quad (4.28)$$

Complex Conjugate Poles

For poles at $s = -\sigma \pm j\omega$:

$$\frac{N(s)}{(s+\sigma)^2 + \omega^2} = \frac{As+B}{(s+\sigma)^2 + \omega^2} \quad (4.29)$$

Example 0.3: Partial Fraction Expansion with Distinct Real Poles

Find the inverse Laplace transform of:

$$F(s) = \frac{2s + 3}{(s + 1)(s + 2)} \quad (4.30)$$

Solution:

$$F(s) = \frac{A}{s + 1} + \frac{B}{s + 2} \quad (4.31)$$

Residues:

$$A = \lim_{s \rightarrow -1} (s + 1) \cdot \frac{2s + 3}{(s + 1)(s + 2)} = \frac{2(-1) + 3}{-1 + 2} = 1 \quad (4.32)$$

$$B = \lim_{s \rightarrow -2} (s + 2) \cdot \frac{2s + 3}{(s + 1)(s + 2)} = \frac{2(-2) + 3}{-2 + 1} = 1 \quad (4.33)$$

Therefore:

$$f(t) = e^{-t} + e^{-2t}, \quad t \geq 0 \quad (4.34)$$

Example: Partial Fraction Expansion with Complex Poles

Find the inverse Laplace transform of:

$$F(s) = \frac{s + 3}{s^2 + 2s + 5} \quad (4.35)$$

Solution: First, identify the poles by factoring the denominator:

$$s^2 + 2s + 5 = (s + 1)^2 + 4 \quad (4.36)$$

The poles are at $s = -1 \pm 2j$ (complex conjugates with $\sigma = 1$, $\omega = 2$).

Rewrite the numerator to match standard forms. Complete the square in the numerator relative to $(s + 1)$:

$$s + 3 = (s + 1) + 2 \quad (4.37)$$

Therefore:

$$F(s) = \frac{(s + 1) + 2}{(s + 1)^2 + 4} \quad (4.38)$$

$$= \frac{s + 1}{(s + 1)^2 + 4} + \frac{2}{(s + 1)^2 + 4} \quad (4.39)$$

$$= \frac{s + 1}{(s + 1)^2 + 2^2} + \frac{2}{(s + 1)^2 + 2^2} \quad (4.40)$$

Using transform pairs for damped cosine and damped sine:

- $\frac{s+a}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \cos \omega t$
- $\frac{\omega}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \sin \omega t$

Therefore:

$$\boxed{f(t) = e^{-t} \cos 2t + e^{-t} \sin 2t = e^{-t}(\cos 2t + \sin 2t), \quad t \geq 0} \quad (4.41)$$

Alternative form: Using the identity $A \cos \theta + B \sin \theta = \sqrt{A^2 + B^2} \cos(\theta - \phi)$ where $\phi = \arctan(B/A)$:

$$f(t) = \sqrt{2}e^{-t} \cos(2t - \pi/4), \quad t \geq 0 \quad (4.42)$$

4.6 Inverse Laplace Transform Techniques

While the formal inverse Laplace transform involves a complex contour integral, practical methods rely on partial fractions and transform tables.

4.6.1 The Cover-Up Method (Heaviside Expansion)

For distinct poles, the **cover-up method** (also called Heaviside's expansion theorem) provides a quick way to find residues.

Cover-Up Method

For $F(s) = \frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$ with distinct poles, the residue at pole $-p_i$ is:

$$A_i = \left. \frac{N(s)}{(s+p_1)\cdots\cancel{(s+p_i)}\cdots(s+p_n)} \right|_{s=-p_i} \quad (4.43)$$

That is: “cover up” the factor $(s+p_i)$ in the denominator and evaluate the rest at $s = -p_i$.

Example: Cover-Up Method

Find $\mathcal{L}^{-1} \left\{ \frac{5s+7}{(s+1)(s+2)(s+3)} \right\}$.

Solution: Using the cover-up method:

$$A_1 = \left. \frac{5s+7}{(s+2)(s+3)} \right|_{s=-1} = \frac{5(-1)+7}{(-1+2)(-1+3)} = \frac{2}{1 \cdot 2} = 1 \quad (4.44)$$

$$A_2 = \left. \frac{5s+7}{(s+1)(s+3)} \right|_{s=-2} = \frac{5(-2)+7}{(-2+1)(-2+3)} = \frac{-3}{(-1)(1)} = 3 \quad (4.45)$$

$$A_3 = \left. \frac{5s+7}{(s+1)(s+2)} \right|_{s=-3} = \frac{5(-3)+7}{(-3+1)(-3+2)} = \frac{-8}{(-2)(-1)} = -4 \quad (4.46)$$

Therefore:

$$F(s) = \frac{1}{s+1} + \frac{3}{s+2} - \frac{4}{s+3} \quad (4.47)$$

And the inverse transform is:

$$\boxed{f(t) = e^{-t} + 3e^{-2t} - 4e^{-3t}, \quad t \geq 0} \quad (4.48)$$

4.6.2 Completing the Square for Complex Poles

When the denominator has complex poles, completing the square converts the quadratic to a form matching standard transform pairs.

Completing the Square

For a quadratic $s^2 + bs + c$, complete the square:

$$s^2 + bs + c = \left(s + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) = (s + \sigma)^2 + \omega^2 \quad (4.49)$$

where $\sigma = b/2$ and $\omega = \sqrt{c - b^2/4}$ (assuming $c > b^2/4$ for complex poles).

Example: Completing the Square

Find $\mathcal{L}^{-1}\left\{\frac{3}{s^2+4s+13}\right\}$.

Solution: Complete the square in the denominator:

$$s^2 + 4s + 13 = (s + 2)^2 - 4 + 13 = (s + 2)^2 + 9 = (s + 2)^2 + 3^2 \quad (4.50)$$

Rewrite $F(s)$:

$$F(s) = \frac{3}{(s + 2)^2 + 3^2} \quad (4.51)$$

This matches the damped sine form $\frac{\omega}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \sin \omega t$, but we have 3 in the numerator and need $\omega = 3$.

$$F(s) = \frac{3}{(s + 2)^2 + 3^2} = 1 \cdot \frac{3}{(s + 2)^2 + 3^2} \quad (4.52)$$

Therefore:

$$\boxed{f(t) = e^{-2t} \sin 3t, \quad t \geq 0} \quad (4.53)$$

4.6.3 Using the Frequency Shifting Theorem

The frequency shifting property $\mathcal{L}\{e^{-at}f(t)\} = F(s + a)$ allows us to handle damped signals systematically.

Inverse Frequency Shift

If $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}^{-1}\{F(s + a)\} = e^{-at}f(t) \quad (4.54)$$

To use this: identify the shift a , find the unshifted inverse transform, then multiply by e^{-at} .

Example: Frequency Shifting

Find $\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^3}\right\}$.

Solution: Recognize this as a shifted version of $\frac{1}{s^3}$:

$$F(s) = \frac{1}{(s+3)^3} = G(s+3) \quad \text{where} \quad G(s) = \frac{1}{s^3} \quad (4.55)$$

The inverse of $G(s) = \frac{1}{s^3}$ is:

$$g(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} = \frac{t^2}{2} \quad (4.56)$$

By the frequency shift theorem:

$$f(t) = e^{-3t} \cdot \frac{t^2}{2} = \frac{t^2 e^{-3t}}{2}, \quad t \geq 0 \quad (4.57)$$

4.7 Solving ODEs with Laplace Transforms

The Laplace transform provides a systematic method for solving linear ODEs with constant coefficients.

Procedure for Solving ODEs with Laplace Transforms

1. Take the Laplace transform of both sides of the ODE
2. Use the differentiation property to handle derivatives, substituting initial conditions
3. Solve algebraically for $Y(s)$
4. Find $y(t)$ by taking the inverse Laplace transform

Key formulas for incorporating initial conditions:

$$\mathcal{L}\{y'\} = sY(s) - y(0) \quad (4.58)$$

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) \quad (4.59)$$

$$\mathcal{L}\{y'''\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0) \quad (4.60)$$

Example: Solving a Second-Order ODE

Solve the initial value problem:

$$y'' + 3y' + 2y = e^{-t}, \quad y(0) = 1, \quad y'(0) = 0 \quad (4.61)$$

Solution:

Step 1: Take the Laplace transform of both sides.

$$\mathcal{L}\{y'' + 3y' + 2y\} = \mathcal{L}\{e^{-t}\} \quad (4.62)$$

Step 2: Apply differentiation property with initial conditions.

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - s - 0 = s^2Y(s) - s \quad (4.63)$$

$$\mathcal{L}\{y'\} = sY(s) - y(0) = sY(s) - 1 \quad (4.64)$$

$$\mathcal{L}\{y\} = Y(s) \quad (4.65)$$

Substituting:

$$[s^2Y(s) - s] + 3[sY(s) - 1] + 2Y(s) = \frac{1}{s+1} \quad (4.66)$$

Step 3: Solve for $Y(s)$.

$$s^2Y(s) - s + 3sY(s) - 3 + 2Y(s) = \frac{1}{s+1} \quad (4.67)$$

$$(s^2 + 3s + 2)Y(s) = \frac{1}{s+1} + s + 3 \quad (4.68)$$

$$(s+1)(s+2)Y(s) = \frac{1}{s+1} + \frac{(s+3)(s+1)}{s+1} \quad (4.69)$$

$$(s+1)(s+2)Y(s) = \frac{1 + (s+3)(s+1)}{s+1} \quad (4.70)$$

$$(s+1)(s+2)Y(s) = \frac{1 + s^2 + 4s + 3}{s+1} = \frac{s^2 + 4s + 4}{s+1} = \frac{(s+2)^2}{s+1} \quad (4.71)$$

Therefore:

$$Y(s) = \frac{(s+2)^2}{(s+1)^2(s+2)} = \frac{s+2}{(s+1)^2} \quad (4.72)$$

Step 4: Find the inverse Laplace transform.

Perform partial fraction expansion:

$$Y(s) = \frac{s+2}{(s+1)^2} = \frac{(s+1)+1}{(s+1)^2} = \frac{1}{s+1} + \frac{1}{(s+1)^2} \quad (4.73)$$

Using transform pairs:

- $\frac{1}{s+1} \leftrightarrow e^{-t}$
- $\frac{1}{(s+1)^2} \leftrightarrow te^{-t}$

Final Answer:

$$y(t) = e^{-t} + te^{-t} = (1+t)e^{-t}, \quad t \geq 0 \quad (4.74)$$

Verification: We can verify by checking initial conditions:

- $y(0) = (1+0)e^0 = 1 \checkmark$
- $y'(t) = e^{-t} - (1+t)e^{-t} = -te^{-t}$, so $y'(0) = 0 \checkmark$

4.8 Transfer Functions

Definition 0.3: Transfer Function

The **transfer function** $G(s)$ of a linear time-invariant system is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions:

$$G(s) = \frac{Y(s)}{U(s)} \quad (4.75)$$

Transfer functions are typically written as:

$$G(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \cdots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0} = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - p_j)} \quad (4.76)$$

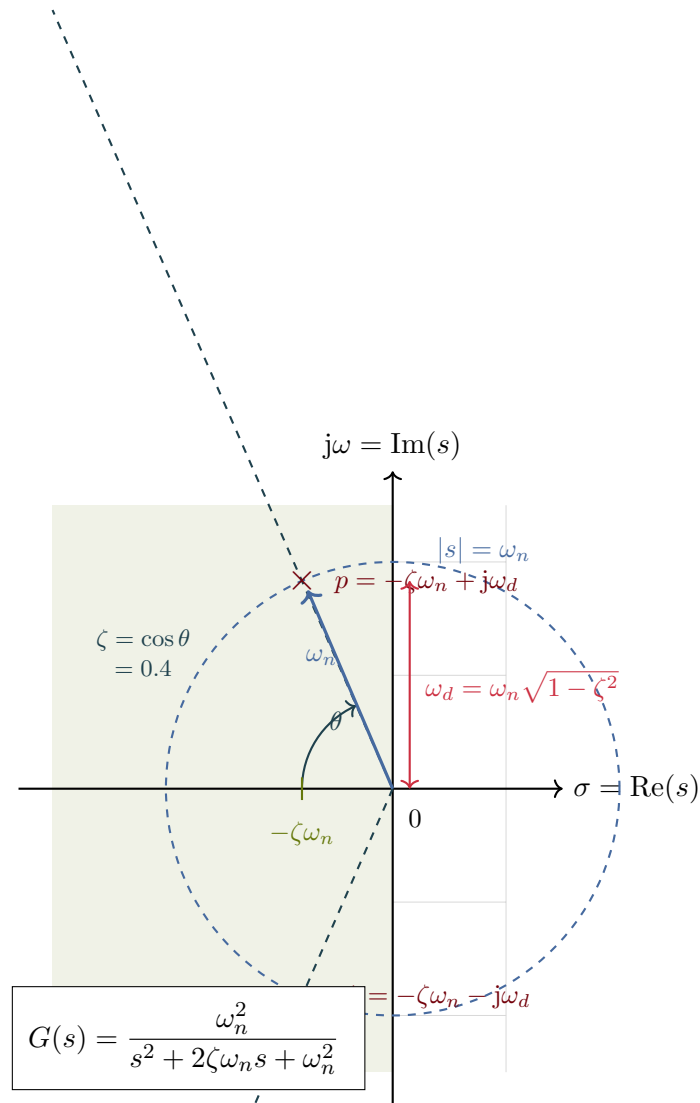
where z_i are the **zeros** and p_j are the **poles** of the transfer function.

Poles and System Behavior

The **poles** of a transfer function determine the system's natural response:

- **Real negative poles** ($s = -a, a > 0$): Decaying exponentials e^{-at}
- **Complex poles** ($s = -\sigma \pm j\omega, \sigma > 0$): Damped oscillations $e^{-\sigma t} \sin(\omega t + \phi)$
- **Poles on imaginary axis** ($s = \pm j\omega$): Sustained oscillations
- **Positive real poles** ($s = a, a > 0$): Unstable exponential growth e^{at}

The **zeros** affect the relative magnitude of different response modes and the transient shape, but do not directly determine stability.

**Pole Locations:**

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Natural Frequency:

$$\omega_n = |s| = \sqrt{\sigma^2 + \omega_d^2}$$

Damping Ratio:

$$\zeta = \cos \theta = \frac{|\sigma|}{\omega_n}$$

System Response:**Underdamped**

$$0 < \zeta < 1$$

Complex poles

Physical Interpretation: $\sigma = -\zeta\omega_n$: decay rate $\omega_d = \omega_n\sqrt{1-\zeta^2}$: damped frequency $\frac{1}{\zeta\omega_n}$: time constant**Time Response:**

$$y(t) \propto e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$

Chapter 5

Fourier Transform and Frequency Analysis

5.1 Fourier Series

Any periodic function $f(t)$ with period T can be expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right) \right] \quad (5.1)$$

where:

$$a_0 = \frac{1}{T} \int_0^T f(t) dt \quad (5.2)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi nt}{T}\right) dt \quad (5.3)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi nt}{T}\right) dt \quad (5.4)$$

Complex exponential form:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T} \quad (5.5)$$

where $c_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$

Example: Fourier Series of a Square Wave

Consider a square wave with period $T = 2\pi$ and amplitude A :

$$f(t) = \begin{cases} A & 0 < t < \pi \\ -A & \pi < t < 2\pi \end{cases}$$

Step 1: Compute a_0

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(t) dt = \frac{1}{2\pi} \left[\int_0^{\pi} A dt + \int_{\pi}^{2\pi} (-A) dt \right] = \frac{1}{2\pi} (A\pi - A\pi) = 0$$

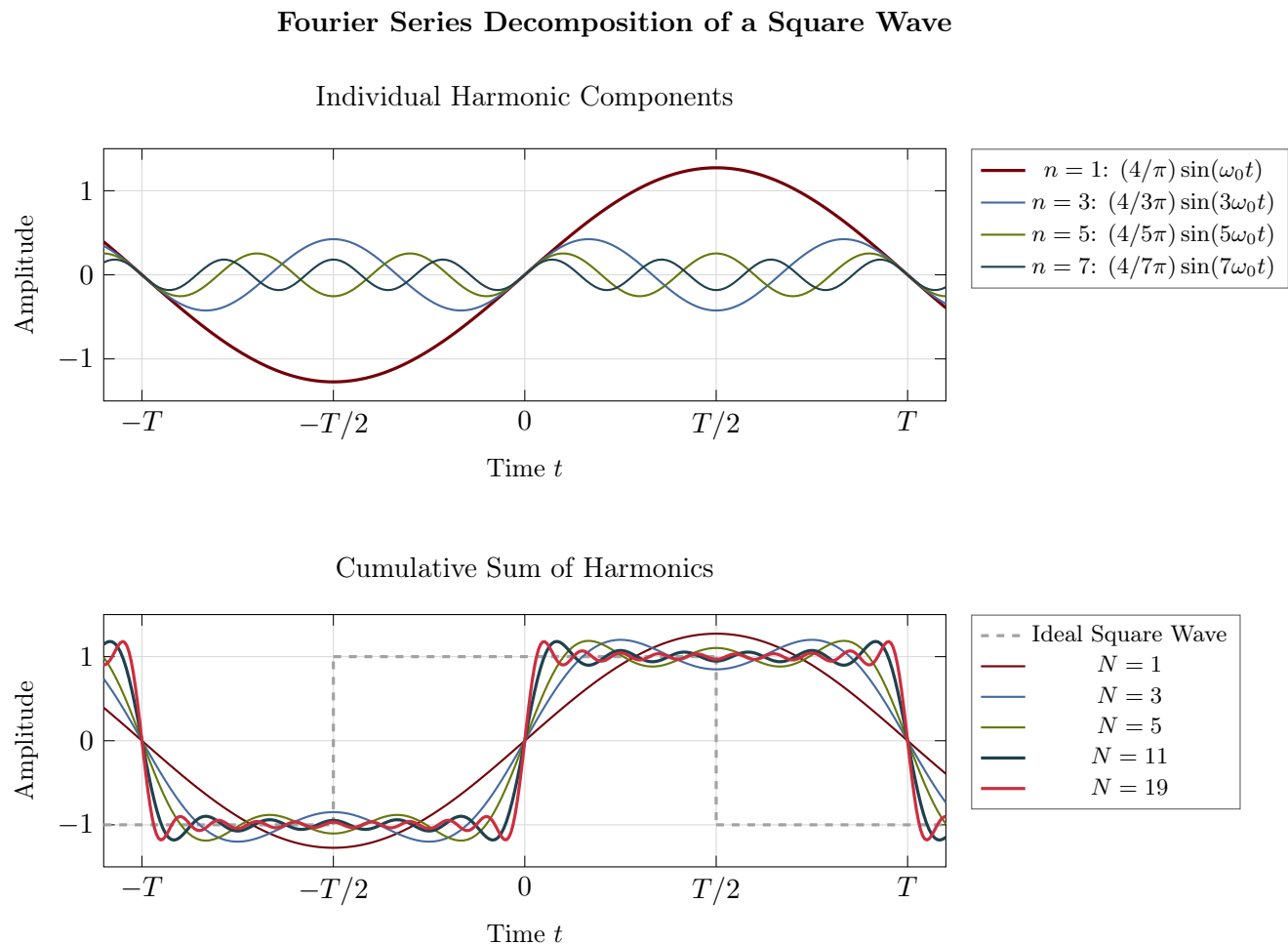


Figure 5.1: Fourier series decomposition of a square wave. **Top:** Individual odd harmonic components ($n = 1, 3, 5, 7$) with amplitudes $4/(n\pi)$. A symmetric square wave contains only odd harmonics. **Bottom:** Cumulative sum showing how adding more harmonics improves the approximation. The dashed gray line shows the ideal square wave. Note the Gibbs phenomenon (overshoot near discontinuities) which persists regardless of how many terms are included. The Fourier series formula is: $f(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n\pi} \sin(n\omega_0 t)$ where $\omega_0 = 2\pi/T$.

Step 2: Compute a_n

Since $f(t)$ is an odd function about $t = \pi$, all cosine coefficients vanish: $a_n = 0$ for all n .

Step 3: Compute b_n

$$\begin{aligned}
 b_n &= \frac{2}{2\pi} \int_0^{2\pi} f(t) \sin(nt) dt \\
 &= \frac{1}{\pi} \left[\int_0^\pi A \sin(nt) dt + \int_\pi^{2\pi} (-A) \sin(nt) dt \right] \\
 &= \frac{A}{\pi} \left[-\frac{\cos(nt)}{n} \Big|_0^\pi + \frac{\cos(nt)}{n} \Big|_\pi^{2\pi} \right] \\
 &= \frac{A}{n\pi} [(-\cos(n\pi) + 1) + (\cos(2n\pi) - \cos(n\pi))] \\
 &= \frac{A}{n\pi} [2 - 2\cos(n\pi)] = \frac{2A}{n\pi} (1 - (-1)^n)
 \end{aligned}$$

For n odd: $b_n = \frac{4A}{n\pi}$. For n even: $b_n = 0$.

Result: The Fourier series of the square wave is:

$$f(t) = \frac{4A}{\pi} \left(\sin(t) + \frac{1}{3} \sin(3t) + \frac{1}{5} \sin(5t) + \cdots \right) = \frac{4A}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\sin(nt)}{n} \quad (5.6)$$

This shows that a square wave contains only odd harmonics, with amplitudes decreasing as $1/n$.

Gibbs Phenomenon

When approximating a discontinuous function with a finite Fourier series, overshoot occurs near discontinuities. This overshoot is approximately 9% of the jump magnitude and does not diminish as more terms are added—it only becomes narrower. This is known as the **Gibbs phenomenon**.

5.2 Fourier Transform

For non-periodic functions, the Fourier transform extends the series to continuous frequencies:

Definition 5.1: Fourier Transform

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \quad (5.7)$$

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad (5.8)$$

Frequency Spectrum of a Damped Sinusoid

Time Domain: $f(t) = e^{-\sigma t} \cos(\omega_0 t)$ ($\sigma = 0.5$, $\omega_0 = 4\pi$ rad/s)

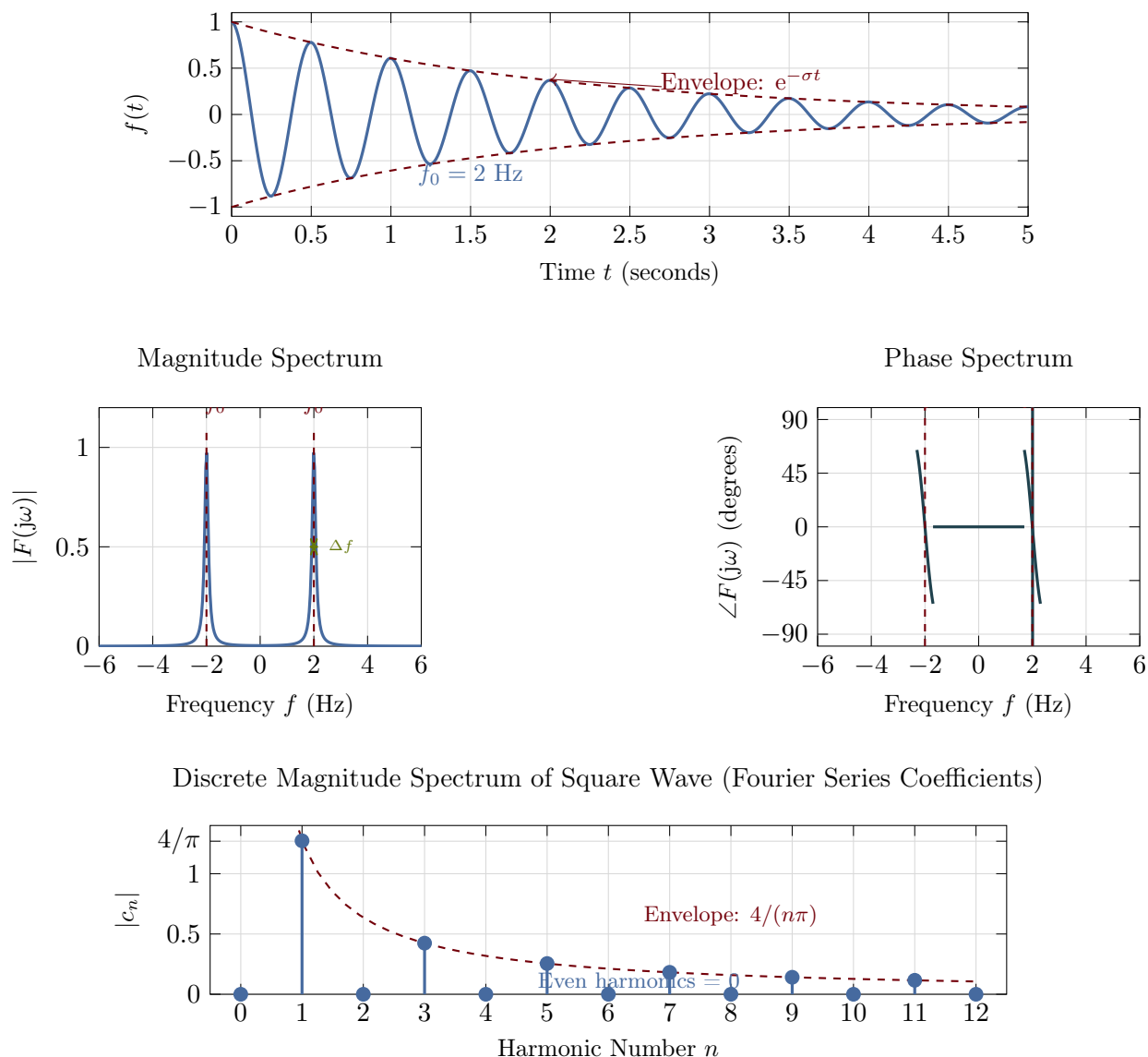


Figure 5.2: Frequency domain representation of signals. **Top:** Time-domain damped sinusoid $f(t) = e^{-\sigma t} \cos(\omega_0 t)$ with decay rate $\sigma = 0.5 \text{ s}^{-1}$ and frequency $f_0 = 2 \text{ Hz}$. The dashed Garnet lines show the exponential envelope. **Middle:** Magnitude spectrum (Atlantic blue) shows peaks at $\pm f_0$ with bandwidth Δf proportional to σ . Phase spectrum (Congaree teal) shows $\pm 90^\circ$ phase transitions at the resonance frequencies. **Bottom:** Discrete spectrum for a periodic square wave showing only odd harmonics with magnitude decreasing as $4/(n\pi)$.

Example: Fourier Transform of a Decaying Exponential

Find the Fourier transform of $f(t) = e^{-at}u(t)$ where $a > 0$ and $u(t)$ is the unit step function.

Solution:

$$\begin{aligned}
 F(\omega) &= \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t} dt \\
 &= \int_0^{\infty} e^{-at}e^{-j\omega t} dt \\
 &= \int_0^{\infty} e^{-(a+j\omega)t} dt \\
 &= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}
 \end{aligned}$$

Since $a > 0$, the exponential vanishes as $t \rightarrow \infty$:

$$F(\omega) = 0 - \frac{1}{-(a+j\omega)} = \frac{1}{a+j\omega}$$

Result:

$$\boxed{\mathcal{F}\{e^{-at}u(t)\} = \frac{1}{a+j\omega}} \quad (5.9)$$

Magnitude and Phase:

$$|F(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}, \quad \angle F(\omega) = -\arctan\left(\frac{\omega}{a}\right)$$

This result shows that a decaying exponential has a low-pass frequency spectrum: low frequencies pass through with little attenuation, while high frequencies are attenuated.

5.2.1 Fourier Transform Properties

The Fourier transform possesses several important properties that make it a powerful tool for signal analysis:

Convolution vs. Modulation

Note the duality between convolution and modulation:

- Convolution in time $\Leftrightarrow$ Multiplication in frequency
- Multiplication in time $\Leftrightarrow$ Convolution in frequency (scaled by $1/2\pi$)

This duality is fundamental to understanding filtering, modulation, and signal processing.

Relationship to Laplace Transform

For causal signals, $F(\omega) = F(s)|_{s=j\omega}$ when the Laplace transform converges on the imaginary axis.

Table 5.1: Fourier Transform Properties

Property	Time Domain	Frequency Domain
Linearity	$af(t) + bg(t)$	$aF(\omega) + bG(\omega)$
Time Shifting	$f(t - t_0)$	$F(\omega)e^{-j\omega t_0}$
Frequency Shifting	$f(t)e^{j\omega_0 t}$	$F(\omega - \omega_0)$
Time Scaling	$f(at)$	$\frac{1}{ a }F\left(\frac{\omega}{a}\right)$
Time Reversal	$f(-t)$	$F(-\omega)$
Differentiation	$\frac{df}{dt}$	$j\omega F(\omega)$
Integration	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$
Convolution	$f(t) * g(t)$	$F(\omega) \cdot G(\omega)$
Modulation	$f(t) \cdot g(t)$	$\frac{1}{2\pi}F(\omega) * G(\omega)$
Parseval's Theorem	$\int_{-\infty}^{\infty} f(t) ^2 dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) ^2 d\omega$
Duality	$F(t)$	$2\pi f(-\omega)$

When Does the Fourier Transform Exist?

The Fourier transform of $f(t)$ exists if:

1. $f(t)$ is absolutely integrable: $\int_{-\infty}^{\infty} |f(t)| dt < \infty$, or
2. $f(t)$ is square integrable: $\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty$

Functions like $\sin(\omega_0 t)$ or the unit step $u(t)$ do not satisfy these conditions, but their Fourier transforms can be defined using generalized functions (distributions) involving impulses $\delta(\omega)$.

5.3 Frequency Response

For a stable LTI system with transfer function $G(s)$, the **frequency response** is:

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)} \quad (5.10)$$

- $|G(j\omega)|$ = magnitude (gain) at frequency ω
- $\angle G(j\omega)$ = phase shift at frequency ω

For a sinusoidal input $u(t) = A \sin(\omega t)$, the steady-state output is:

$$y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (5.11)$$

5.4 Bode Plots

Bode plots display magnitude (in dB) and phase versus frequency (on a log scale):

$$|G(j\omega)|_{\text{dB}} = 20 \log_{10} |G(j\omega)| \quad (5.12)$$

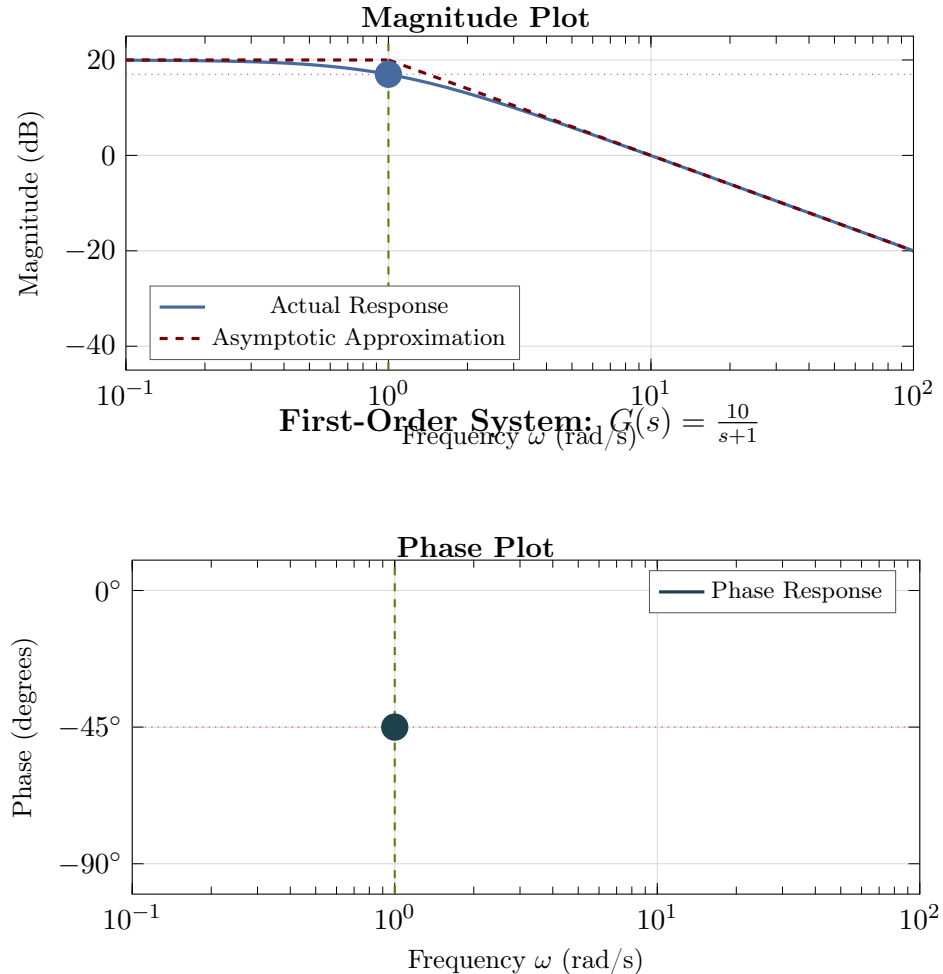


Figure 5.3: Bode plot for a first-order system $G(s) = \frac{10}{s+1}$. The magnitude plot (top, Atlantic blue) shows the steady-state gain of 20 dB below the corner frequency $\omega_c = 1$ rad/s, with a -20 dB/decade slope afterward. The phase plot (bottom, Congaree teal) shows -45° phase lag at the corner frequency. Dashed Garnet lines show asymptotic approximations. The UofSCRose accent color highlights key features: the -3 dB magnitude point and the -45° phase point at the corner frequency.

Example: Drawing a Bode Plot from Frequency Response

Construct the Bode plot for the first-order low-pass filter:

$$G(s) = \frac{K}{1 + s/\omega_c}$$

where $K = 10$ (DC gain) and $\omega_c = 100$ rad/s (corner frequency).

Step 1: Find the frequency response

Substitute $s = j\omega$:

$$G(j\omega) = \frac{K}{1 + j\omega/\omega_c} = \frac{10}{1 + j\omega/100}$$

Step 2: Calculate magnitude

$$|G(j\omega)| = \frac{K}{\sqrt{1 + (\omega/\omega_c)^2}} = \frac{10}{\sqrt{1 + (\omega/100)^2}}$$

In decibels:

$$|G(j\omega)|_{\text{dB}} = 20 \log_{10}(K) - 20 \log_{10} \sqrt{1 + (\omega/\omega_c)^2} = 20 - 10 \log_{10}(1 + \omega^2/10000)$$

Step 3: Calculate phase

$$\angle G(j\omega) = -\arctan\left(\frac{\omega}{\omega_c}\right) = -\arctan\left(\frac{\omega}{100}\right)$$

Step 4: Asymptotic approximations

Magnitude:

- For $\omega \ll \omega_c$: $|G|_{\text{dB}} \approx 20 \log_{10}(K) = 20$ dB (flat)
- For $\omega \gg \omega_c$: $|G|_{\text{dB}} \approx 20 \log_{10}(K) - 20 \log_{10}(\omega/\omega_c)$ (slope: -20 dB/decade)
- At $\omega = \omega_c$: $|G|_{\text{dB}} = 20 - 3 = 17$ dB (corner point, 3 dB below)

Phase:

- For $\omega \ll \omega_c/10$: $\angle G \approx 0^\circ$
- At $\omega = \omega_c$: $\angle G = -45^\circ$
- For $\omega \gg 10\omega_c$: $\angle G \approx -90^\circ$

Result: The Bode plot shows a flat response at low frequencies with 20 dB gain, rolling off at -20 dB/decade above the corner frequency, with phase shifting from 0° to -90° .

Standard Bode Plot Building Blocks

Any transfer function can be decomposed into these basic elements:

1. **Constant gain K :** Flat magnitude at $20 \log_{10} |K|$ dB, phase 0° or $\pm 180^\circ$
2. **Poles/zeros at origin $(s)^{\pm n}$:** $\pm 20n$ dB/decade slope through 0 dB at $\omega = 1$, phase $\pm 90n^\circ$
3. **Real pole/zero $(1 + s/\omega_c)^{\pm 1}$:** ± 20 dB/decade above ω_c , phase change of $\pm 90^\circ$
4. **Complex poles/zeros:** Resonant peak/notch near ω_n , phase change of $\pm 180^\circ$

5.5 Discrete Fourier Transform and FFT

When working with sampled data, we use the Discrete Fourier Transform (DFT) instead of the continuous Fourier transform.

Definition 5.2: Discrete Fourier Transform (DFT)

For a sequence of N samples $\{x[0], x[1], \dots, x[N-1]\}$, the DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1 \quad (5.13)$$

The inverse DFT (IDFT) recovers the original sequence:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N-1 \quad (5.14)$$

5.5.1 Connection to Continuous Fourier Transform

The DFT approximates the continuous Fourier transform at discrete frequencies. If the continuous signal $f(t)$ is sampled at rate $f_s = 1/T_s$ to produce N samples:

- The DFT frequency bins are spaced at $\Delta f = f_s/N$ Hz
- The k -th DFT coefficient corresponds to frequency $f_k = k \cdot f_s/N$
- The relationship is: $X[k] \approx T_s \cdot F(2\pi f_k)$ for band-limited signals

Nyquist Sampling Theorem

To avoid aliasing, the sampling frequency must satisfy:

$$f_s > 2f_{\max}$$

where $f_{\max}$ is the highest frequency component in the signal. The frequency $f_s/2$ is called the **Nyquist frequency**. Frequency components above the Nyquist frequency will “fold back” and appear as lower frequencies (aliasing).

5.5.2 Fast Fourier Transform (FFT)

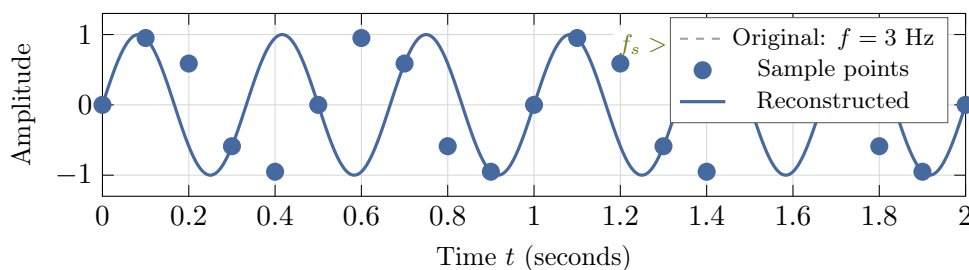
Direct computation of the DFT requires $\mathcal{O}(N^2)$ operations. The **Fast Fourier Transform (FFT)** is an efficient algorithm that computes the DFT in $\mathcal{O}(N \log N)$ operations by exploiting symmetry and periodicity of the complex exponentials.

FFT Key Points

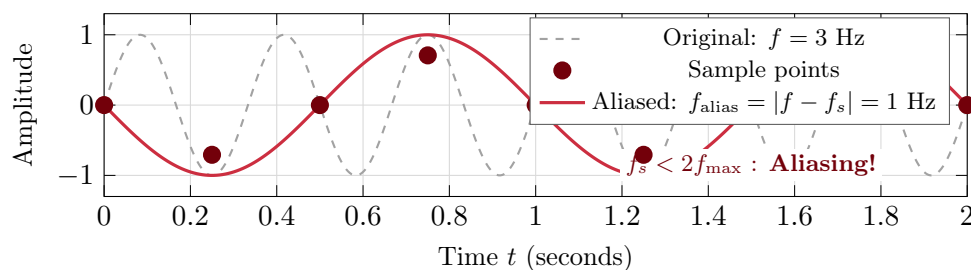
- The FFT computes the *exact same result* as the DFT—it is not an approximation
- Most efficient when N is a power of 2 (e.g., 256, 512, 1024, 2048)

Sampling and the Nyquist Frequency

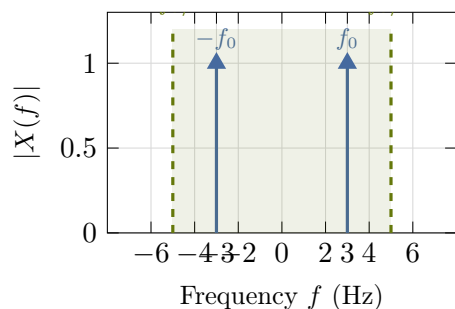
Adequate Sampling: $f_s = 10 \text{ Hz} > 2f_{\text{signal}} = 6 \text{ Hz}$ (Nyquist criterion satisfied)



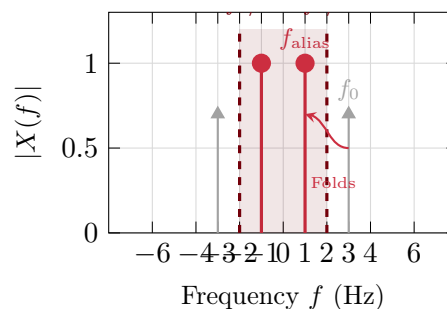
Undersampling: $f_s = 4 \text{ Hz} < 2f_{\text{signal}} = 6 \text{ Hz}$ (Aliasing occurs!)



Adequate Sampling Spectrum



Undersampled Spectrum (Aliasing)



Nyquist-Shannon Sampling Theorem: A bandlimited signal with maximum frequency f_{max} can be perfectly reconstructed from samples if $f_s > 2f_{\text{max}}$.

Nyquist Frequency: $f_N = f_s/2$ **Nyquist Rate:** $f_s = 2f_{\text{max}}$

Figure 5.4: Nyquist sampling and aliasing in the Fourier domain.

- Speedup factor: For $N = 1024$, FFT is about $100\times$ faster than direct DFT
- The Cooley-Tukey algorithm (1965) is the most common FFT implementation
- Modern libraries (FFTW, NumPy, MATLAB) use highly optimized FFT routines

Using FFT in Practice

When using FFT for spectral analysis:

1. **Zero-padding:** Add zeros to increase frequency resolution (interpolation in frequency domain)
2. **Windowing:** Apply a window function (Hanning, Hamming, Blackman) to reduce spectral leakage
3. **Averaging:** Use Welch's method (overlapping windowed segments) for noise reduction
4. **Scaling:** Normalize properly to get correct magnitude units

In MATLAB/Python:

```
1 import numpy as np
2 from scipy.fft import fft, fftfreq
3
4 # Sample signal
5 fs = 1000 # Sampling frequency (Hz)
6 t = np.arange(0, 1, 1/fs)
7 x = np.sin(2*np.pi*50*t) + 0.5*np.sin(2*np.pi*120*t)
8
9 # Compute FFT
10 N = len(x)
11 X = fft(x)
12 freq = fftfreq(N, 1/fs)
13
14 # Magnitude spectrum (single-sided)
15 X_mag = 2*np.abs(X[:N//2])/N
```

Chapter 6

Linear Algebra

Linear algebra is essential for state-space control, multi-input multi-output (MIMO) systems, and machine learning.

6.1 Vectors and Matrices

Vector Operations

- Addition: $\mathbf{x} + \mathbf{y} = [x_1 + y_1, x_2 + y_2, \dots, x_n + y_n]^\top$
- Scalar multiplication: $\alpha \mathbf{x} = [\alpha x_1, \alpha x_2, \dots, \alpha x_n]^\top$
- Dot product: $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$
- Norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$ (Euclidean norm)

Matrix Operations

- Addition: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$
- Multiplication: $(\mathbf{AB})_{ij} = \sum_k a_{ik} b_{kj}$
- Transpose: $(\mathbf{A}^\top)_{ij} = a_{ji}$
- Inverse: $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (if exists)

6.2 Special Matrices

- **Identity matrix:** $\mathbf{I}$ with ones on diagonal, zeros elsewhere
- **Diagonal matrix:** $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$
- **Symmetric matrix:** $\mathbf{A} = \mathbf{A}^\top$
- **Positive definite:** $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- **Orthogonal matrix:** $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$

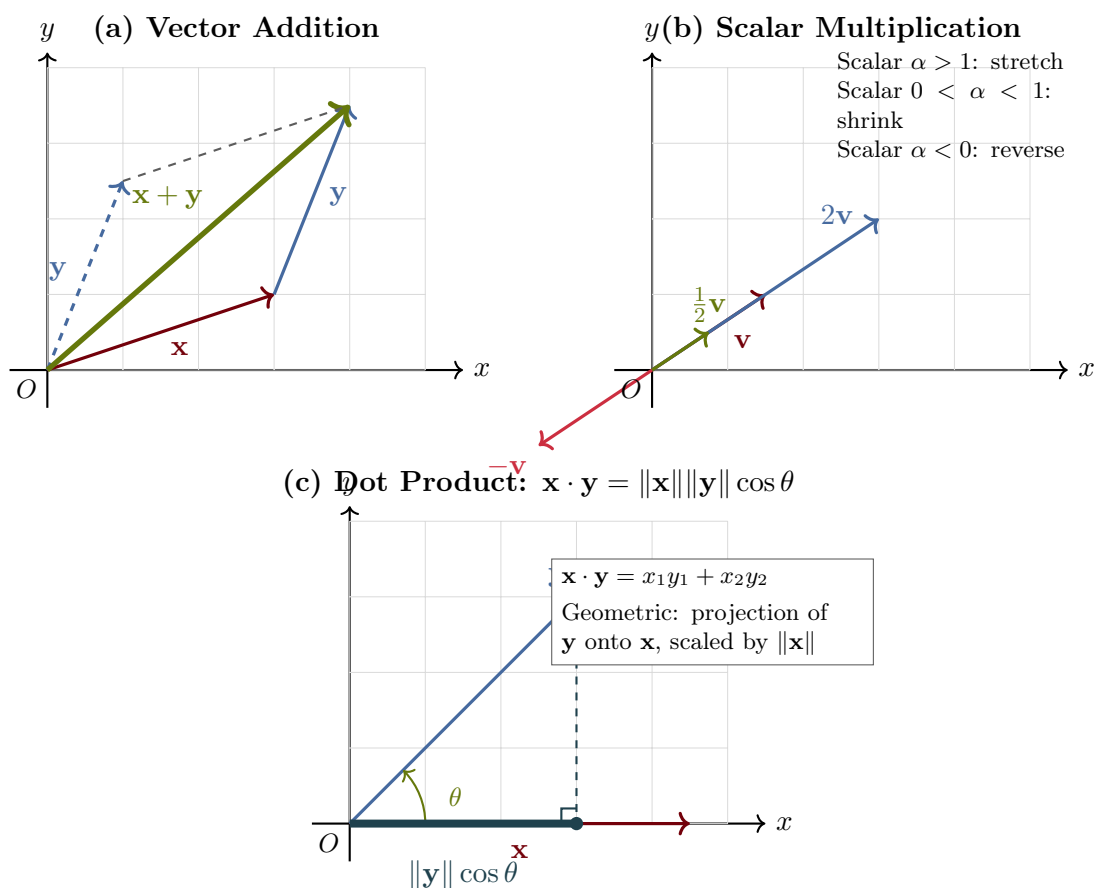


Figure 6.1: Geometric interpretation of fundamental vector operations in $\mathbb{R}^2$. (a) Vector addition follows the parallelogram law: $\mathbf{x} + \mathbf{y}$ is the diagonal of the parallelogram formed by $\mathbf{x}$ and $\mathbf{y}$. (b) Scalar multiplication scales vector magnitude; positive scalars preserve direction while negative scalars reverse it. (c) The dot product $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$ equals the product of $\|\mathbf{x}\|$ with the projection of $\mathbf{y}$ onto $\mathbf{x}$. The projection (Congaree) shows the geometric meaning of the inner product.

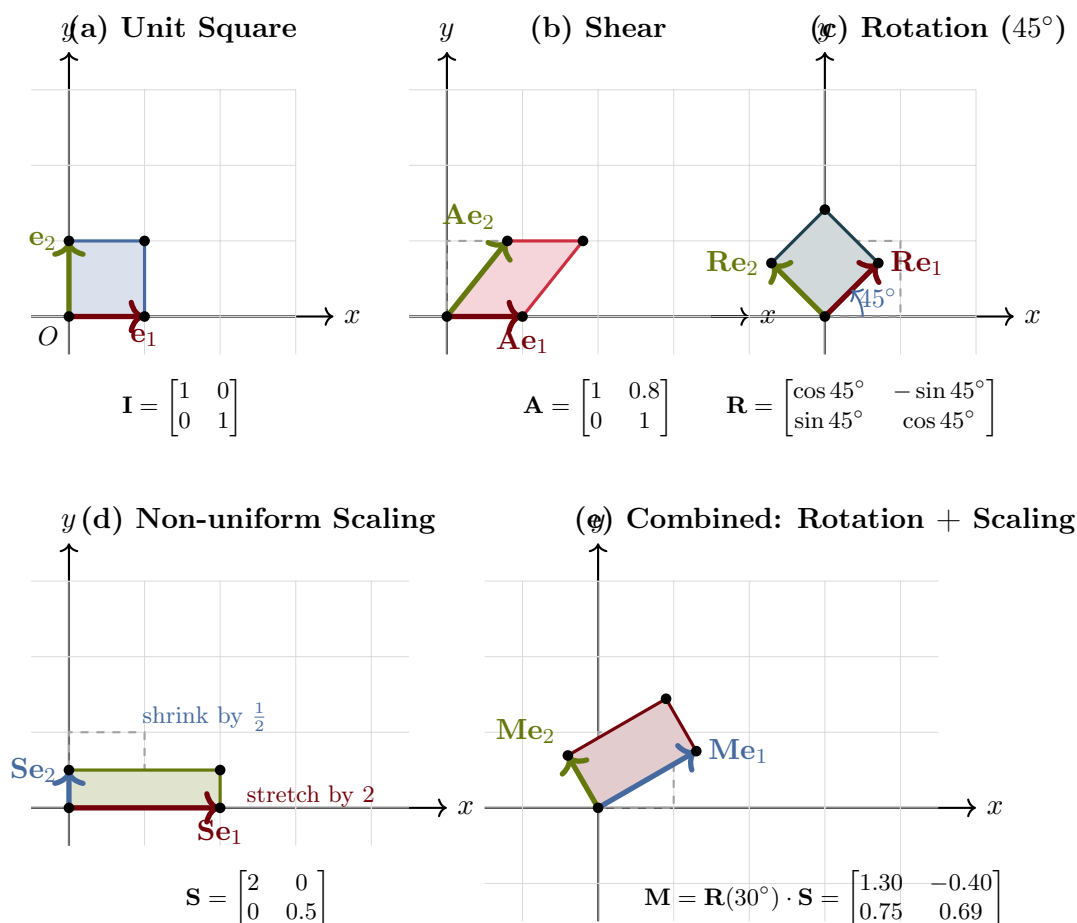


Figure 6.2: Geometric interpretation of 2×2 matrix transformations acting on the unit square. (a) The identity matrix preserves all vectors. (b) Shear transforms the square into a parallelogram by sliding points parallel to one axis. (c) Rotation preserves lengths and angles (orthogonal transformation). (d) Non-uniform scaling stretches or compresses along each axis independently. (e) Combined transformations compose: here scaling followed by rotation. In each case, the matrix columns represent the transformed basis vectors $A\mathbf{e}_1$ and $A\mathbf{e}_2$. The dashed squares show the original unit square for reference.

6.3 Determinant and Rank

The **determinant** of a square matrix:

$$\det(\mathbf{A}) = |\mathbf{A}| = \sum_{j=1}^n (-1)^{1+j} a_{1j} M_{1j} \quad (6.1)$$

where M_{1j} is the minor (determinant of submatrix).

Properties:

- $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
- $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$

The **rank** is the number of linearly independent rows (or columns):

$$\text{rank}(\mathbf{A}) \leq \min(m, n) \text{ for } \mathbf{A} \in \mathbb{R}^{m \times n} \quad (6.2)$$

6.4 Eigenvalues and Eigenvectors

Definition 0.5: Eigenvalue Problem

For a square matrix $\mathbf{A}$, scalar λ is an **eigenvalue** and non-zero vector $\mathbf{v}$ is the corresponding **eigenvector** if:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (6.3)$$

Eigenvalues are roots of the **characteristic equation**:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \quad (6.4)$$

Properties

- Sum of eigenvalues = $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$
- Product of eigenvalues = $\det(\mathbf{A})$
- Eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i$
- Eigenvalues of $\mathbf{A}^n$ are λ_i^n

Stability and Eigenvalues

A continuous-time system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is:

- **Stable** if all eigenvalues of $\mathbf{A}$ have negative real parts
- **Marginally stable** if eigenvalues have non-positive real parts with simple imaginary-axis eigenvalues

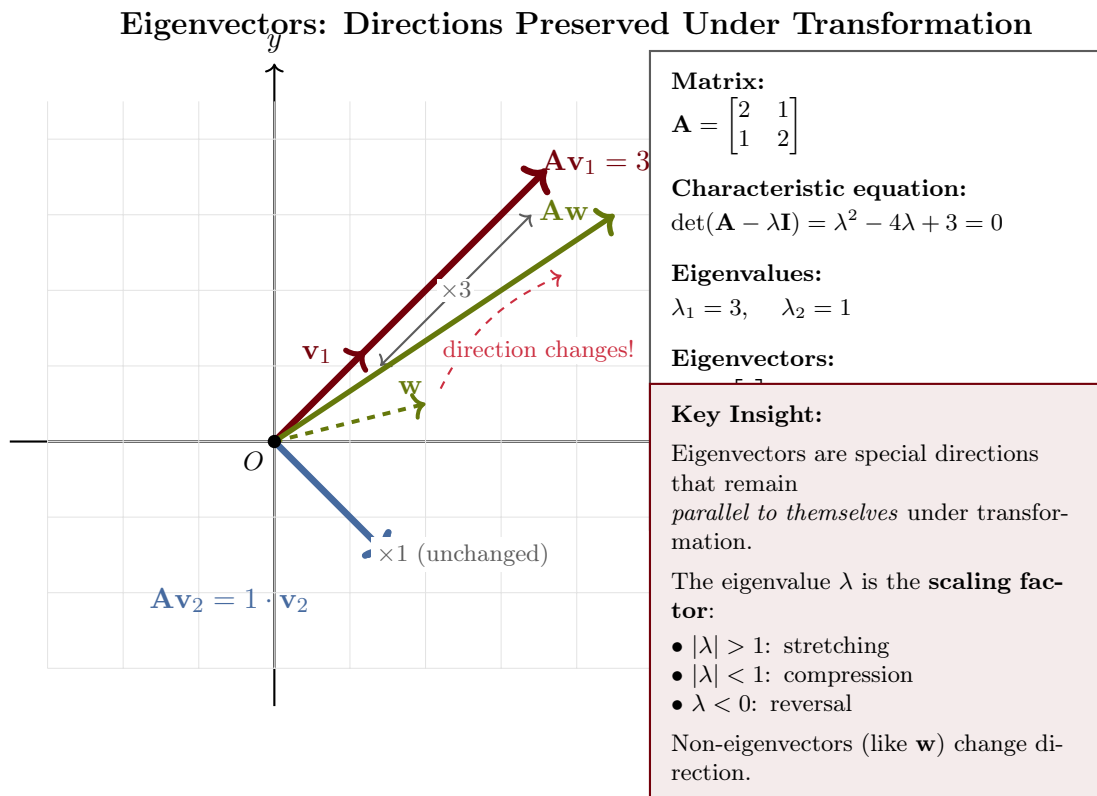


Figure 6.3: Geometric visualization of eigenvalues and eigenvectors. For matrix $\mathbf{A}$, eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ (dashed) are special directions that remain parallel to themselves after transformation (solid). The eigenvalue is the scaling factor: $\mathbf{v}_1$ is scaled by $\lambda_1 = 3$ (stretched), while $\mathbf{v}_2$ is scaled by $\lambda_2 = 1$ (unchanged). In contrast, a non-eigenvector $\mathbf{w}$ both rotates and scales under transformation—its direction is not preserved. This geometric property is fundamental to understanding system stability: eigenvalues determine how state components grow or decay along eigenvector directions.

- **Unstable** if any eigenvalue has positive real part

Example: Eigenvalues and Eigenvectors of a 2x2 Matrix

Find the eigenvalues and eigenvectors of:

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Solution:

Step 1: Find eigenvalues by solving $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$:

$$\det \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix} = (4 - \lambda)(3 - \lambda) - 2 = 0$$

$$\lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 5)(\lambda - 2) = 0 \quad \Rightarrow \quad \lambda_1 = 5, \quad \lambda_2 = 2$$

Step 2: Find eigenvector for $\lambda_1 = 5$:

$$(\mathbf{A} - 5\mathbf{I})\mathbf{v}_1 = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \mathbf{v}_1 = \mathbf{0}$$

From $-v_1 + v_2 = 0$, we get $v_2 = v_1$. Choosing $v_1 = 1$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 3: Find eigenvector for $\lambda_2 = 2$:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v}_2 = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \mathbf{v}_2 = \mathbf{0}$$

From $2v_1 + v_2 = 0$, we get $v_2 = -2v_1$. Choosing $v_1 = 1$:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Verification: $\mathbf{A}\mathbf{v}_1 = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5\mathbf{v}_1 \quad \checkmark \quad \text{and} \quad \mathbf{A}\mathbf{v}_2 = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2\mathbf{v}_2 \quad \checkmark$

Cayley-Hamilton Theorem

Every square matrix satisfies its own characteristic equation. If the characteristic polynomial of $\mathbf{A}$ is:

$$p(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

then:

$$p(\mathbf{A}) = \mathbf{A}^n + a_{n-1}\mathbf{A}^{n-1} + \cdots + a_1\mathbf{A} + a_0\mathbf{I} = \mathbf{0}$$

Applications:

- Computing $\mathbf{A}^{-1}$ from powers of $\mathbf{A}$
- Expressing high powers of $\mathbf{A}$ in terms of lower powers
- Computing the matrix exponential $e^{\mathbf{A}t}$

6.5 Matrix Decompositions

Eigendecomposition

For diagonalizable matrix $\mathbf{A}$:

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} \quad (6.5)$$

where $\mathbf{V}$ contains eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Singular Value Decomposition (SVD)

Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top} \quad (6.6)$$

where $\mathbf{U}, \mathbf{V}$ are orthogonal and $\mathbf{\Sigma}$ contains singular values.

Example: Computing SVD for a Simple Matrix

Find the SVD of:

$$\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Solution:

Step 1: Compute $\mathbf{A}^{\top}\mathbf{A}$:

$$\mathbf{A}^{\top}\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$$

Step 2: Find eigenvalues of $\mathbf{A}^{\top}\mathbf{A}$ (these are σ_i^2):

Since this is diagonal, eigenvalues are $\sigma_1^2 = 9$ and $\sigma_2^2 = 4$.

Singular values: $\sigma_1 = 3$, $\sigma_2 = 2$.

Step 3: Form $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Step 4: Find $\mathbf{V}$ (eigenvectors of $\mathbf{A}^{\top}\mathbf{A}$):

For diagonal matrix, eigenvectors are standard basis vectors:

$$\mathbf{V} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

Step 5: Find $\mathbf{U}$ from $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{\Sigma}^{-1}$:

$$\mathbf{U} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/3 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

Result: $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{I} \cdot \mathbf{\Sigma} \cdot \mathbf{I} = \mathbf{\Sigma}$ (as expected for a diagonal matrix).
 For non-diagonal matrices, the procedure involves more computation but follows the same principles.

QR Decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \quad (6.7)$$

where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is upper triangular.

LU Decomposition

$$\mathbf{A} = \mathbf{L}\mathbf{U} \quad (6.8)$$

where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

6.6 Matrix Exponential

The matrix exponential is crucial for solving state-space equations:

Definition 0.6: Matrix Exponential

$$\mathbf{e}^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} \quad (6.9)$$

Properties

- $\mathbf{e}^0 = \mathbf{I}$
- $\frac{d}{dt}\mathbf{e}^{\mathbf{A}t} = \mathbf{A}\mathbf{e}^{\mathbf{A}t}$
- $\mathbf{e}^{\mathbf{A}(t_1+t_2)} = \mathbf{e}^{\mathbf{A}t_1}\mathbf{e}^{\mathbf{A}t_2}$
- $(\mathbf{e}^{\mathbf{A}t})^{-1} = \mathbf{e}^{-\mathbf{A}t}$

Computation via Laplace

$$\mathbf{e}^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\} \quad (6.10)$$

Computation via eigendecomposition

If $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$:

$$\mathbf{e}^{\mathbf{A}t} = \mathbf{V}\mathbf{e}^{\mathbf{\Lambda}t}\mathbf{V}^{-1} = \mathbf{V} \text{diag}(\mathbf{e}^{\lambda_1 t}, \dots, \mathbf{e}^{\lambda_n t})\mathbf{V}^{-1} \quad (6.11)$$

Example: Computing Matrix Exponential for a Simple System

Compute $\mathbf{e}^{\mathbf{A}t}$ for the system matrix:

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

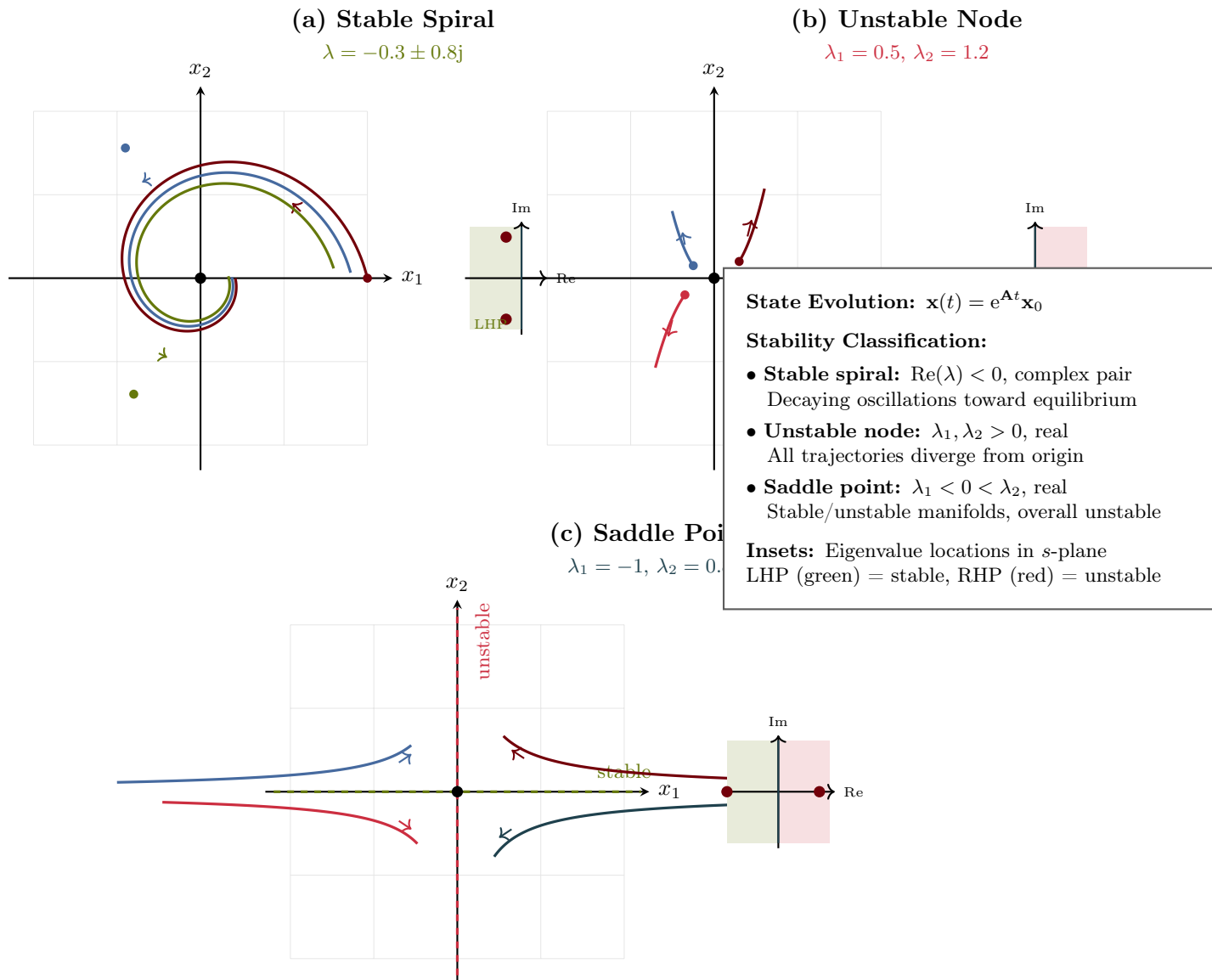


Figure 6.4: Phase portraits showing time evolution of the autonomous system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ via the matrix exponential $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0$ for three characteristic eigenvalue configurations. (a) **Stable spiral:** complex conjugate eigenvalues with negative real part produce decaying oscillatory trajectories spiraling toward the origin. (b) **Unstable node:** two positive real eigenvalues cause all trajectories to diverge exponentially from the equilibrium. (c) **Saddle point:** one stable and one unstable eigenvalue create invariant manifolds; trajectories approach the stable manifold but ultimately diverge along the unstable direction. Insets show eigenvalue locations in the s -plane; the imaginary axis separates stable (LHP) from unstable (RHP) behavior.

Solution:

Method 1: Direct computation (diagonal matrix)

For a diagonal matrix, the matrix exponential is simply:

$$e^{\mathbf{A}t} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix}$$

Method 2: Using the Laplace transform

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{bmatrix} s+1 & 0 \\ 0 & s+2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

Taking the inverse Laplace transform:

$$e^{\mathbf{A}t} = \mathcal{L}^{-1} \left\{ \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix} \right\} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix}$$

Interpretation: The state transition matrix $e^{\mathbf{A}t}$ shows that both modes decay exponentially, with eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -2$ determining the decay rates. The system is stable since both eigenvalues have negative real parts.

6.7 Linear Systems of Equations

Solving $\mathbf{Ax} = \mathbf{b}$:

- **Unique solution** if $\mathbf{A}$ is square and invertible: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$
- **No solution** if $\mathbf{b} \notin \text{range}(\mathbf{A})$
- **Infinite solutions** if $\text{null}(\mathbf{A}) \neq \{\mathbf{0}\}$

Example: Solving a System Using Gaussian Elimination

Solve the system $\mathbf{Ax} = \mathbf{b}$ where:

$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix}$$

Solution:

Step 1: Form augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

Step 2: Forward elimination

$R_2 \leftarrow R_2 + \frac{3}{2}R_1$:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

$$R_3 \leftarrow R_3 + R_1:$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 2 & 1 & 5 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 4R_2:$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

Step 3: Back substitution

From Row 3: $-x_3 = 1 \Rightarrow x_3 = -1$

From Row 2: $\frac{1}{2}x_2 + \frac{1}{2}(-1) = 1 \Rightarrow x_2 = 3$

From Row 1: $2x_1 + 3 - (-1) = 8 \Rightarrow 2x_1 = 4 \Rightarrow x_1 = 2$

Solution: $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$

Verification: $\mathbf{Ax} = \begin{bmatrix} 4 + 3 + 1 \\ -6 - 3 - 2 \\ -4 + 3 - 2 \end{bmatrix} = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix} = \mathbf{b} \checkmark$

Least Squares

When $\mathbf{Ax} = \mathbf{b}$ has no exact solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}^\dagger \mathbf{b} \quad (6.12)$$

where $\mathbf{A}^\dagger$ is the Moore-Penrose pseudoinverse.

6.8 Controllability and Observability

Two fundamental concepts in state-space control theory determine whether a system can be fully controlled and whether its internal states can be determined from measurements.

Definition 0.7: Controllability Matrix

For a linear time-invariant system $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$, the **controllability matrix** is:

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] \quad (6.13)$$

where n is the dimension of the state vector $\mathbf{x}$.

Controllability Condition

A system is **controllable** (or reachable) if and only if the controllability matrix has full row rank:

$$\text{rank}(\mathcal{C}) = n$$

Physical interpretation: The system is controllable if the input $\mathbf{u}$ can drive the state from any initial condition to any desired final state in finite time.

Definition 0.8: Observability Matrix

For a linear time-invariant system with output $\mathbf{y} = \mathbf{C}\mathbf{x}$, the **observability matrix** is:

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \quad (6.14)$$

Observability Condition

A system is **observable** if and only if the observability matrix has full column rank:

$$\text{rank}(\mathcal{O}) = n$$

Physical interpretation: The system is observable if the initial state $\mathbf{x}(0)$ can be uniquely determined from knowledge of the input $\mathbf{u}(t)$ and output $\mathbf{y}(t)$ over a finite time interval.

Connection to State-Space Control**Importance for Control Design**

- **Pole placement** (state feedback): Requires controllability. If the system is not controllable, some eigenvalues cannot be moved by state feedback.
- **Observer design**: Requires observability. If the system is not observable, some states cannot be estimated from output measurements.
- **Kalman decomposition**: Any LTI system can be decomposed into controllable/uncontrollable and observable/unobservable subsystems.

The duality between controllability and observability is a fundamental result:

- The pair $(\mathbf{A}, \mathbf{B})$ is controllable if and only if $(\mathbf{A}^\top, \mathbf{B}^\top)$ is observable.
- The pair $(\mathbf{A}, \mathbf{C})$ is observable if and only if $(\mathbf{A}^\top, \mathbf{C}^\top)$ is controllable.

This duality is essential for understanding the relationship between controller design and observer design in modern control theory.

Chapter 7

Probability and Statistics

Probability theory is essential for stochastic control, Kalman filtering, and machine learning. This chapter provides the mathematical foundations needed to understand uncertainty quantification, state estimation, and statistical inference in control systems.

7.1 Probability Fundamentals

Definition: Probability Space

A **probability space** is a triple $(\Omega, \mathcal{F}, P)$ where:

- Ω is the **sample space** (set of all possible outcomes)
- $\mathcal{F}$ is the **σ -algebra** (collection of events)
- $P : \mathcal{F} \rightarrow [0, 1]$ is the **probability measure**

Axioms of Probability

1. $P(A) \geq 0$ for all events A
2. $P(\Omega) = 1$ where Ω is the sample space
3. For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Conditional Probability

Definition: Conditional Probability

The **conditional probability** of event A given event B (with $P(B) > 0$) is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (7.1)$$

Bayes' Theorem

Bayes' Theorem

For events A and B with $P(B) > 0$:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (7.2)$$

In the context of estimation:

- $P(A)$ is the **prior** probability
- $P(B|A)$ is the **likelihood**
- $P(A|B)$ is the **posterior** probability
- $P(B)$ is the **evidence** (normalizing constant)

The denominator can be expanded using the law of total probability:

$$P(B) = \sum_i P(B|A_i)P(A_i) \quad (7.3)$$

where $\{A_i\}$ forms a partition of the sample space.

Example: Computing Conditional Probability with Bayes' Theorem

A sensor for detecting system faults has the following characteristics:

- The sensor correctly detects a fault (true positive) 95% of the time: $P(\text{alarm}|\text{fault}) = 0.95$
- The sensor has a false alarm rate of 3%: $P(\text{alarm}|\text{no fault}) = 0.03$
- From historical data, the probability of a fault occurring is 1%: $P(\text{fault}) = 0.01$

Question: If the alarm triggers, what is the probability that an actual fault occurred?

Solution:

We seek $P(\text{fault}|\text{alarm})$. Using Bayes' theorem:

$$P(\text{fault}|\text{alarm}) = \frac{P(\text{alarm}|\text{fault})P(\text{fault})}{P(\text{alarm})}$$

First, compute $P(\text{alarm})$ using total probability:

$$\begin{aligned} P(\text{alarm}) &= P(\text{alarm}|\text{fault})P(\text{fault}) + P(\text{alarm}|\text{no fault})P(\text{no fault}) \\ &= (0.95)(0.01) + (0.03)(0.99) \\ &= 0.0095 + 0.0297 = 0.0392 \end{aligned}$$

Now apply Bayes' theorem:

$$P(\text{fault}|\text{alarm}) = \frac{(0.95)(0.01)}{0.0392} = \frac{0.0095}{0.0392} \approx 0.242$$

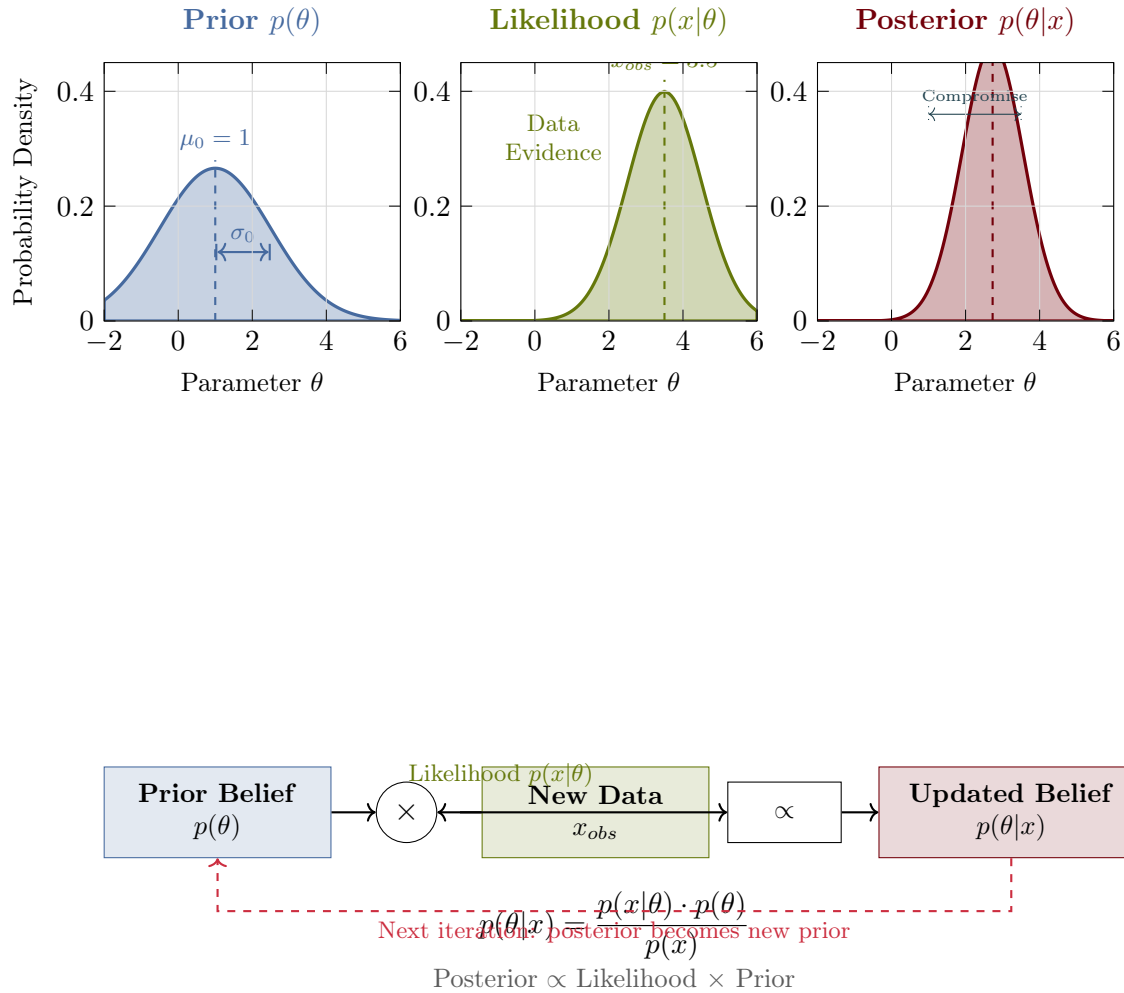


Figure 7.1: Bayesian updating process showing how prior beliefs are combined with data to form posterior estimates. **Top row:** The prior distribution (Atlantic blue, $\mu_0 = 1$, broad uncertainty) represents initial knowledge before observing data. The likelihood function (Horseshoe green) quantifies how probable the observed data $x_{obs} = 3.5$ is for each possible parameter value. The posterior distribution (Garnet) is the normalized product, representing updated knowledge—narrower than the prior (reduced uncertainty) with mean shifted toward the data. **Bottom:** Schematic of Bayes' theorem. The dashed Rose arrow shows sequential updating: each posterior becomes the prior for the next observation. This framework underlies Kalman filtering, recursive estimation, and probabilistic inference in control systems.

Interpretation: Despite the sensor having a 95% detection rate, when an alarm triggers, there is only about a 24% chance of an actual fault. This counterintuitive result (the *base rate fallacy*) occurs because faults are rare events. The false alarms from the large population of non-fault cases dominate.

Engineering implication: This analysis is critical for designing alert systems in control applications. A second confirmation measurement or a more stringent threshold may be needed to reduce false positives.

7.2 Random Variables

Discrete Random Variables

- Probability mass function: $p(x) = P(X = x)$
- Expected value: $E[X] = \sum_x x \cdot p(x)$
- Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Continuous Random Variables

- Probability density function: $f(x)$ where $P(a \leq X \leq b) = \int_a^b f(x) dx$
- Expected value: $E[X] = \int_{-\infty}^{\infty} x f(x) dx$
- Cumulative distribution: $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

Properties of Expectation and Variance

For random variables X and Y , and constants a, b :

$$E[aX + b] = aE[X] + b \quad (7.4)$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad (7.5)$$

$$E[X + Y] = E[X] + E[Y] \quad (\text{always true}) \quad (7.6)$$

If X and Y are independent:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad (7.7)$$

7.3 Common Distributions

Gaussian (Normal) Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (7.8)$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

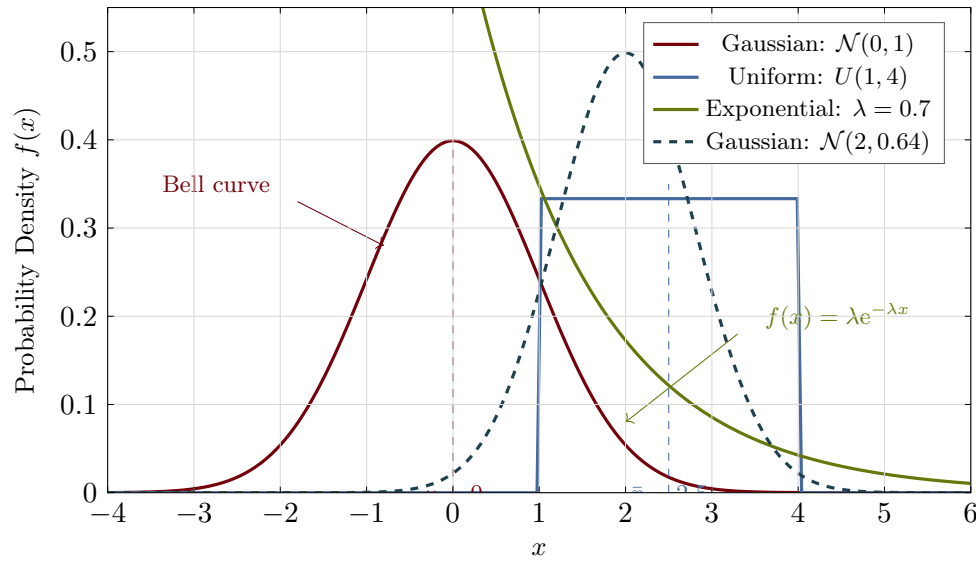


Figure 7.2: Comparison of common probability distributions used in control systems and stochastic modeling. The Gaussian (normal) distribution (Garnet, solid) centered at $\mu = 0$ with $\sigma = 1$ is the most prevalent in estimation and filtering. The Uniform distribution (Atlantic blue) models bounded uncertainty with constant probability over $[1, 4]$. The Exponential distribution (Horseshoe green) characterizes waiting times and failure rates. A second Gaussian (Congaree teal, dashed) shows how shifting the mean and adjusting variance changes the distribution shape. Note how each distribution integrates to unity over its support.

Multivariate Gaussian

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (7.9)$$

where $\boldsymbol{\mu}$ is the mean vector and Σ is the covariance matrix.

Other Important Distributions

- **Uniform:** $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$
- **Exponential:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$
- **Bernoulli:** $P(X = 1) = p$, $P(X = 0) = 1 - p$
- **Poisson:** $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

Example: Mean and Variance of a Mixture of Distributions

A manufacturing process has two operational modes. In Mode A (70% of the time), output follows $\mathcal{N}(10, 1)$. In Mode B (30% of the time), output follows $\mathcal{N}(15, 4)$. Find the overall mean and variance.

Solution:

Let X be the output and M be the mode indicator. The mixture distribution has PDF:

$$f_X(x) = 0.7 \cdot f_A(x) + 0.3 \cdot f_B(x)$$

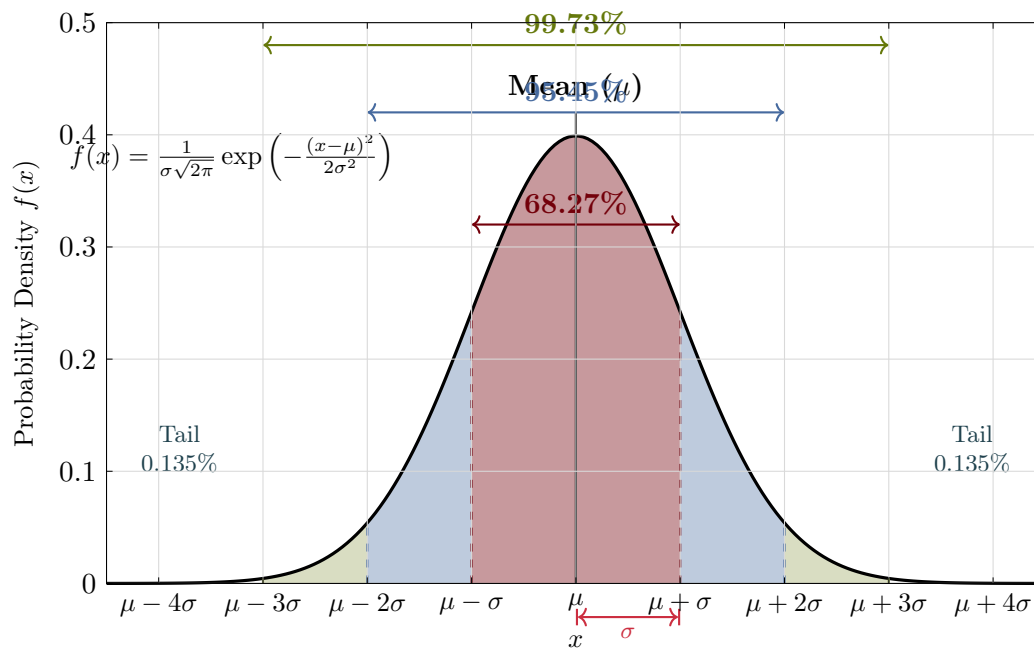


Figure 7.3: The Gaussian (normal) distribution illustrating the 68-95-99.7 empirical rule. The probability density function is shown with shaded regions indicating the probability mass within one (Garnet, 68.27%), two (Atlantic blue, 95.45%), and three (Horseshoe green, 99.73%) standard deviations of the mean. This rule is fundamental in statistical quality control, confidence interval estimation, and Kalman filter design. In control systems, sensor noise is often modeled as Gaussian with known μ and σ , making these probability regions essential for uncertainty quantification.

Mean: Using the law of total expectation:

$$\begin{aligned} E[X] &= E[E[X|M]] = P(M = A)E[X|M = A] + P(M = B)E[X|M = B] \\ &= 0.7 \times 10 + 0.3 \times 15 = 7 + 4.5 = 11.5 \end{aligned}$$

Variance: Using the law of total variance:

$$\text{Var}(X) = E[\text{Var}(X|M)] + \text{Var}(E[X|M])$$

First term (expected conditional variance):

$$E[\text{Var}(X|M)] = 0.7 \times 1 + 0.3 \times 4 = 0.7 + 1.2 = 1.9$$

Second term (variance of conditional means):

$$\begin{aligned} \text{Var}(E[X|M]) &= E[(E[X|M])^2] - (E[E[X|M]])^2 \\ &= (0.7 \times 10^2 + 0.3 \times 15^2) - 11.5^2 \\ &= (70 + 67.5) - 132.25 = 137.5 - 132.25 = 5.25 \end{aligned}$$

Total variance:

$$\text{Var}(X) = 1.9 + 5.25 = 7.15$$

Key insight: The variance of a mixture (7.15) is larger than both component variances (1 and 4) because the variance of conditional means contributes significantly when the modes have different means.

7.4 Covariance and Correlation

Understanding the relationship between multiple random variables is fundamental for multivariate analysis, state estimation, and sensor fusion in control systems.

Definition: Covariance

The **covariance** between two random variables X and Y measures their linear association:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y] \quad (7.10)$$

Properties of Covariance

1. $\text{Cov}(X, X) = \text{Var}(X)$
2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ (symmetry)
3. $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$ (linearity)
4. $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$ (distributive)
5. If X and Y are independent, then $\text{Cov}(X, Y) = 0$

Covariance Zero Does Not Imply Independence

The converse of property 5 is **not** true. Two random variables can have zero covariance but still be dependent. Covariance only measures *linear* relationships.

Example: Let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$. Then $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0 - 0 = 0$, but X and Y are clearly dependent.

Definition: Correlation Coefficient

The **Pearson correlation coefficient** normalizes covariance to a dimensionless measure:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \quad (7.11)$$

where $\rho_{XY} \in [-1, 1]$.

Interpretation:

- $\rho = 1$: Perfect positive linear relationship
- $\rho = -1$: Perfect negative linear relationship
- $\rho = 0$: No linear relationship (uncorrelated)
- $|\rho| \approx 1$: Strong linear relationship

Covariance Matrices

For a random vector $\mathbf{x} = [X_1, X_2, \dots, X_n]^\top$, the covariance matrix captures all pairwise covariances:

Definition: Covariance Matrix

The **covariance matrix** Σ of a random vector $\mathbf{x}$ is:

$$\Sigma = E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top] \quad (7.12)$$

where $\boldsymbol{\mu} = E[\mathbf{x}]$. The (i, j) element is $\Sigma_{ij} = \text{Cov}(X_i, X_j)$.

Explicitly, the covariance matrix is:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n) \end{bmatrix} \quad (7.13)$$

Properties of Covariance Matrices

For any covariance matrix Σ :

1. **Symmetric:** $\Sigma = \Sigma^\top$
2. **Positive semi-definite:** $\mathbf{v}^\top \Sigma \mathbf{v} \geq 0$ for all $\mathbf{v}$
3. **Diagonal elements are variances:** $\Sigma_{ii} = \text{Var}(X_i) \geq 0$

4. Off-diagonal bounds: $|\Sigma_{ij}| \leq \sqrt{\Sigma_{ii}\Sigma_{jj}}$ (follows from $|\rho| \leq 1$)

Linear transformation of covariance: If $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$, then:

$$\Sigma_{\mathbf{y}} = \mathbf{A}\Sigma_{\mathbf{x}}\mathbf{A}^{\top} \quad (7.14)$$

This result is fundamental in Kalman filtering when propagating uncertainty through system dynamics.

7.5 Stochastic Processes

A **stochastic process** is a collection of random variables indexed by time: $\{X(t) : t \in T\}$

White Noise

Uncorrelated random sequence with:

- $E[w(t)] = 0$
- $E[w(t)w(\tau)] = \sigma^2\delta(t - \tau)$

Markov Process

Future states depend only on present state:

$$P(X_{n+1}|X_n, X_{n-1}, \dots, X_0) = P(X_{n+1}|X_n) \quad (7.15)$$

7.6 Estimation Theory

Maximum Likelihood Estimation (MLE)

Definition: Maximum Likelihood Estimator

Given observations $\mathbf{x} = (x_1, x_2, \dots, x_n)$ from a distribution $f(x|\theta)$, the **maximum likelihood estimator** is:

$$\hat{\theta}_{ML} = \arg \max_{\theta} \prod_{i=1}^n f(x_i|\theta) = \arg \max_{\theta} \sum_{i=1}^n \log f(x_i|\theta) \quad (7.16)$$

The second form (log-likelihood) is typically more convenient for computation.

Example: Maximum Likelihood Estimation for Gaussian Parameters

Given n independent observations $x_1, x_2, \dots, x_n$ from a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$, find the MLEs for μ and σ^2 .

Solution:

Step 1: Write the likelihood function.

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Step 2: Take the log-likelihood.

$$\begin{aligned} \ell(\mu, \sigma^2) &= \log L(\mu, \sigma^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

Step 3: Find the MLE for μ .

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

Solving: $\sum_{i=1}^n x_i = n\mu$, so:

$$\hat{\mu}_{ML} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Step 4: Find the MLE for σ^2 .

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

Solving:

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Note: The MLE for variance is biased (expected value is $\frac{n-1}{n}\sigma^2$). The unbiased estimator uses $n - 1$ in the denominator:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Bayesian Estimation

$$p(\theta|\mathbf{x}) \propto p(\mathbf{x}|\theta)p(\theta) \quad (7.17)$$

(posterior $\propto$ likelihood $\times$ prior)

Example: Bayesian Parameter Estimation

Suppose we want to estimate the mean μ of a Gaussian distribution with known variance $\sigma^2 = 1$. We have a prior belief that $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$ where $\mu_0 = 0$ and $\tau_0^2 = 4$. After observing a single measurement $x = 2.5$, find the posterior distribution.

Solution:

Step 1: Identify the components.

- Prior: $p(\mu) \propto \exp\left(-\frac{(\mu-0)^2}{2 \times 4}\right) = \exp\left(-\frac{\mu^2}{8}\right)$
- Likelihood: $p(x|\mu) \propto \exp\left(-\frac{(x-\mu)^2}{2}\right) = \exp\left(-\frac{(2.5-\mu)^2}{2}\right)$

Step 2: Apply Bayes' theorem.

$$\begin{aligned} p(\mu|x) &\propto p(x|\mu) \cdot p(\mu) \\ &\propto \exp\left(-\frac{(2.5-\mu)^2}{2}\right) \cdot \exp\left(-\frac{\mu^2}{8}\right) \end{aligned}$$

Step 3: Combine the exponents.

$$\begin{aligned} -\frac{(2.5-\mu)^2}{2} - \frac{\mu^2}{8} &= -\frac{4(2.5-\mu)^2 + \mu^2}{8} \\ &= -\frac{4(6.25 - 5\mu + \mu^2) + \mu^2}{8} \\ &= -\frac{25 - 20\mu + 4\mu^2 + \mu^2}{8} \\ &= -\frac{5\mu^2 - 20\mu + 25}{8} \end{aligned}$$

Step 4: Complete the square.

$$\begin{aligned} 5\mu^2 - 20\mu + 25 &= 5(\mu^2 - 4\mu) + 25 \\ &= 5(\mu^2 - 4\mu + 4 - 4) + 25 \\ &= 5(\mu - 2)^2 + 5 \end{aligned}$$

So:

$$p(\mu|x) \propto \exp\left(-\frac{5(\mu-2)^2}{8}\right) = \exp\left(-\frac{(\mu-2)^2}{2 \times 0.8}\right)$$

Result: The posterior is $\mu|x \sim \mathcal{N}(\mu_n, \tau_n^2)$ where:

$$\begin{aligned} \tau_n^2 &= \frac{1}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} = \frac{1}{\frac{1}{4} + 1} = \frac{4}{5} = 0.8 \\ \mu_n &= \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{x}{\sigma^2} \right) = 0.8 \left(\frac{0}{4} + \frac{2.5}{1} \right) = 2.0 \end{aligned}$$

Interpretation:

- The posterior mean (2.0) is between the prior mean (0) and the observation (2.5), weighted by their respective precisions.
- The posterior variance (0.8) is smaller than both the prior variance (4) and the likelihood variance (1), reflecting increased certainty after incorporating data.
- This “Gaussian-Gaussian” conjugate pair is the foundation of the Kalman filter.

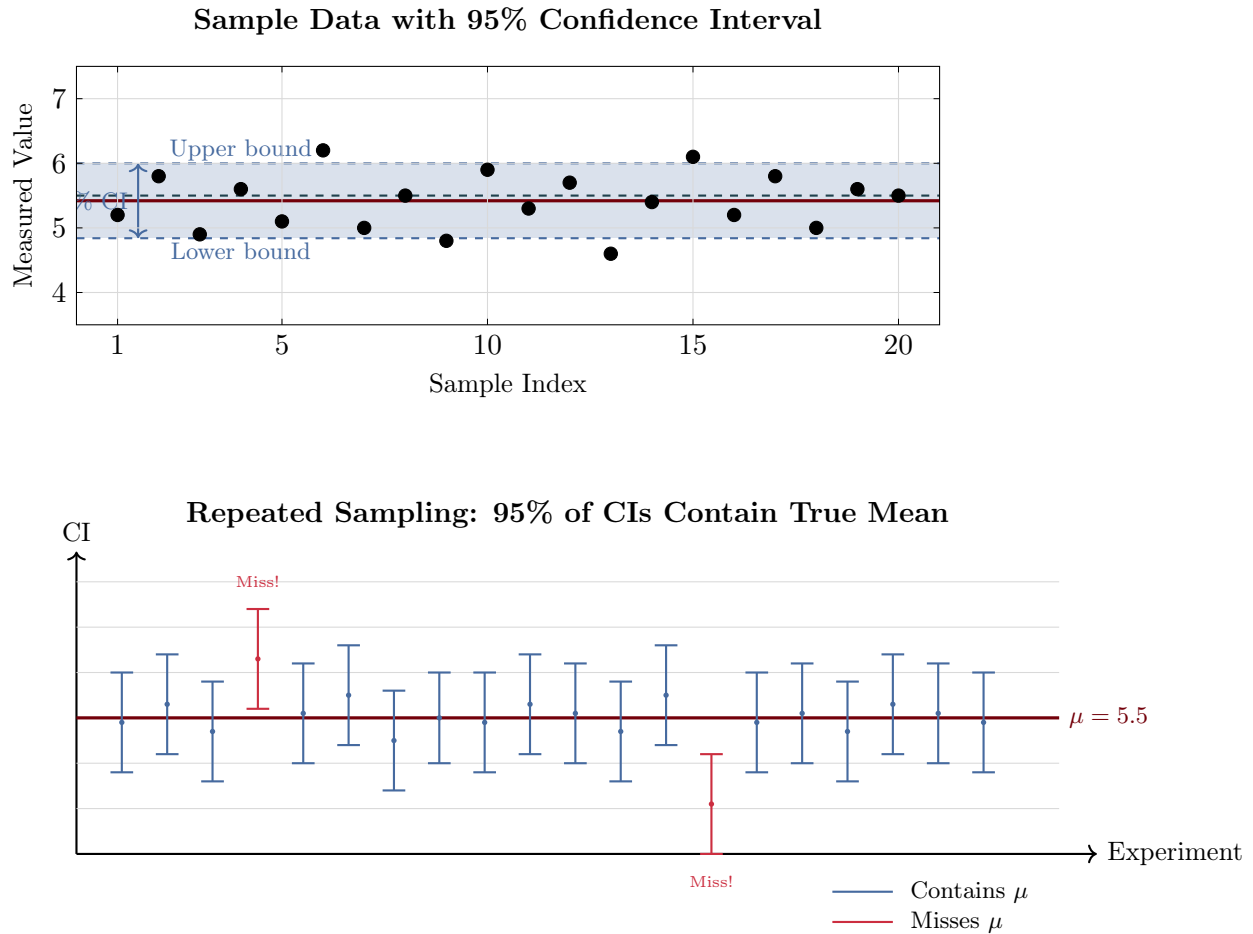


Figure 7.4: Confidence intervals for parameter estimation. **Top:** A sample of $n = 20$ measurements (black dots) with sample mean $\bar{x} = 5.42$ (Garnet line) and 95% confidence interval (Atlantic blue shaded region). The true population mean μ (Congaree dashed) falls within the CI. **Bottom:** Interpretation of confidence level through repeated sampling. Each vertical bar represents a 95% CI from an independent experiment. Approximately 95% of intervals (Atlantic blue) contain the true mean, while 5% (Rose) do not.

Minimum Mean Square Error (MMSE)

$$\hat{\theta}_{MMSE} = E[\theta|\mathbf{x}] \quad (7.18)$$

Comparison of Estimators

Estimator	Criterion	Result
MLE	Maximize $p(\mathbf{x} \theta)$	Mode of likelihood
MAP	Maximize $p(\theta \mathbf{x})$	Mode of posterior
MMSE	Minimize $E[(\theta - \hat{\theta})^2 \mathbf{x}]$	Mean of posterior

For symmetric, unimodal posteriors (like Gaussians), MAP and MMSE coincide.

7.7 Hypothesis Testing Basics

Hypothesis testing provides a framework for making decisions under uncertainty, which is essential in fault detection, change detection, and quality control in control systems.

Definition: Hypothesis Test

A **hypothesis test** is a procedure for deciding between two competing claims:

- H_0 : The **null hypothesis** (default assumption, often “no effect” or “no change”)
- H_1 (or H_a): The **alternative hypothesis** (the claim we’re testing for)

Example hypotheses in control systems:

- H_0 : System is operating normally; H_1 : Fault has occurred
- H_0 : Controller gain equals design value; H_1 : Gain has changed
- H_0 : Sensor is calibrated; H_1 : Sensor has drifted

Types of Errors

Definition: Type I and Type II Errors

	H_0 True	H_1 True
Reject H_0	Type I Error (α)	Correct Decision
Fail to reject H_0	Correct Decision	Type II Error (β)

- **Type I Error (False Positive):** Rejecting H_0 when it is true. Probability: α (significance level)
- **Type II Error (False Negative):** Failing to reject H_0 when H_1 is true. Probability: β
- **Power:** $1 - \beta$ = probability of correctly rejecting H_0 when H_1 is true

Trade-off Between Error Types

There is an inherent trade-off between Type I and Type II errors:

- Decreasing α (fewer false alarms) typically increases β (more missed detections)
- The only way to reduce both simultaneously is to collect more data

In control applications, the relative costs of each error type should guide the choice of α .

The p-value

Definition: p-value

The **p-value** is the probability of observing a test statistic as extreme as, or more extreme than, the observed value, assuming the null hypothesis is true:

$$p\text{-value} = P(\text{test statistic} \geq \text{observed value} \mid H_0 \text{ true}) \quad (7.19)$$

(For two-sided tests, consider both tails.)

Interpretation:

- Small p-value ($< \alpha$): Strong evidence against H_0 ; reject H_0
- Large p-value ($\geq \alpha$): Insufficient evidence against H_0 ; fail to reject H_0
- Common significance levels: $\alpha = 0.05$ (5%), $\alpha = 0.01$ (1%)

Common Misinterpretations of p-values

The p-value is **NOT**:

- The probability that H_0 is true
- The probability that H_1 is false
- The probability of making an error

The p-value is the probability of the *data* (or more extreme), given H_0 is true—not the probability of H_0 given the data.

Example: Z-test for a Mean

Setup: Test whether a population mean μ differs from a hypothesized value μ_0 when the population variance σ^2 is known.

Hypotheses: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ (two-sided)

Test statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad (7.20)$$

Under H_0 , $Z \sim \mathcal{N}(0, 1)$.

Decision rule: Reject H_0 if $|Z| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the critical value from the standard normal distribution.

Connection to Confidence Intervals

Hypothesis testing and confidence intervals are dual concepts:

- A $100(1 - \alpha)\%$ confidence interval contains all parameter values that would *not* be rejected at significance level α
- Rejecting $H_0 : \mu = \mu_0$ at level α is equivalent to μ_0 being outside the $(1 - \alpha)$ confidence interval

Chapter 8

Optimization

Optimization is central to optimal control, parameter estimation, and machine learning. This chapter introduces the fundamental concepts and methods that underpin modern control system design and analysis.

8.1 Unconstrained Optimization

Unconstrained optimization seeks to find the minimum (or maximum) of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ without any restrictions on the decision variable $\mathbf{x}$.

Definition: Local and Global Minima

A point $\mathbf{x}^*$ is a **local minimum** of f if there exists $\epsilon > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x}$ with $\|\mathbf{x} - \mathbf{x}^*\| < \epsilon$.

A point $\mathbf{x}^*$ is a **global minimum** if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$.

8.1.1 Optimality Conditions

Necessary Condition (First-Order)

At a local minimum $\mathbf{x}^*$ of a differentiable function:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad (8.1)$$

Points satisfying this condition are called **stationary points** or **critical points**.

Sufficient Condition (Second-Order)

If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}(\mathbf{x}^*) \succ 0$ (positive definite Hessian), then $\mathbf{x}^*$ is a **strict local minimum**.

The Hessian Matrix

The Hessian matrix $\mathbf{H}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the $n \times n$ matrix of second partial derivatives:

$$\mathbf{H}_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (8.2)$$

For twice continuously differentiable functions, the Hessian is symmetric: $\mathbf{H} = \mathbf{H}^\top$.

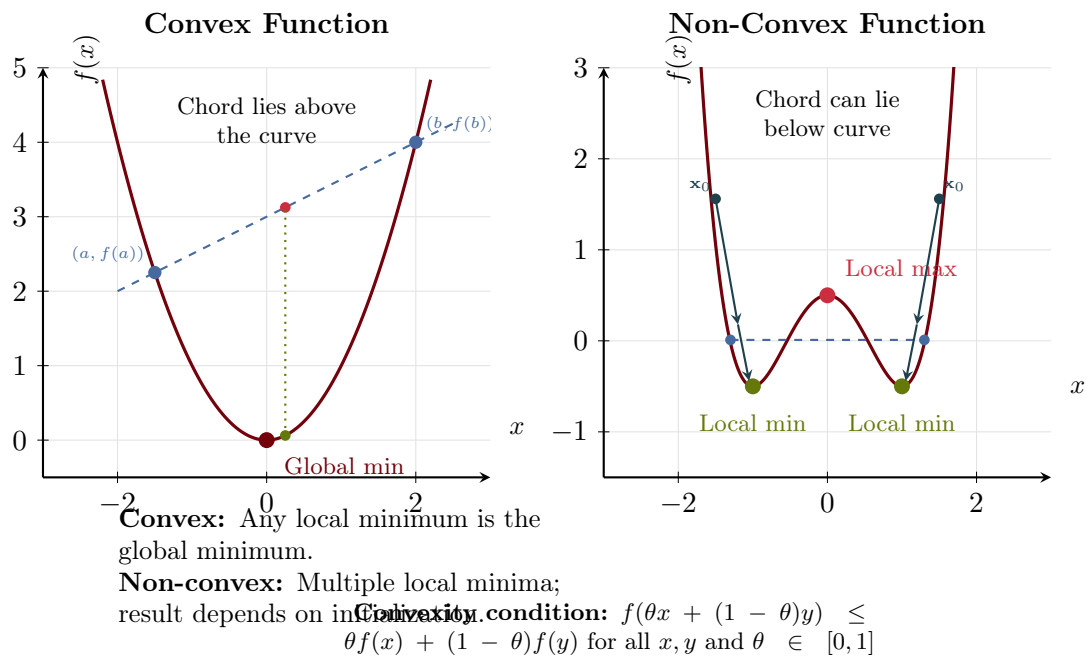


Figure 8.1: Comparison of convex and non-convex functions. **Left:** A convex function $f(x) = x^2$ (Garnet curve). The chord connecting any two points on the curve (Atlantic dashed line) lies above the curve, satisfying the convexity definition. There is a unique global minimum (Garnet dot). **Right:** A non-convex function $f(x) = x^4 - 2x^2 + 0.5$ (double-well potential). The chord can lie below portions of the curve. Multiple local minima (Horseshoe dots) exist, separated by a local maximum (Rose dot). Gradient descent paths (Congaree arrows) from different initial points converge to different local minima, illustrating why non-convex optimization is challenging.

Example: Finding the Minimum of a Quadratic Function

Find the minimum of the function $f(x, y) = x^2 + 4y^2 - 2x - 8y + 5$.

Solution:

Step 1: Compute the gradient.

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x - 2 \\ 8y - 8 \end{bmatrix} \quad (8.3)$$

Step 2: Set the gradient to zero (necessary condition).

$$2x - 2 = 0 \quad \Rightarrow \quad x^* = 1 \quad (8.4)$$

$$8y - 8 = 0 \quad \Rightarrow \quad y^* = 1 \quad (8.5)$$

The critical point is $\mathbf{x}^* = (1, 1)$.

Step 3: Compute the Hessian.

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix} \quad (8.6)$$

Step 4: Check positive definiteness (sufficient condition).

The eigenvalues of $\mathbf{H}$ are $\lambda_1 = 2 > 0$ and $\lambda_2 = 8 > 0$, so $\mathbf{H} \succ 0$.

Alternatively, check that all leading principal minors are positive:

- $\det([2]) = 2 > 0$
- $\det(\mathbf{H}) = 2 \cdot 8 - 0 = 16 > 0$

Conclusion: The point $(1, 1)$ is a strict local minimum (and global minimum, since f is convex).

The minimum value is $f(1, 1) = 1 + 4 - 2 - 8 + 5 = 0$.

8.1.2 Gradient Descent

Gradient descent is an iterative optimization algorithm that moves in the direction of steepest descent:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) \quad (8.7)$$

where $\alpha > 0$ is the **step size** (or **learning rate**).

Intuition Behind Gradient Descent

The negative gradient $-\nabla f(\mathbf{x})$ points in the direction of steepest descent of f at point $\mathbf{x}$. By taking small steps in this direction, we progressively reduce the function value, eventually converging to a local minimum (under appropriate conditions).

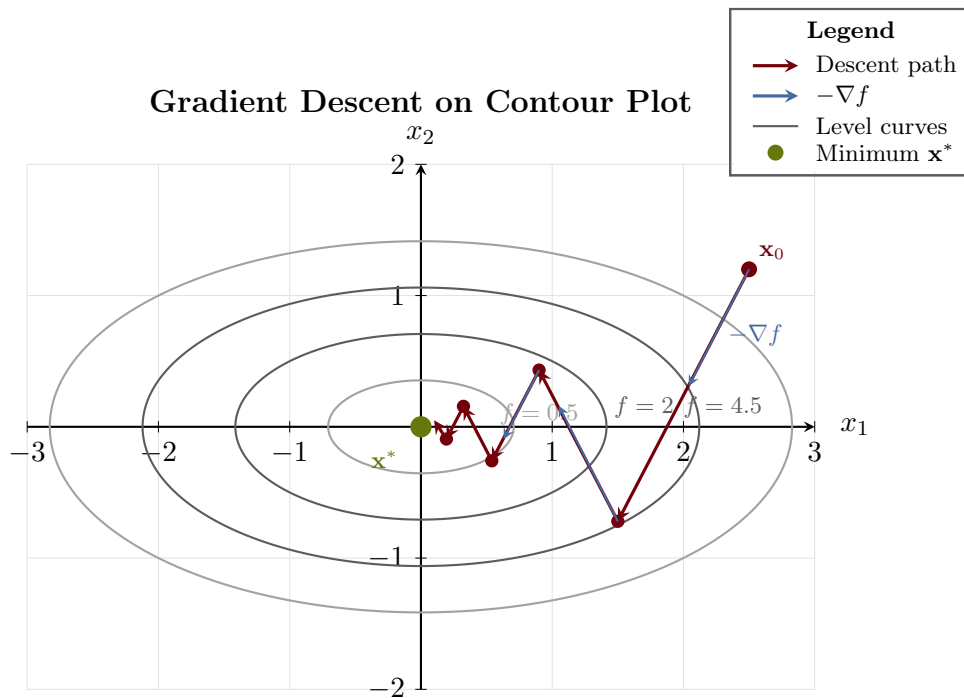


Figure 8.2: Gradient descent on a quadratic function $f(\mathbf{x}) = x_1^2 + 4x_2^2$. The elliptical contours represent level curves of constant function value. Starting from initial point $\mathbf{x}_0 = (2.5, 1.2)$, gradient descent iteratively moves in the direction of steepest descent (negative gradient, shown in Atlantic blue). The zigzag pattern (Garnet path) is characteristic of gradient descent on ill-conditioned problems where the contours are elongated ellipses. Each step moves perpendicular to the local level curve, eventually converging to the global minimum at $\mathbf{x}^* = (0, 0)$ (Horseshoe green).

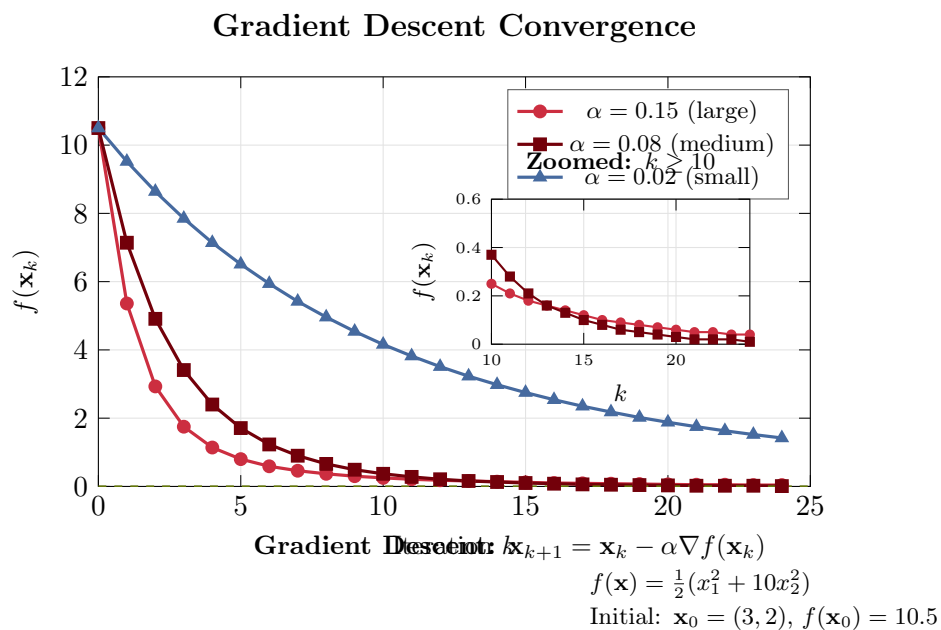


Figure 8.3: Convergence of gradient descent for a quadratic function $f(\mathbf{x}) = \frac{1}{2}(x_1^2 + 10x_2^2)$ with different step sizes α . A large step size (Rose, $\alpha = 0.15$) converges fastest initially but may oscillate. A medium step size (Garnet, $\alpha = 0.08$) provides stable, efficient convergence. A small step size (Atlantic blue, $\alpha = 0.02$) guarantees convergence but is slow. The inset shows that after $k = 10$ iterations, larger step sizes reach lower function values. The optimal step size balances speed and stability, typically $\alpha < 2/L$ where L is the Lipschitz constant of the gradient.

Example: Gradient Descent on the Rosenbrock Function

The **Rosenbrock function** is a classic test problem for optimization algorithms:

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2 \quad (8.8)$$

This function has a global minimum at $(x^*, y^*) = (1, 1)$ where $f(1, 1) = 0$. The minimum lies in a narrow, curved valley, making it challenging for gradient-based methods.

Solution:

Step 1: Compute the gradient.

$$\frac{\partial f}{\partial x} = -2(1 - x) - 400x(y - x^2) = 2x - 2 + 400x^3 - 400xy \quad (8.9)$$

$$\frac{\partial f}{\partial y} = 200(y - x^2) \quad (8.10)$$

Step 2: Apply gradient descent with $\alpha = 0.001$ and initial point $(x_0, y_0) = (-1, 1)$.

Computing the first few iterations:

k	x_k	y_k	$f(x_k, y_k)$
0	-1.000	1.000	4.000
1	-0.996	1.000	3.985
2	-0.992	1.000	3.970
5	-0.980	1.000	3.925
10	-0.960	0.999	3.852
50	-0.810	0.937	3.416
100	-0.631	0.684	2.871
500	0.382	0.158	0.395
1000	0.723	0.524	0.077
5000	0.987	0.974	0.00017

Key observations:

- Convergence is slow due to the ill-conditioned nature of the problem (the eigenvalues of the Hessian vary widely).
- The algorithm first moves quickly toward the valley floor, then slowly along the valley toward the minimum.
- A very small step size is needed to prevent divergence near the minimum.

Practical lesson: The Rosenbrock function illustrates why more sophisticated methods (conjugate gradient, quasi-Newton, Adam) are often preferred over vanilla gradient descent.

8.1.3 Newton's Method

Newton's method uses second-order information to achieve faster convergence:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\mathbf{H}(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) \quad (8.11)$$

Computational Considerations

Newton's method requires computing and inverting the Hessian at each iteration, which is $O(n^3)$ for an n -dimensional problem. For large-scale problems, **quasi-Newton methods** (such as BFGS or L-BFGS) approximate the Hessian using gradient information, achieving a balance between convergence speed and computational cost.

8.2 Convergence Analysis

Understanding how quickly optimization algorithms converge is crucial for selecting appropriate methods.

Definition: Convergence Rate

Let $\{\mathbf{x}_k\}$ be a sequence converging to $\mathbf{x}^*$. The sequence has:

- **Linear convergence** if there exists $c \in (0, 1)$ such that:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq c \|\mathbf{x}_k - \mathbf{x}^*\| \quad (8.12)$$

The error decreases by a constant factor at each iteration.

- **Superlinear convergence** if:

$$\lim_{k \rightarrow \infty} \frac{\|\mathbf{x}_{k+1} - \mathbf{x}^*\|}{\|\mathbf{x}_k - \mathbf{x}^*\|} = 0 \quad (8.13)$$

- **Quadratic convergence** if there exists $M > 0$ such that:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq M \|\mathbf{x}_k - \mathbf{x}^*\|^2 \quad (8.14)$$

The number of correct digits roughly doubles at each iteration.

Convergence Rate Comparison

Method	Rate	Cost per iteration
Gradient descent	Linear	$O(n)$
Conjugate gradient	Superlinear	$O(n)$
Newton's method	Quadratic	$O(n^3)$
Quasi-Newton (BFGS)	Superlinear	$O(n^2)$

8.2.1 Learning Rate Selection Guidelines

The choice of learning rate (step size) α critically affects convergence:

- **Too large:** Algorithm may diverge or oscillate around the minimum.
- **Too small:** Convergence is extremely slow, wasting computational resources.

- **Optimal (for quadratic functions):** $\alpha = 2/(\lambda_{\max} + \lambda_{\min})$ where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest eigenvalues of the Hessian.

Condition Number and Convergence

For gradient descent on a quadratic function with Hessian $\mathbf{H}$, the convergence rate is:

$$c = \frac{\kappa - 1}{\kappa + 1} \quad (8.15)$$

where $\kappa = \lambda_{\max}/\lambda_{\min}$ is the **condition number**. A large condition number (ill-conditioned problem) leads to slow convergence.

Practical learning rate strategies:

1. **Line search:** At each iteration, choose α to minimize $f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k))$ along the search direction.
2. **Backtracking:** Start with a large α and reduce it until sufficient decrease is achieved (Armijo condition).
3. **Adaptive methods:** Algorithms like Adam, RMSprop, and AdaGrad automatically adjust learning rates based on gradient history.
4. **Learning rate schedules:** Decrease α over time (e.g., $\alpha_k = \alpha_0/(1 + \beta k)$).

8.3 Constrained Optimization

Many practical problems have constraints on the decision variables. We consider both equality and inequality constraints.

8.3.1 Equality Constraints: Lagrange Multipliers

Consider minimizing $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$ where $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $m < n$.

The **Lagrangian function** is:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) \quad (8.16)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^m$ is the vector of **Lagrange multipliers**.

Necessary Conditions

At a constrained local minimum, the following conditions hold:

$$\nabla_{\mathbf{x}} L = \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) = \mathbf{0} \quad (8.17)$$

$$\nabla_{\boldsymbol{\lambda}} L = \mathbf{g}(\mathbf{x}^*) = \mathbf{0} \quad (8.18)$$

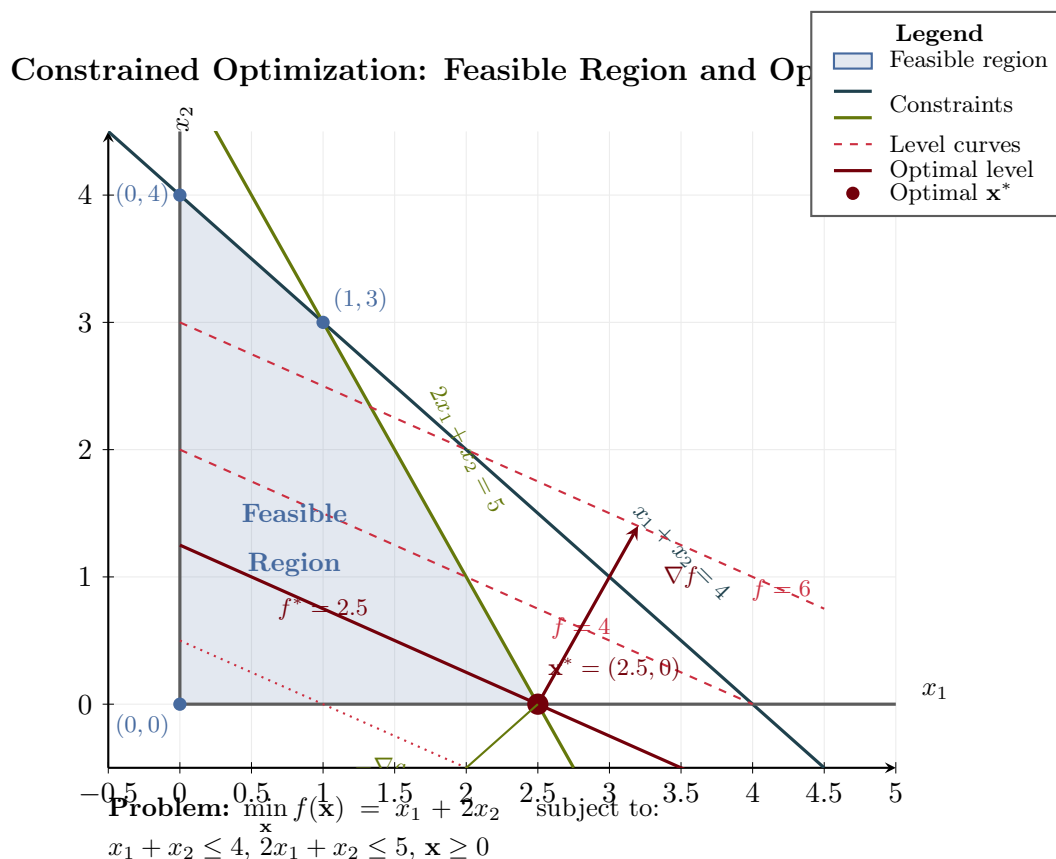


Figure 8.4: Visualization of a constrained optimization problem. The feasible region (shaded in Atlantic blue) is the intersection of half-planes defined by the linear inequality constraints. The dashed Rose curves are level sets of the linear objective function $f(\mathbf{x}) = x_1 + 2x_2$, with lower values toward the origin. The optimal point $\mathbf{x}^* = (2.5, 0)$ (Garnet) occurs where the lowest level curve touches the feasible region boundary. At this vertex, the gradient ∇f points outward from the feasible region, satisfying the KKT conditions. The constraint $2x_1 + x_2 \leq 5$ is active at the optimum.

Example: Lagrange Multipliers for Constrained Optimization

Find the minimum of $f(x, y) = x^2 + y^2$ subject to $g(x, y) = x + y - 4 = 0$.

Problem interpretation: Find the point on the line $x + y = 4$ closest to the origin.

Solution:

Step 1: Form the Lagrangian.

$$L(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 4) \quad (8.19)$$

Step 2: Compute the necessary conditions.

$$\frac{\partial L}{\partial x} = 2x + \lambda = 0 \quad (8.20)$$

$$\frac{\partial L}{\partial y} = 2y + \lambda = 0 \quad (8.21)$$

$$\frac{\partial L}{\partial \lambda} = x + y - 4 = 0 \quad (8.22)$$

Step 3: Solve the system.

From the first two equations: $2x + \lambda = 2y + \lambda \Rightarrow x = y$.

Substituting into the constraint: $x + x = 4 \Rightarrow x = 2$.

Therefore: $x^* = 2$, $y^* = 2$, and $\lambda^* = -4$.

Verification:

- Constraint satisfied: $2 + 2 - 4 = 0 \checkmark$
- $\nabla f = (2x, 2y) = (4, 4)$ and $\nabla g = (1, 1)$
- $\nabla f = -\lambda^* \nabla g$: $(4, 4) = -(-4)(1, 1) = (4, 4) \checkmark$

Interpretation of λ^* : The Lagrange multiplier $\lambda^* = -4$ represents the rate of change of the optimal value with respect to the constraint bound. If the constraint were $x + y = 4 + \epsilon$ for small ϵ , the minimum value would be approximately $f^* + (-4)(-\epsilon) = 8 + 4\epsilon$.

The minimum distance to the origin is $\sqrt{f(2, 2)} = \sqrt{8} = 2\sqrt{2}$.

8.3.2 Inequality Constraints: KKT Conditions

For problems with inequality constraints:

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad \mathbf{h}(\mathbf{x}) \leq \mathbf{0} \quad (8.23)$$

The **Karush-Kuhn-Tucker (KKT) conditions** are necessary for optimality (and sufficient for convex problems):

$$\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla h_i(\mathbf{x}^*) = \mathbf{0} \quad (\text{Stationarity}) \quad (8.24)$$

$$h_i(\mathbf{x}^*) \leq 0 \quad (\text{Primal feasibility}) \quad (8.25)$$

$$\mu_i \geq 0 \quad (\text{Dual feasibility}) \quad (8.26)$$

$$\mu_i h_i(\mathbf{x}^*) = 0 \quad (\text{Complementary slackness}) \quad (8.27)$$

Complementary Slackness

The condition $\mu_i h_i(\mathbf{x}^*) = 0$ means that for each constraint:

- Either $h_i(\mathbf{x}^*) = 0$ (constraint is **active**/binding), or
- $\mu_i = 0$ (constraint has no influence on the solution)

This elegant condition encodes the intuition that only constraints that are “tight” at the optimum affect the solution.

8.4 Linear Programming

Linear programming (LP) is the optimization of a linear objective function subject to linear constraints.

Definition: Linear Program

A **linear program** in standard form is:

$$\min_{\mathbf{x}} \quad \mathbf{c}^\top \mathbf{x} \quad (8.28)$$

$$\text{subject to} \quad \mathbf{Ax} = \mathbf{b} \quad (8.29)$$

$$\mathbf{x} \geq \mathbf{0} \quad (8.30)$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, and $\mathbf{b} \in \mathbb{R}^m$.

Converting to Standard Form

Any LP can be converted to standard form:

- **Maximization:** $\max \mathbf{c}^\top \mathbf{x} = -\min(-\mathbf{c}^\top \mathbf{x})$
- **Inequality constraints:** $\mathbf{Ax} \leq \mathbf{b}$ becomes $\mathbf{Ax} + \mathbf{s} = \mathbf{b}$ with slack variable $\mathbf{s} \geq \mathbf{0}$
- **Free variables:** x_i unrestricted becomes $x_i = x_i^+ - x_i^-$ with $x_i^+, x_i^- \geq 0$

8.4.1 Graphical Solution Method (2D)

For two-variable LPs, the solution can be found graphically:

1. **Plot the constraint region:** Each linear constraint defines a half-plane. The **feasible region** is the intersection of all half-planes.
2. **Identify vertices:** The feasible region is a convex polygon (if bounded and non-empty).
3. **Find the optimal vertex:** The optimal solution occurs at a vertex of the feasible region. Either:
 - Evaluate the objective at each vertex and select the best, or
 - Draw level curves of the objective and find the extreme feasible point.

Example: Linear Programming (Graphical Solution)

Solve the following linear program:

$$\begin{aligned} \max_{x_1, x_2} \quad & 3x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Solution:

Step 1: Identify the feasible region.

The constraints define the following half-planes:

- $x_1 + x_2 \leq 4$: region below the line $x_1 + x_2 = 4$
- $2x_1 + x_2 \leq 6$: region below the line $2x_1 + x_2 = 6$
- $x_1 \geq 0$: right of the x_2 -axis
- $x_2 \geq 0$: above the x_1 -axis

Step 2: Find the vertices of the feasible region.

The vertices are found by solving pairs of constraint equations:

- $A = (0, 0)$: intersection of $x_1 = 0$ and $x_2 = 0$
- $B = (3, 0)$: intersection of $2x_1 + x_2 = 6$ and $x_2 = 0$
- $C = (2, 2)$: intersection of $x_1 + x_2 = 4$ and $2x_1 + x_2 = 6$
- $D = (0, 4)$: intersection of $x_1 + x_2 = 4$ and $x_1 = 0$

To find C : Subtract the first equation from the second:

$$\begin{aligned} (2x_1 + x_2) - (x_1 + x_2) &= 6 - 4 \\ x_1 &= 2 \\ x_2 &= 4 - 2 = 2 \end{aligned}$$

Step 3: Evaluate the objective function at each vertex.

Vertex	(x_1, x_2)	$3x_1 + 2x_2$
A	$(0, 0)$	0
B	$(3, 0)$	9
C	$(2, 2)$	10
D	$(0, 4)$	8

Conclusion: The maximum value is 10 at the point $(x_1^*, x_2^*) = (2, 2)$.

Geometric interpretation: The objective function $3x_1 + 2x_2 = k$ represents a family of parallel lines. We seek the line with the largest k that still touches the feasible region. This occurs at vertex C .

8.4.2 The Simplex Method

For problems with more than two variables, the graphical method is impractical. The **simplex method**, developed by George Dantzig in 1947, is an algebraic algorithm that efficiently solves LPs.

Simplex Method Overview

The simplex method exploits a fundamental property of linear programs: *if an optimal solution exists, it occurs at a vertex (basic feasible solution) of the feasible region.*

The algorithm:

1. Start at a basic feasible solution (vertex)
2. Move along an edge to an adjacent vertex that improves the objective
3. Repeat until no improving direction exists (optimality)

Although the feasible region may have exponentially many vertices, the simplex method typically visits only a small number before finding the optimum.

Computational Complexity

While the simplex method works extremely well in practice, its worst-case complexity is exponential. For guaranteed polynomial-time performance, **interior-point methods** (such as the barrier method) are preferred for very large-scale problems.

8.5 Convex Optimization

Convex optimization problems have the remarkable property that any local minimum is also a global minimum.

Definition: Convex Function

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if for all $\mathbf{x}, \mathbf{y}$ and $\theta \in [0, 1]$:

$$f(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y}) \quad (8.31)$$

Geometrically, the function lies below (or on) the line segment connecting any two points on its graph.

Why Convexity Matters

For convex optimization problems:

1. **No local minima trap:** Any local minimum is global
2. **Efficient algorithms:** Polynomial-time algorithms exist with guaranteed convergence
3. **Strong duality:** Under mild conditions, primal and dual problems have equal optimal values

4. **Robust solutions:** Small perturbations lead to small changes in the optimal solution

Second-Order Characterization

For a twice-differentiable function f :

$$f \text{ is convex} \iff \mathbf{H}(\mathbf{x}) \succeq 0 \text{ for all } \mathbf{x} \quad (8.32)$$

where $\mathbf{H} \succeq 0$ denotes positive semidefiniteness.

Examples of Convex Problems

- **Linear programming:** $\min \mathbf{c}^\top \mathbf{x}$ subject to $\mathbf{Ax} \leq \mathbf{b}$
- **Quadratic programming:** $\min \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}$ subject to $\mathbf{Ax} \leq \mathbf{b}$ (when $\mathbf{Q} \succeq 0$)
- **Least squares:** $\min \|\mathbf{Ax} - \mathbf{b}\|^2$
- **Semidefinite programming:** Optimization over positive semidefinite matrices

8.6 Dynamic Programming

For sequential decision problems, **Bellman's Principle of Optimality** provides a powerful decomposition:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

The **Bellman equation** for discrete-time optimal control:

$$V^*(x_k) = \min_{u_k} [L(x_k, u_k) + V^*(f(x_k, u_k))] \quad (8.33)$$

where:

- $V^*(x)$ is the **optimal value function** (minimum cost-to-go from state x)
- $L(x_k, u_k)$ is the stage cost at time k
- $f(x_k, u_k)$ is the state transition function

Dynamic Programming in Control

Dynamic programming is foundational to:

- **Model Predictive Control (MPC):** Solve finite-horizon optimal control problems online
- **Linear Quadratic Regulator (LQR):** Closed-form solution via the Riccati equation
- **Reinforcement Learning:** Learn optimal policies through interaction with the environment

Chapter 9

Discrete-Time Systems

Digital control systems operate on sampled data, requiring discrete-time analysis. This chapter develops the mathematical foundations for analyzing signals and systems that exist only at discrete time instants, forming the basis for digital control system design.

9.1 Sampling and Reconstruction

The transition from continuous-time to discrete-time systems begins with **sampling**—the process of converting a continuous signal into a sequence of values at discrete time instants.

9.1.1 Ideal Sampling

Definition: Ideal Sampling

Ideal sampling of a continuous-time signal $x(t)$ produces a sequence of impulses:

$$x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \quad (9.1)$$

where T is the **sampling period** and $f_s = 1/T$ is the **sampling frequency** (in Hz).

In the frequency domain, multiplication by an impulse train produces spectral replications:

$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s) \quad (9.2)$$

The spectrum of the sampled signal consists of the original spectrum repeated at intervals of f_s .

9.1.2 The Nyquist-Shannon Sampling Theorem

Nyquist-Shannon Sampling Theorem

A bandlimited continuous-time signal with maximum frequency $f_{\max}$ can be **perfectly reconstructed** from its samples if and only if the sampling frequency satisfies:

$$f_s > 2f_{\max} \quad (9.3)$$

The minimum sampling rate $f_N = 2f_{\max}$ is called the **Nyquist rate**, and $f_s/2$ is called the **Nyquist frequency** (the highest frequency that can be represented without aliasing).

Aliasing

When the sampling theorem is violated ($f_s < 2f_{\max}$), **aliasing** occurs: high-frequency components appear as spurious low-frequency components in the sampled signal. A signal at frequency f will appear at the aliased frequency:

$$f_{\text{alias}} = |f - kf_s| \quad (9.4)$$

where k is chosen so that $f_{\text{alias}} \leq f_s/2$. Aliasing is **irreversible**—once aliased, the original frequency cannot be recovered.

9.1.3 Anti-Aliasing Filters

To prevent aliasing, an **anti-aliasing filter** (a low-pass filter with cutoff frequency below $f_s/2$) must be applied before sampling:

Anti-Aliasing Filter Design

For practical systems:

1. Determine the maximum signal frequency of interest, $f_{\max}$
2. Choose $f_s \geq 2.5f_{\max}$ to $10f_{\max}$ (oversample to allow for non-ideal filter rolloff)
3. Design an analog low-pass filter with cutoff $f_c < f_s/2$
4. The filter should provide sufficient attenuation (> 40 dB typical) at $f_s/2$

Common anti-aliasing filters include Butterworth (maximally flat passband), Chebyshev (sharper rolloff), and Bessel (linear phase for minimal distortion).

9.1.4 Sample-and-Hold Operation

Practical analog-to-digital converters (ADCs) cannot sample instantaneously. Instead, they use a **sample-and-hold** circuit that captures the signal value and holds it constant during conversion.

Definition: Zero-Order Hold (ZOH)

A **zero-order hold** maintains each sample value constant until the next sample:

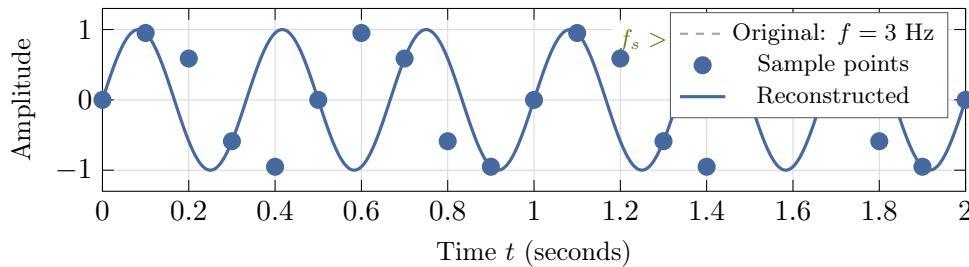
$$x_{\text{ZOH}}(t) = x[n], \quad nT \leq t < (n+1)T \quad (9.5)$$

The ZOH has frequency response:

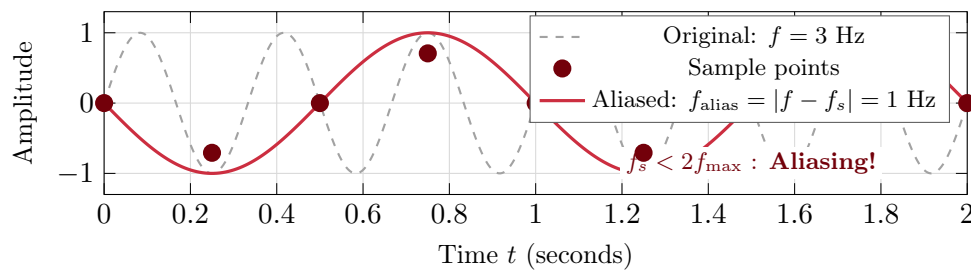
$$H_{\text{ZOH}}(f) = T \cdot \text{sinc}(\pi fT) \cdot e^{-j\pi fT} = T \frac{\sin(\pi fT)}{\pi fT} e^{-j\pi fT} \quad (9.6)$$

Sampling and the Nyquist Frequency

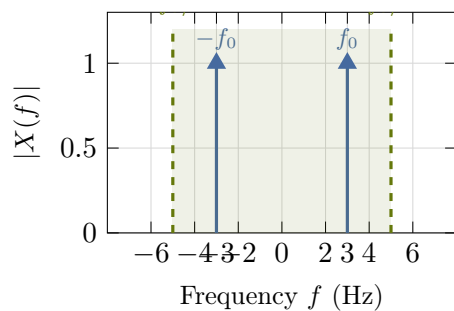
Adequate Sampling: $f_s = 10 \text{ Hz} > 2f_{\text{signal}} = 6 \text{ Hz}$ (Nyquist criterion satisfied)



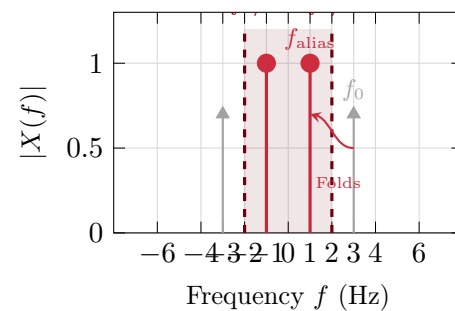
Undersampling: $f_s = 4 \text{ Hz} < 2f_{\text{signal}} = 6 \text{ Hz}$ (Aliasing occurs!)



Adequate Sampling Spectrum



Undersampled Spectrum (Aliasing)



Nyquist-Shannon Sampling Theorem: A bandlimited signal with maximum frequency f_{max} can be perfectly reconstructed from samples if $f_s > 2f_{\text{max}}$.

Nyquist Frequency: $f_N = f_s/2$ **Nyquist Rate:** $f_s = 2f_{\text{max}}$

Figure 9.1: Nyquist sampling and aliasing for discrete-time systems.

The ZOH introduces a half-sample delay and acts as a low-pass filter, attenuating high frequencies. This must be accounted for in digital controller design.

9.2 Discrete-Time Signals

Definition: Discrete-Time Signal

A **discrete-time signal** (or **sequence**) is a function defined only at integer values of the independent variable:

$$x[n], \quad n \in \mathbb{Z} \quad (9.7)$$

where n is the **sample index**. If derived from sampling a continuous signal at period T , then $x[n] = x(nT)$.

9.2.1 Elementary Sequences

The fundamental building blocks of discrete-time signal analysis include:

Unit Impulse (Kronecker Delta)

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (9.8)$$

Any sequence can be expressed as a sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \quad (9.9)$$

Unit Step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (9.10)$$

Related to the impulse by: $u[n] = \sum_{k=0}^{\infty} \delta[n - k]$ and $\delta[n] = u[n] - u[n - 1]$.

Unit Ramp

$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (9.11)$$

Exponential Sequence

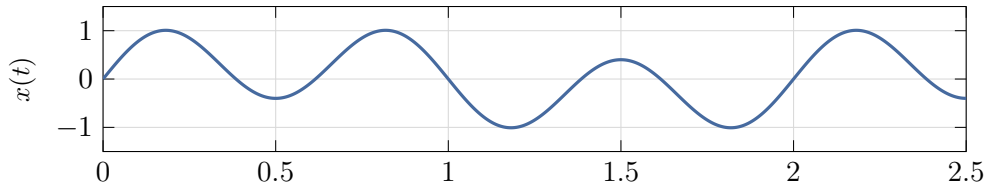
$$x[n] = a^n u[n] \quad (9.12)$$

where a can be real or complex. The sequence behavior depends on $|a|$:

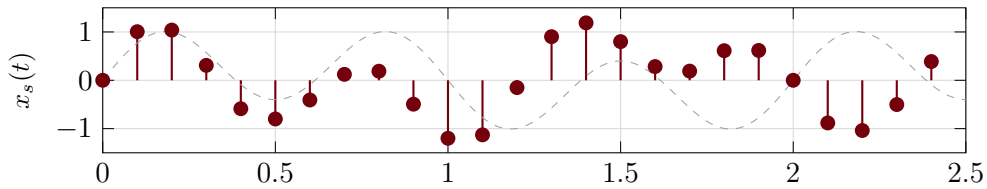
- $|a| < 1$: Decaying exponential (stable)
- $|a| = 1$: Constant magnitude (marginally stable)
- $|a| > 1$: Growing exponential (unstable)

Sampling and Reconstruction Process

(a) Continuous-Time Signal $x(t)$



(b) Ideal Sampling: $x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)$, $T = 0.1$ s ($f_s = 10$ Hz)



(c) Zero-Order Hold (ZOH) Reconstruction: Hold Each Sample for Duration T

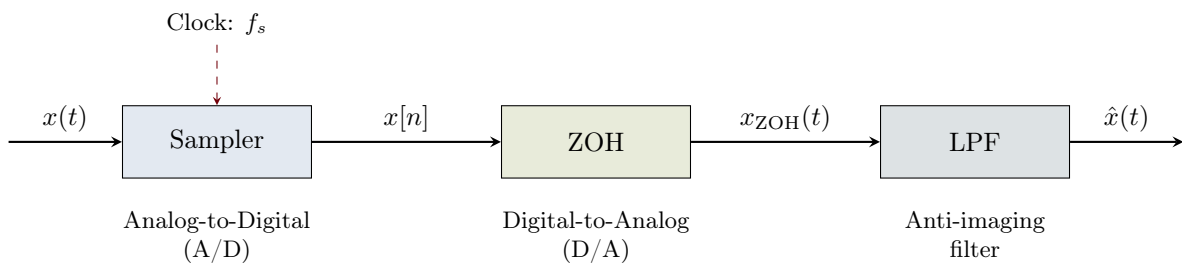
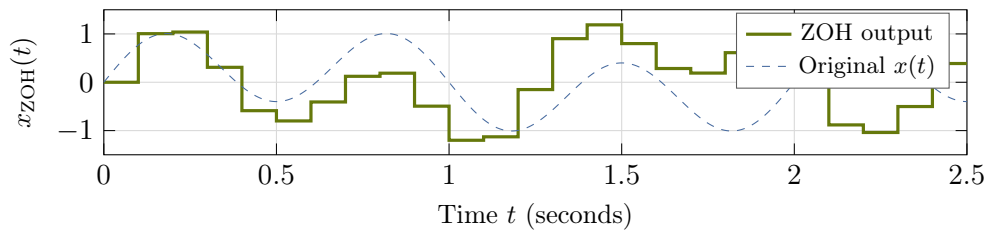


Figure 9.2: The sampling and reconstruction process. (a) A continuous-time signal $x(t)$ composed of multiple frequency components. (b) Ideal sampling multiplies the signal by an impulse train, producing discrete samples at intervals $T = 1/f_s$. (c) Zero-order hold (ZOH) reconstruction holds each sample constant until the next sample arrives, creating a staircase approximation. The block diagram shows the complete digital signal processing chain: sampling, hold, and low-pass filtering for reconstruction.

Common Discrete-Time Sequences

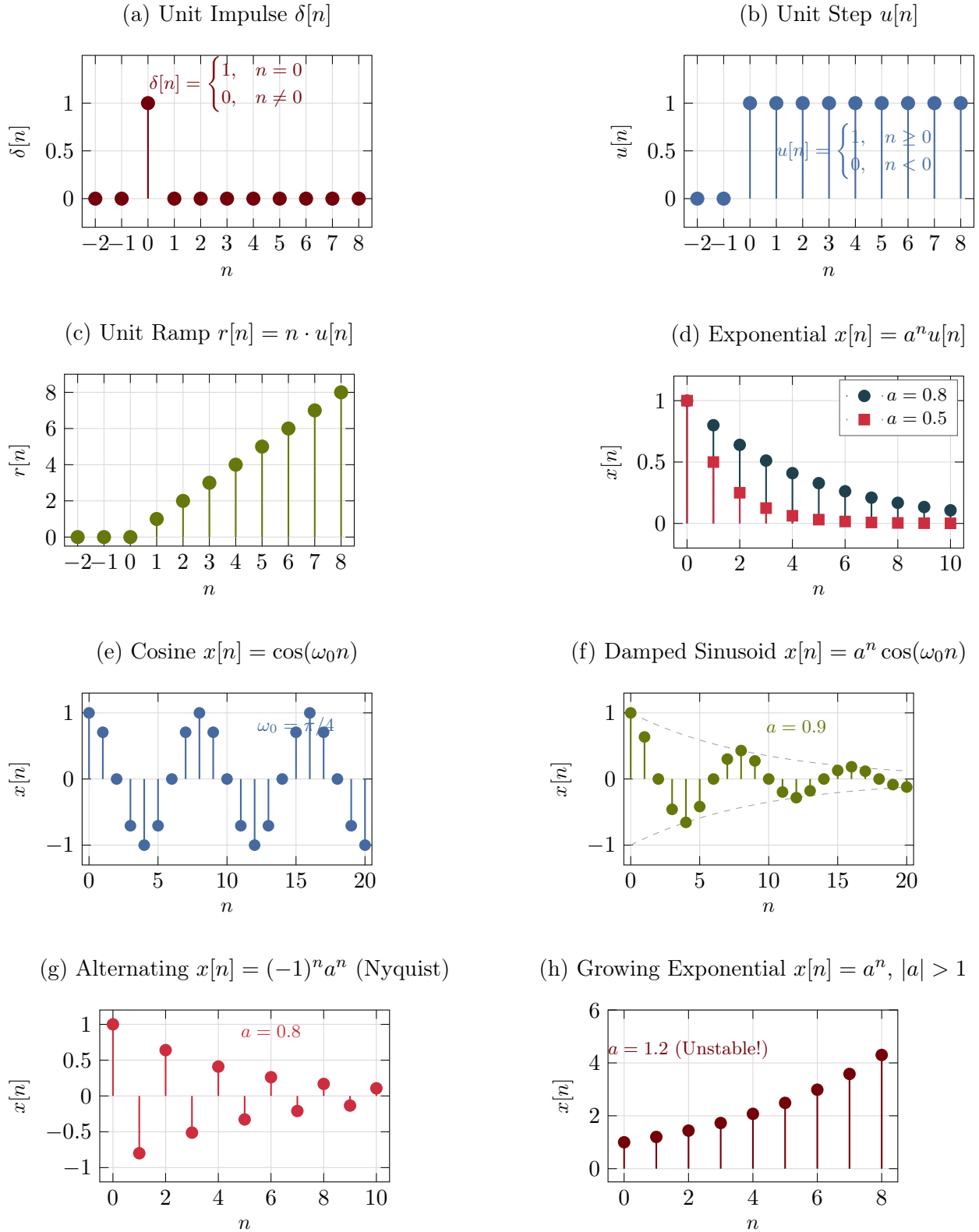


Figure 9.3: Common discrete-time sequences. (a) Unit impulse $\delta[n]$: a single nonzero value at $n = 0$. (b) Unit step $u[n]$: constant value of 1 for $n \geq 0$. (c) Unit ramp $r[n] = nu[n]$: linearly increasing for $n \geq 0$. (d) Decaying exponentials a^n with $|a| < 1$: smaller $|a|$ gives faster decay. (e) Discrete cosine: periodic with period $2\pi/\omega_0$ samples. (f) Damped sinusoid: exponential envelope times sinusoid. (g) Alternating sequence: corresponds to pole at negative real axis ($z < 0$). (h) Growing exponential with $|a| > 1$: represents unstable system response.

Sinusoidal Sequences

$$x[n] = A \cos(\omega_0 n + \phi) \quad (9.13)$$

$$x[n] = A \sin(\omega_0 n + \phi) \quad (9.14)$$

where ω_0 is the **discrete-time frequency** in radians per sample. Note that discrete-time sinusoids are periodic only if $\omega_0/(2\pi)$ is rational.

Complex Exponential

$$x[n] = r^n e^{j\omega_0 n} = r^n (\cos \omega_0 n + j \sin \omega_0 n) \quad (9.15)$$

This combines exponential growth/decay (r) with oscillation (ω_0).

9.2.2 Signal Operations

Time Shifting

$$y[n] = x[n - n_0] \quad (9.16)$$

Shifting by $n_0 > 0$ delays the signal; $n_0 < 0$ advances it.

Time Reversal

$$y[n] = x[-n] \quad (9.17)$$

Scaling

$$y[n] = c \cdot x[n] \quad (9.18)$$

Addition

$$y[n] = x_1[n] + x_2[n] \quad (9.19)$$

Multiplication

$$y[n] = x_1[n] \cdot x_2[n] \quad (9.20)$$

Convolution Sum

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad (9.21)$$

Convolution in the time domain corresponds to multiplication in the z -domain.

9.3 The z -Transform

The z -transform is to discrete-time systems what the Laplace transform is to continuous-time systems—it converts difference equations into algebraic equations.

Definition: z -Transform

The **z -transform** of a sequence $x[n]$ is:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (9.22)$$

where z is a complex variable. For causal sequences ($x[n] = 0$ for $n < 0$), the **unilateral z -transform** is:

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (9.23)$$

9.3.1 Region of Convergence (ROC)**Definition: Region of Convergence**

The **Region of Convergence (ROC)** is the set of all complex values z for which the z -transform sum converges absolutely:

$$\sum_{n=-\infty}^{\infty} |x[n]z^{-n}| < \infty \quad (9.24)$$

The ROC is always an annular region (a ring) in the z -plane:

- **Right-sided sequences** ($x[n] = 0$ for $n < N_1$): ROC is $|z| > r_{\min}$ (exterior of a circle)
- **Left-sided sequences** ($x[n] = 0$ for $n > N_2$): ROC is $|z| < r_{\max}$ (interior of a circle)
- **Two-sided sequences**: ROC is an annulus $r_{\min} < |z| < r_{\max}$
- **Finite-duration sequences**: ROC is entire z -plane except possibly $z = 0$ and/or $z = \infty$

ROC and Causality/Stability

For a **causal** system, the ROC extends outward from the outermost pole. For the system to also be **stable**, the ROC must include the unit circle $|z| = 1$. This means:

A causal LTI system is stable if and only if all poles lie inside the unit circle.

9.3.2 Common z -Transform Pairs**Example: Deriving the z -Transform of $a^n u[n]$**

Find the z -transform of $x[n] = a^n u[n]$.

Solution: Applying the definition for a causal sequence:

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n \quad (9.25)$$

This is a geometric series with ratio $r = a/z$. The series converges when $|a/z| < 1$, i.e.,

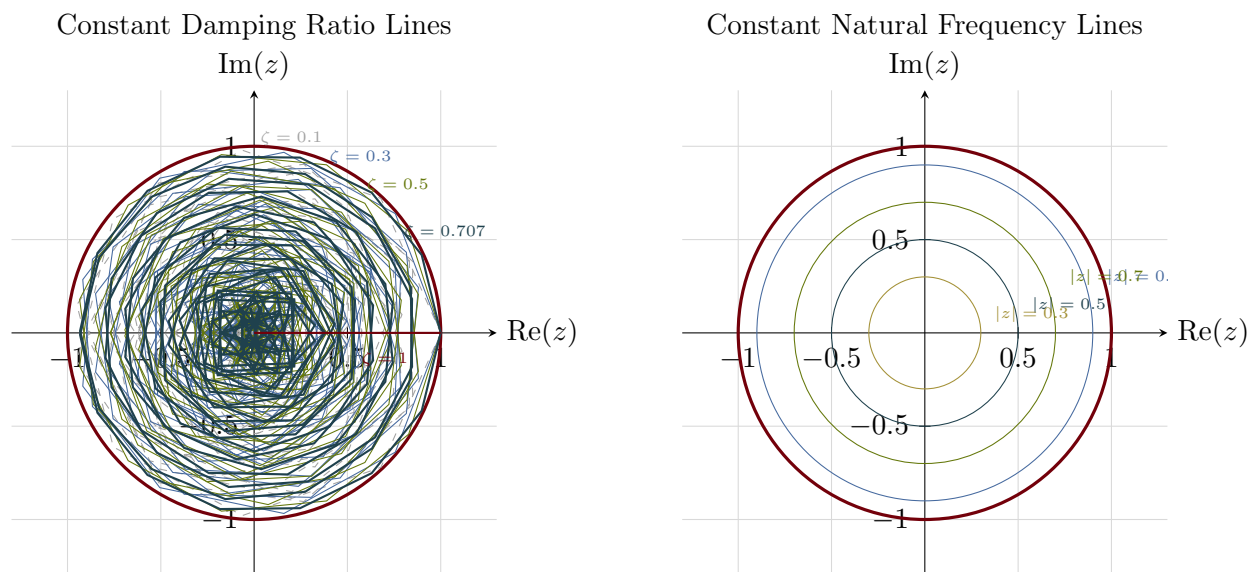
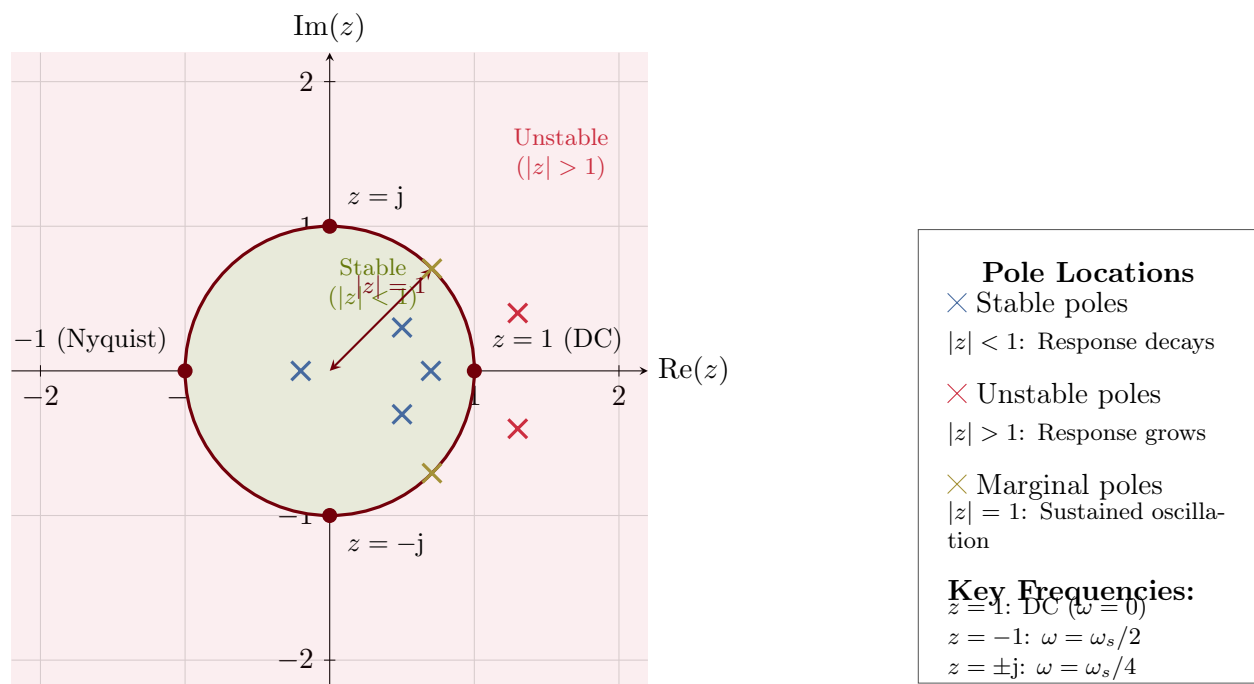
The z -Plane and Stability Regions

Figure 9.4: The z -plane for discrete-time system analysis. **Top left:** The unit circle divides stable (Horseshoe shading, $|z| < 1$) from unstable (Rose shading, $|z| > 1$) regions. Atlantic crosses mark stable poles, Rose crosses mark unstable poles, and Honeycomb crosses mark marginally stable poles on the unit circle. **Bottom left:** Lines of constant damping ratio ζ in the z -plane are logarithmic spirals. Higher damping corresponds to trajectories closer to the positive real axis. **Bottom right:** Circles of constant magnitude $|z|$ correspond to constant settling time—smaller $|z|$ gives faster decay.

Signal Name	$x[n]$ for $n \geq 0$	$X(z)$	ROC
Unit impulse	$\delta[n]$	1	All z
Unit step	$u[n] = 1$	$\frac{z}{z-1}$	$ z > 1$
Ramp	n	$\frac{z}{(z-1)^2}$	$ z > 1$
Quadratic	n^2	$\frac{z(z+1)}{(z-1)^3}$	$ z > 1$
Exponential	a^n	$\frac{z}{z-a}$	$ z > a $
Exponential ramp	na^n	$\frac{az}{(z-a)^2}$	$ z > a $
Exponential quadratic	$n^2 a^n$	$\frac{az(z+a)}{(z-a)^3}$	$ z > a $
Sine	$\sin(\omega_0 n)$	$\frac{z \sin \omega_0}{z^2 - 2z \cos \omega_0 + 1}$	$ z > 1$
Cosine	$\cos(\omega_0 n)$	$\frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}$	$ z > 1$
Damped sine	$a^n \sin(\omega_0 n)$	$\frac{az \sin \omega_0}{z^2 - 2az \cos \omega_0 + a^2}$	$ z > a $
Damped cosine	$a^n \cos(\omega_0 n)$	$\frac{z(z - a \cos \omega_0)}{z^2 - 2az \cos \omega_0 + a^2}$	$ z > a $

Table 9.1: Common z -transform pairs for causal sequences ($x[n] = 0$ for $n < 0$). $|z| > |a|$:

$$X(z) = \frac{1}{1 - a/z} = \frac{z}{z - a}, \quad |z| > |a| \quad (9.26)$$

Therefore:

$$\mathcal{Z}\{a^n u[n]\} = \frac{z}{z - a}, \quad \text{ROC: } |z| > |a| \quad (9.27)$$

9.3.3 z -Transform Properties

Property	Time Domain	z -Domain
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$
Time delay	$x[n - k], k > 0$	$z^{-k}X(z)$
Time advance	$x[n + k], k > 0$	$z^k \left(X(z) - \sum_{n=0}^{k-1} x[n]z^{-n} \right)$
Scaling in z	$a^n x[n]$	$X(z/a)$
Differentiation in z	$nx[n]$	$-z \frac{dX(z)}{dz}$
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{z}{z-1}X(z)$
Initial value	$x[0]$	$\lim_{z \rightarrow \infty} X(z)$
Final value	$\lim_{n \rightarrow \infty} x[n]$	$\lim_{z \rightarrow 1} (z-1)X(z)$

Table 9.2: Properties of the z -transform (assuming causal sequences).

Final Value Theorem Conditions

The Final Value Theorem for the z -transform is only valid if:

1. The limit $\lim_{n \rightarrow \infty} x[n]$ exists

2. $(z - 1)X(z)$ has all poles inside the unit circle

If there are poles on or outside the unit circle (except possibly a simple pole at $z = 1$), the theorem gives incorrect results.

9.3.4 Inverse z -Transform Methods

The inverse z -transform can be computed using several methods:

Partial Fraction Expansion

The most common approach: decompose $X(z)$ into simpler terms that appear in transform tables.

Partial Fraction for z -Transforms

When using partial fractions with z -transforms, first divide by z to get $X(z)/z$, perform the expansion, then multiply back by z :

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \cdots \quad (9.28)$$

Then:

$$X(z) = \frac{A_1 z}{z - p_1} + \frac{A_2 z}{z - p_2} + \cdots \quad (9.29)$$

Each term $\frac{A_i z}{z - p_i}$ corresponds to $A_i p_i^n u[n]$.

Example: Inverse z -Transform by Partial Fractions

Find the inverse z -transform of:

$$X(z) = \frac{z(2z - 1)}{(z - 1)(z - 0.5)} \quad (9.30)$$

Solution: First, divide by z :

$$\frac{X(z)}{z} = \frac{2z - 1}{(z - 1)(z - 0.5)} \quad (9.31)$$

Perform partial fraction expansion:

$$\frac{2z - 1}{(z - 1)(z - 0.5)} = \frac{A}{z - 1} + \frac{B}{z - 0.5} \quad (9.32)$$

Using the cover-up method:

$$A = \left. \frac{2z - 1}{z - 0.5} \right|_{z=1} = \frac{2(1) - 1}{1 - 0.5} = \frac{1}{0.5} = 2 \quad (9.33)$$

$$B = \left. \frac{2z - 1}{z - 1} \right|_{z=0.5} = \frac{2(0.5) - 1}{0.5 - 1} = \frac{0}{-0.5} = 0 \quad (9.34)$$

Therefore:

$$\frac{X(z)}{z} = \frac{2}{z - 1} \quad (9.35)$$

Multiplying by z :

$$X(z) = \frac{2z}{z-1} \quad (9.36)$$

From the transform table, $\frac{z}{z-1} \leftrightarrow u[n]$:

$$x[n] = 2 \cdot 1^n \cdot u[n] = 2u[n] \quad (9.37)$$

Power Series Expansion

Express $X(z)$ as a power series in z^{-1} by long division:

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots \quad (9.38)$$

The coefficients of z^{-n} give $x[n]$ directly.

Contour Integration

The formal inverse z -transform integral:

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (9.39)$$

where C is a counterclockwise contour in the ROC encircling the origin. This is typically evaluated using residues.

9.4 Difference Equations

Definition: Linear Constant-Coefficient Difference Equation

A **linear constant-coefficient difference equation (LCCDE)** of order N has the form:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m] \quad (9.40)$$

or equivalently:

$$a_0 y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M] \quad (9.41)$$

where $x[n]$ is the input and $y[n]$ is the output.

9.4.1 First-Order Difference Equations

The first-order LCCDE:

$$y[n] - ay[n-1] = bx[n] \quad (9.42)$$

Homogeneous Solution

Setting $x[n] = 0$:

$$y_h[n] = C \cdot a^n \quad (9.43)$$

where C is determined by initial conditions. The value a is the **characteristic root**.

Solution Using z -Transform

Taking the z -transform with initial condition $y[-1]$:

$$Y(z) - a(z^{-1}Y(z) + y[-1]) = bX(z) \quad (9.44)$$

$$Y(z)(1 - az^{-1}) = bX(z) + ay[-1] \quad (9.45)$$

$$Y(z) = \frac{bX(z)}{1 - az^{-1}} + \frac{ay[-1]}{1 - az^{-1}} \quad (9.46)$$

Example: First-Order Difference Equation

Solve: $y[n] - 0.8y[n-1] = x[n]$ with $y[-1] = 0$ and $x[n] = u[n]$.

Solution: Taking the z -transform:

$$Y(z) - 0.8z^{-1}Y(z) = \frac{z}{z-1} \quad (9.47)$$

Solving for $Y(z)$:

$$Y(z)(1 - 0.8z^{-1}) = \frac{z}{z-1} \quad (9.48)$$

$$Y(z) = \frac{z}{(z-1)(1-0.8z^{-1})} = \frac{z^2}{(z-1)(z-0.8)} \quad (9.49)$$

Using partial fractions on $Y(z)/z$:

$$\frac{Y(z)}{z} = \frac{z}{(z-1)(z-0.8)} = \frac{A}{z-1} + \frac{B}{z-0.8} \quad (9.50)$$

Cover-up method:

$$A = \left. \frac{z}{z-0.8} \right|_{z=1} = \frac{1}{0.2} = 5 \quad (9.51)$$

$$B = \left. \frac{z}{z-1} \right|_{z=0.8} = \frac{0.8}{-0.2} = -4 \quad (9.52)$$

Therefore:

$$Y(z) = \frac{5z}{z-1} - \frac{4z}{z-0.8} \quad (9.53)$$

Taking the inverse z -transform:

$$\boxed{y[n] = 5(1)^n u[n] - 4(0.8)^n u[n] = (5 - 4(0.8)^n) u[n]} \quad (9.54)$$

The steady-state value is $\lim_{n \rightarrow \infty} y[n] = 5$.

9.4.2 Second-Order Difference Equations

The general second-order LCCDE:

$$y[n] + a_1y[n-1] + a_2y[n-2] = b_0x[n] + b_1x[n-1] + b_2x[n-2] \quad (9.55)$$

The **characteristic equation**:

$$\lambda^2 + a_1\lambda + a_2 = 0 \quad (9.56)$$

has roots λ_1, λ_2 that determine the homogeneous solution form:

- **Distinct real roots:** $y_h[n] = C_1\lambda_1^n + C_2\lambda_2^n$
- **Repeated roots** ($\lambda_1 = \lambda_2 = \lambda$): $y_h[n] = (C_1 + C_2n)\lambda^n$
- **Complex conjugate roots** ($\lambda = re^{\pm j\theta}$): $y_h[n] = r^n(C_1 \cos n\theta + C_2 \sin n\theta)$

Example: Second-Order System with Complex Poles

Solve: $y[n] - 1.2 \cos(0.5)y[n-1] + 0.36y[n-2] = x[n]$ with zero initial conditions and $x[n] = \delta[n]$.

Solution: The characteristic equation is:

$$\lambda^2 - 1.2 \cos(0.5)\lambda + 0.36 = 0 \quad (9.57)$$

Using the quadratic formula with $\cos(0.5) \approx 0.8776$:

$$\lambda = \frac{1.2(0.8776) \pm \sqrt{1.2^2(0.8776)^2 - 4(0.36)}}{2} \quad (9.58)$$

This gives complex conjugate roots $\lambda_{1,2} = 0.6e^{\pm j0.5}$ (magnitude 0.6, angle ± 0.5 rad).

Taking the z -transform of the difference equation:

$$Y(z) (1 - 1.2 \cos(0.5)z^{-1} + 0.36z^{-2}) = 1 \quad (9.59)$$

$$Y(z) = \frac{z^2}{z^2 - 1.2 \cos(0.5)z + 0.36} \quad (9.60)$$

From the transform table for damped sinusoids with $a = 0.6$ and $\omega_0 = 0.5$:

$$y[n] = \frac{(0.6)^n \sin(0.5(n+1))}{\sin(0.5)} u[n] \quad (9.61)$$

This is a damped oscillation that decays as $(0.6)^n$.

9.4.3 Stability from the Characteristic Equation

Stability Criterion for Difference Equations

A linear constant-coefficient difference equation is **stable** (BIBO stable) if and only if all roots of its characteristic equation have magnitude less than 1:

$$|\lambda_i| < 1 \quad \text{for all } i \quad (9.62)$$

This is equivalent to requiring all poles of the system's transfer function to lie inside the unit circle.

For a second-order system with characteristic equation $\lambda^2 + a_1\lambda + a_2 = 0$, the stability conditions are:

$$|a_2| < 1 \quad (9.63)$$

$$1 + a_1 + a_2 > 0 \quad (9.64)$$

$$1 - a_1 + a_2 > 0 \quad (9.65)$$

These form a **stability triangle** in the (a_1, a_2) parameter space.

9.5 Discrete-Time Systems

9.5.1 Transfer Functions in the z-Domain

Definition: Discrete-Time Transfer Function

For a discrete-time LTI system described by an LCCDE, the **transfer function** is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^M b_m z^{-m}}{\sum_{k=0}^N a_k z^{-k}} = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{a_0 z^N + a_1 z^{N-1} + \dots + a_N} \quad (9.66)$$

assuming zero initial conditions.

The transfer function can be written in pole-zero form:

$$H(z) = K \frac{(z - z_1)(z - z_2) \dots (z - z_M)}{(z - p_1)(z - p_2) \dots (z - p_N)} = K \frac{\prod_{i=1}^M (z - z_i)}{\prod_{j=1}^N (z - p_j)} \quad (9.67)$$

where z_i are the **zeros** and p_j are the **poles**.

9.5.2 Stability in the z-Domain

BIBO Stability in the z-Domain

A causal discrete-time LTI system with transfer function $H(z)$ is **Bounded-Input Bounded-Output (BIBO) stable** if and only if all poles lie strictly inside the unit circle:

$$|p_i| < 1 \quad \text{for all poles } p_i \quad (9.68)$$

The relationship between pole location and time response:

- **Real pole at $z = p$ ($0 < p < 1$):** Decaying exponential p^n
- **Real pole at $z = p$ ($-1 < p < 0$):** Alternating decaying sequence p^n
- **Complex poles at $z = re^{\pm j\theta}$ ($r < 1$):** Damped oscillation $r^n \cos(n\theta + \phi)$
- **Pole at $z = 1$:** Constant (DC component)
- **Pole at $z = -1$:** Alternating constant ($\pm 1, \mp 1, \dots$)
- **Poles with $|z| > 1$:** Unstable growing response

9.5.3 Frequency Response

The frequency response of a discrete-time system is obtained by evaluating the transfer function on the unit circle:

Definition: Discrete-Time Frequency Response

The **frequency response** of a discrete-time system is:

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} \quad (9.69)$$

where ω is the discrete-time frequency in radians per sample. The frequency response is periodic with period 2π .

The magnitude and phase responses are:

$$|H(e^{j\omega})| = \text{magnitude response} \quad (9.70)$$

$$\angle H(e^{j\omega}) = \text{phase response} \quad (9.71)$$

Frequency Relationships

For a system with sampling period T :

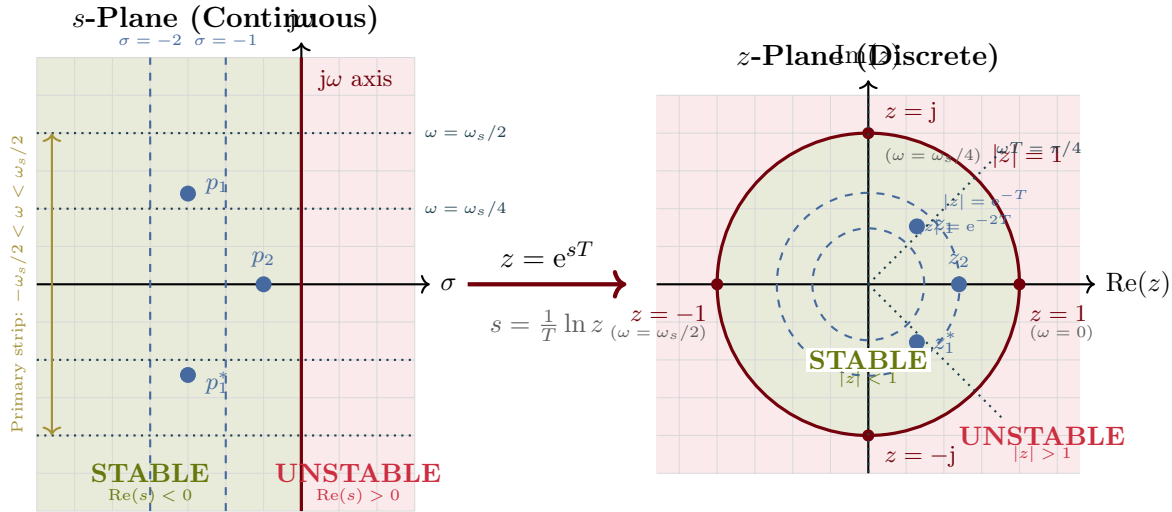
- Discrete frequency ω (rad/sample) relates to continuous frequency Ω (rad/s) by: $\omega = \Omega T$
- $\omega = 0$ corresponds to DC (zero frequency)
- $\omega = \pi$ corresponds to the Nyquist frequency $f_s/2$
- The usable frequency range is $0 \leq \omega \leq \pi$ (or equivalently, $0 \leq f \leq f_s/2$)

Example: Frequency Response of a First-Order System

Find the frequency response of $H(z) = \frac{1-a}{z-a}$ where $|a| < 1$.

Solution: Substitute $z = e^{j\omega}$:

$$H(e^{j\omega}) = \frac{1-a}{e^{j\omega}-a} \quad (9.72)$$

Mapping from the s -Plane to the z -Plane: $z = e^{sT}$ 

Key Mapping Properties ($z = e^{sT}$ where T is the sampling period, $\omega_s = 2\pi/T$):

s -Plane Feature	z -Plane Feature	Implication
Imaginary axis ($\sigma = 0$)	Unit circle ($ z = 1$)	Stability boundary preserved
Left half-plane ($\sigma < 0$)	Inside unit circle ($ z < 1$)	Stable poles map to stable poles
Right half-plane ($\sigma > 0$)	Outside unit circle ($ z > 1$)	Unstable poles map to unstable poles
Vertical line ($\sigma = \text{const}$)	Circle of radius $e^{\sigma T}$	Decay rate $\rightarrow$ pole magnitude
Horizontal line ($\omega = \text{const}$)	Ray at angle ωT	Frequency $\rightarrow$ angle on unit circle
$s = 0$ (DC)	$z = 1$	Zero frequency
$s = \pm j\omega_s/2$ (Nyquist)	$z = -1$	Highest representable frequency
Strip $-\omega_s/2 < \omega < \omega_s/2$	Entire z -plane	Frequency warping: entire strip maps once

Figure 9.5: Mapping from the continuous-time s -plane to the discrete-time z -plane via $z = e^{sT}$. **Left:** The s -plane with stable (Horseshoe) left half-plane and unstable (Rose) right half-plane. Vertical lines of constant σ (Atlantic dashed) and horizontal lines of constant ω (Congaree dotted) are shown. **Right:** The z -plane where the imaginary axis maps to the unit circle (Garnet), vertical lines map to circles of constant magnitude (Atlantic dashed), and horizontal lines map to rays of constant angle (Congaree dotted). The primary strip $-\omega_s/2 < \omega < \omega_s/2$ in the s -plane maps to the entire z -plane, with higher frequency strips overlapping (aliasing effect). Sample poles (Atlantic dots) show how complex conjugate pairs transform.

To find the magnitude, compute:

$$|H(e^{j\omega})|^2 = \frac{(1-a)^2}{|e^{j\omega} - a|^2} = \frac{(1-a)^2}{(\cos \omega - a)^2 + \sin^2 \omega} \quad (9.73)$$

Expanding the denominator:

$$(\cos \omega - a)^2 + \sin^2 \omega = \cos^2 \omega - 2a \cos \omega + a^2 + \sin^2 \omega = 1 - 2a \cos \omega + a^2 \quad (9.74)$$

Therefore:

$$|H(e^{j\omega})| = \frac{1-a}{\sqrt{1-2a \cos \omega + a^2}} \quad (9.75)$$

At DC ($\omega = 0$):

$$|H(1)| = \frac{1-a}{\sqrt{1-2a+a^2}} = \frac{1-a}{|1-a|} = 1 \quad (9.76)$$

At Nyquist ($\omega = \pi$):

$$|H(-1)| = \frac{1-a}{\sqrt{1+2a+a^2}} = \frac{1-a}{1+a} \quad (9.77)$$

This is a **low-pass filter** for $0 < a < 1$ and a **high-pass filter** for $-1 < a < 0$.

9.6 Relationship Between s-Domain and z-Domain

When discretizing a continuous-time system, there is a fundamental mapping between the s -plane and z -plane.

9.6.1 The Mapping $z = e^{sT}$

For a sampled system with sampling period T :

$$z = e^{sT} \quad (9.78)$$

This maps:

- The imaginary axis ($s = j\omega$) to the unit circle ($z = e^{j\omega T}$)
- The left half-plane ($\text{Re}(s) < 0$) to inside the unit circle ($|z| < 1$)
- The right half-plane ($\text{Re}(s) > 0$) to outside the unit circle ($|z| > 1$)

Important correspondences:

$$s = 0 \quad \leftrightarrow \quad z = 1 \quad (9.79)$$

$$s = \pm j \frac{\omega_s}{2} = \pm j \frac{\pi}{T} \quad \leftrightarrow \quad z = -1 \quad (9.80)$$

$$s = -\sigma \quad \leftrightarrow \quad z = e^{-\sigma T} \quad (9.81)$$

Important Note

The mapping $z = e^{sT}$ is **many-to-one**: frequencies differing by multiples of the sampling frequency $\omega_s = 2\pi/T$ map to the same z value. This is the mathematical basis of aliasing.

9.6.2 Discrete-Time State-Space

The continuous-time state-space system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (9.82)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (9.83)$$

can be **exactly discretized** with zero-order hold to:

$$\mathbf{x}[k+1] = \mathbf{A}_d\mathbf{x}[k] + \mathbf{B}_d\mathbf{u}[k] \quad (9.84)$$

$$\mathbf{y}[k] = \mathbf{C}_d\mathbf{x}[k] + \mathbf{D}_d\mathbf{u}[k] \quad (9.85)$$

where:

$$\mathbf{A}_d = e^{\mathbf{A}T} \quad (9.86)$$

$$\mathbf{B}_d = \int_0^T e^{\mathbf{A}\tau} d\tau \cdot \mathbf{B} = \mathbf{A}^{-1}(e^{\mathbf{A}T} - \mathbf{I})\mathbf{B} \quad (\text{if } \mathbf{A} \text{ is invertible}) \quad (9.87)$$

$$\mathbf{C}_d = \mathbf{C} \quad (9.88)$$

$$\mathbf{D}_d = \mathbf{D} \quad (9.89)$$

The matrix exponential $e^{\mathbf{A}T}$ can be computed using:

$$e^{\mathbf{A}T} = \mathbf{I} + \mathbf{A}T + \frac{(\mathbf{A}T)^2}{2!} + \frac{(\mathbf{A}T)^3}{3!} + \dots \quad (9.90)$$

Computing Discrete-Time Matrices

In practice, compute $\mathbf{A}_d$ and $\mathbf{B}_d$ using:

- MATLAB: `[Ad,Bd] = c2d(A,B,T,'zoh')`
- Python (scipy): `Ad, Bd, _, _ = cont2discrete((A,B,C,D), T, method='zoh')`
- For simple systems, eigenvalue decomposition or Padé approximation

Avoid computing $\mathbf{A}^{-1}$ explicitly; use numerically stable methods.

The discrete-time transfer function from state-space is:

$$H(z) = \mathbf{C}_d(z\mathbf{I} - \mathbf{A}_d)^{-1}\mathbf{B}_d + \mathbf{D}_d \quad (9.91)$$

Chapter 10

Special Functions and Signals

Special functions and standard test signals form the foundation of control systems analysis. Understanding these functions allows engineers to characterize system behavior, predict responses to various inputs, and design controllers that meet performance specifications.

10.1 Standard Test Inputs

Control systems are analyzed using a set of canonical test inputs. These inputs are chosen because they represent fundamental building blocks from which more complex signals can be constructed, and because they have straightforward Laplace transforms that simplify analysis.

10.1.1 Impulse Function (Dirac Delta)

Definition 10.1: Dirac Delta Function

The **Dirac delta function** (or unit impulse) $\delta(t)$ is a generalized function defined by:

$$\delta(t) = 0 \quad \text{for } t \neq 0 \quad (10.1)$$

with the property that:

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (10.2)$$

The delta function can be understood as the limit of a sequence of increasingly narrow, tall pulses whose area remains unity. One common representation is:

$$\delta(t) = \lim_{\epsilon \rightarrow 0} \begin{cases} \frac{1}{\epsilon} & 0 \leq t < \epsilon \\ 0 & \text{otherwise} \end{cases} \quad (10.3)$$

Sifting Property

The most important property of the delta function is the **sifting property**:

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0) \quad (10.4)$$

This property “sifts out” the value of $f(t)$ at the point $t = t_0$.

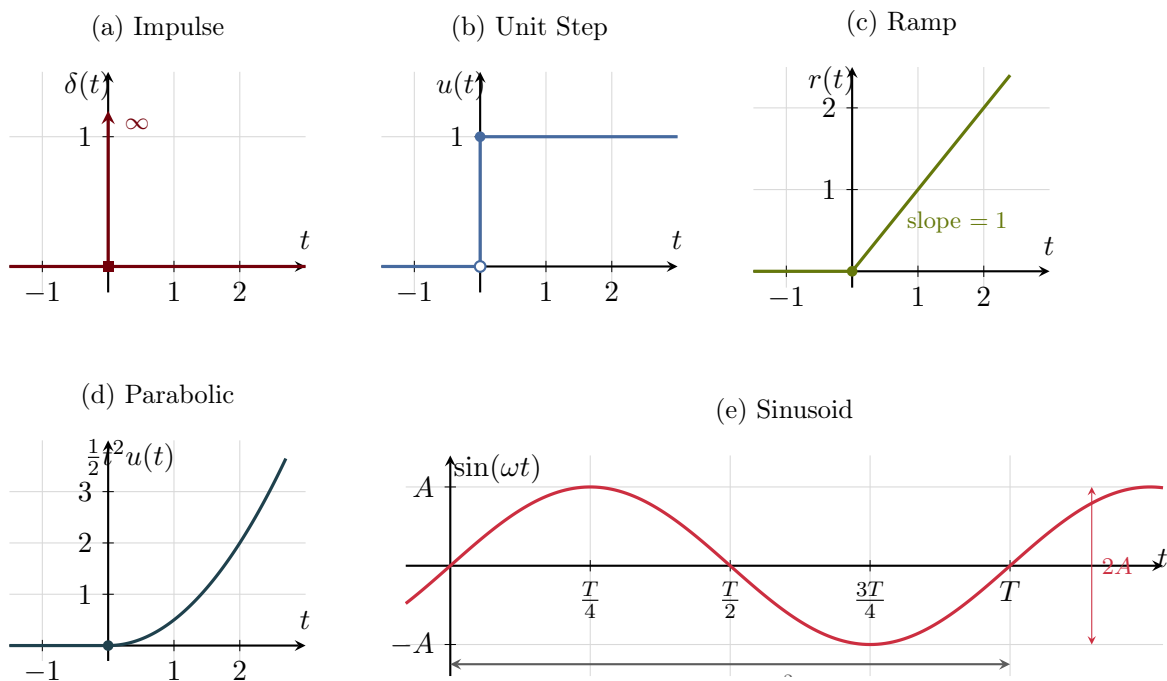


Figure 10.1: Standard test input signals for control systems analysis. (a) **Impulse function** $\delta(t)$: infinite at $t = 0$, zero elsewhere, with unit area—used to characterize system impulse response. (b) **Unit step function** $u(t)$: discontinuous jump from 0 to 1 at $t = 0$ —the most common test signal for transient response. (c) **Ramp function** $r(t) = tu(t)$: linearly increasing signal—tests steady-state tracking of changing references. (d) **Parabolic function** $\frac{1}{2}t^2 u(t)$: quadratic growth—tests acceleration tracking capability. (e) **Sinusoidal function** $A \sin(\omega t)$: periodic oscillation with amplitude A and period $T = 2\pi/\omega$ —fundamental for frequency response analysis.

Properties of the Delta Function

1. **Scaling:** $\delta(at) = \frac{1}{|a|}\delta(t)$ for $a \neq 0$
2. **Symmetry:** $\delta(-t) = \delta(t)$
3. **Multiplication:** $f(t)\delta(t - t_0) = f(t_0)\delta(t - t_0)$
4. **Derivative sifting:** $\int_{-\infty}^{\infty} f(t)\delta'(t) dt = -f'(0)$

Laplace Transform

$$\boxed{\mathcal{L}\{\delta(t)\} = 1} \quad (10.5)$$

Proof. Using the definition of the Laplace transform and the sifting property:

$$\mathcal{L}\{\delta(t)\} = \int_0^{\infty} \delta(t)e^{-st} dt = e^{-s \cdot 0} = 1 \quad (10.6)$$

□

For a delayed impulse:

$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0} \quad \text{for } t_0 \geq 0 \quad (10.7)$$

Note 10.1. The impulse response of a system is the output when the input is a unit impulse $\delta(t)$. Since $\mathcal{L}\{\delta(t)\} = 1$, the impulse response $h(t)$ has Laplace transform $H(s)$, which is the system's transfer function. This makes the impulse response fundamental to system characterization.

10.1.2 Unit Step Function

Definition 10.2: Unit Step Function

The **unit step function** (Heaviside function) is defined as:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \quad (10.8)$$

The unit step function is essential for representing signals that “turn on” at a specific time. A delayed step function $u(t - t_0)$ turns on at time $t = t_0$.

Laplace Transform

$$\boxed{\mathcal{L}\{u(t)\} = \frac{1}{s}} \quad (10.9)$$

Proof.

$$\mathcal{L}\{u(t)\} = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[-\frac{1}{s}e^{-st} \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \right) = \frac{1}{s} \quad (10.10)$$

for $\text{Re}(s) > 0$. □

For a delayed step:

$$\mathcal{L}\{u(t - t_0)\} = \frac{e^{-st_0}}{s} \quad \text{for } t_0 \geq 0 \quad (10.11)$$

Example 10.1: Step Response of a First-Order System

Find the step response of the first-order system $G(s) = \frac{1}{s+a}$.

Solution: The output in the s -domain is:

$$Y(s) = G(s) \cdot U(s) = \frac{1}{s+a} \cdot \frac{1}{s} = \frac{1}{s(s+a)} \quad (10.12)$$

Using partial fraction expansion:

$$\frac{1}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a} \quad (10.13)$$

Applying the cover-up method:

$$A = \left. \frac{1}{s+a} \right|_{s=0} = \frac{1}{a} \quad (10.14)$$

$$B = \left. \frac{1}{s} \right|_{s=-a} = -\frac{1}{a} \quad (10.15)$$

Therefore:

$$Y(s) = \frac{1}{a} \left(\frac{1}{s} - \frac{1}{s+a} \right) \quad (10.16)$$

Taking the inverse Laplace transform:

$$\boxed{y(t) = \frac{1}{a}(1 - e^{-at})u(t)} \quad (10.17)$$

This shows the characteristic exponential approach to a final value of $1/a$ with time constant $\tau = 1/a$.

10.1.3 Ramp Function**Definition 10.3: Ramp Function**

The **ramp function** is defined as:

$$r(t) = t \cdot u(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} \quad (10.18)$$

The ramp function represents a signal that increases linearly with time. It is used to test how well a system can track constantly changing inputs (velocity inputs in position control systems).

Laplace Transform

$$\boxed{\mathcal{L}\{r(t)\} = \mathcal{L}\{t \cdot u(t)\} = \frac{1}{s^2}} \quad (10.19)$$

Proof. Using integration by parts with $u = t$ and $dv = e^{-st} dt$:

$$\mathcal{L}\{t\} = \int_0^{\infty} te^{-st} dt \quad (10.20)$$

$$= \left[-\frac{t}{s} e^{-st} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt \quad (10.21)$$

$$= 0 + \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2} \quad (10.22)$$

□

For a ramp of slope K :

$$\mathcal{L}\{Kt \cdot u(t)\} = \frac{K}{s^2} \quad (10.23)$$

10.1.4 Parabolic Function

Definition 10.4: Parabolic Function

The **parabolic function** (or quadratic ramp) is defined as:

$$p(t) = \frac{1}{2}t^2 \cdot u(t) = \begin{cases} 0 & t < 0 \\ \frac{1}{2}t^2 & t \geq 0 \end{cases} \quad (10.24)$$

The parabolic input represents constant acceleration. It tests a system's ability to track accelerating inputs.

Laplace Transform

$$\mathcal{L}\left\{\frac{1}{2}t^2 \cdot u(t)\right\} = \frac{1}{s^3} \quad (10.25)$$

More generally:

$$\mathcal{L}\{t^n \cdot u(t)\} = \frac{n!}{s^{n+1}} \quad (10.26)$$

Standard Test Input Hierarchy

The standard test inputs form a hierarchy related by differentiation:

$$\text{Parabolic} \xrightarrow{\frac{d}{dt}} \text{Ramp} \xrightarrow{\frac{d}{dt}} \text{Step} \xrightarrow{\frac{d}{dt}} \text{Impulse} \quad (10.27)$$

Equivalently, in the s -domain:

$$\frac{1}{s^3} \xrightarrow{\times s} \frac{1}{s^2} \xrightarrow{\times s} \frac{1}{s} \xrightarrow{\times s} 1 \quad (10.28)$$

10.1.5 Sinusoidal Input

Definition 10.5: Sinusoidal Functions

The fundamental sinusoidal functions are:

$$x(t) = A \sin(\omega t + \phi) \quad (10.29)$$

$$x(t) = A \cos(\omega t + \phi) \quad (10.30)$$

where A is the amplitude, ω is the angular frequency (rad/s), and ϕ is the phase (rad).

Key parameters:

- **Amplitude:** A (peak value)
- **Angular frequency:** $\omega = 2\pi f$ (rad/s)
- **Frequency:** $f = \omega/(2\pi)$ (Hz)
- **Period:** $T = 1/f = 2\pi/\omega$ (s)
- **Phase:** ϕ (rad or degrees)

Laplace Transforms

$$\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2} \quad (10.31)$$

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2} \quad (10.32)$$

For damped sinusoids:

$$\mathcal{L}\{e^{-at} \sin \omega t\} = \frac{\omega}{(s + a)^2 + \omega^2} \quad (10.33)$$

$$\mathcal{L}\{e^{-at} \cos \omega t\} = \frac{s + a}{(s + a)^2 + \omega^2} \quad (10.34)$$

Complex Exponential Representation

Using Euler's formula, sinusoids can be expressed as:

$$\cos \omega t = \operatorname{Re}\{e^{j\omega t}\} = \frac{e^{j\omega t} + e^{-j\omega t}}{2} \quad (10.35)$$

$$\sin \omega t = \operatorname{Im}\{e^{j\omega t}\} = \frac{e^{j\omega t} - e^{-j\omega t}}{2j} \quad (10.36)$$

Note 10.2. Sinusoidal inputs are fundamental for frequency response analysis. When a sinusoidal input $A \sin(\omega t)$ is applied to a stable LTI system with transfer function $G(s)$, the steady-state output is:

$$y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (10.37)$$

The magnitude $|G(j\omega)|$ and phase $\angle G(j\omega)$ characterize the system's frequency response.

Signal	Time Domain	Laplace Transform	Physical Meaning
Impulse	$\delta(t)$	1	Instantaneous shock
Step	$u(t)$	$\frac{1}{s}$	Sudden constant input
Ramp	$t \cdot u(t)$	$\frac{1}{s^2}$	Constant velocity
Parabolic	$\frac{1}{2}t^2 \cdot u(t)$	$\frac{1}{s^3}$	Constant acceleration
Sine	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	Oscillatory input
Cosine	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$	Oscillatory input
Exponential	e^{-at}	$\frac{1}{s+a}$	Decay/growth

Table 10.1: Summary of standard test inputs and their Laplace transforms.

10.1.6 Summary of Standard Test Inputs

10.2 Singularity Functions

The standard test inputs are related through differentiation and integration, forming a family called **singularity functions**. Understanding these relationships provides powerful tools for signal analysis.

Definition 10.6: Singularity Functions

The **singularity functions** are a family of functions related by differentiation:

$$u_{-2}(t) = \delta'(t) \quad (\text{doublet}) \quad (10.38)$$

$$u_{-1}(t) = \delta(t) \quad (\text{impulse}) \quad (10.39)$$

$$u_0(t) = u(t) \quad (\text{step}) \quad (10.40)$$

$$u_1(t) = r(t) = tu(t) \quad (\text{ramp}) \quad (10.41)$$

$$u_2(t) = \frac{t^2}{2}u(t) \quad (\text{parabolic}) \quad (10.42)$$

In general: $u_n(t) = \frac{t^n}{n!}u(t)$ for $n \geq 0$.

10.2.1 Differentiation and Integration Relationships

The singularity functions satisfy:

$$\frac{d}{dt}u_n(t) = u_{n-1}(t) \quad (10.43)$$

$$\int_{-\infty}^t u_n(\tau) d\tau = u_{n+1}(t) \quad (10.44)$$

This gives the important relationships:

$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau \quad (10.45)$$

$$u(t) = \frac{dr(t)}{dt}, \quad r(t) = \int_{-\infty}^t u(\tau) d\tau \quad (10.46)$$

10.2.2 Laplace Transform Relationships

In the s -domain, differentiation corresponds to multiplication by s :

$$\mathcal{L}\{u_n(t)\} = \frac{1}{s^{n+1}} \quad \text{for } n \geq -1 \quad (10.47)$$

This gives:

$$\mathcal{L}\{\delta(t)\} = 1 \quad (10.48)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s} \quad (10.49)$$

$$\mathcal{L}\{r(t)\} = \frac{1}{s^2} \quad (10.50)$$

$$\mathcal{L}\{p(t)\} = \frac{1}{s^3} \quad (10.51)$$

Example 10.2: Using Singularity Function Relationships

Given that the impulse response of a system is $h(t) = e^{-2t}u(t)$, find the step response.

Solution: Since the step function is the integral of the impulse:

$$\text{Step response} = \int_0^t h(\tau) d\tau = \int_0^t e^{-2\tau} d\tau \quad (10.52)$$

Evaluating:

$$y(t) = \left[-\frac{1}{2}e^{-2\tau} \right]_0^t = -\frac{1}{2}e^{-2t} + \frac{1}{2} = \frac{1}{2}(1 - e^{-2t}) \quad (10.53)$$

Alternative approach using Laplace transforms:

The transfer function is $H(s) = \frac{1}{s+2}$.

Step response: $Y(s) = H(s) \cdot \frac{1}{s} = \frac{1}{s(s+2)} = \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right)$

Therefore: $y(t) = \frac{1}{2}(1 - e^{-2t})u(t)$

10.2.3 Use in System Analysis

Singularity functions are used extensively in control systems:

1. **Impulse Response:** The response to $\delta(t)$ is the system's impulse response $h(t)$, whose Laplace transform is the transfer function $H(s)$.
2. **Step Response:** The response to $u(t)$ reveals the system's transient behavior and steady-state characteristics.
3. **Ramp Response:** The response to $r(t)$ shows the system's tracking ability for linearly varying inputs.
4. **Error Constants:** The position, velocity, and acceleration error constants (K_p , K_v , K_a) are defined using step, ramp, and parabolic inputs respectively.

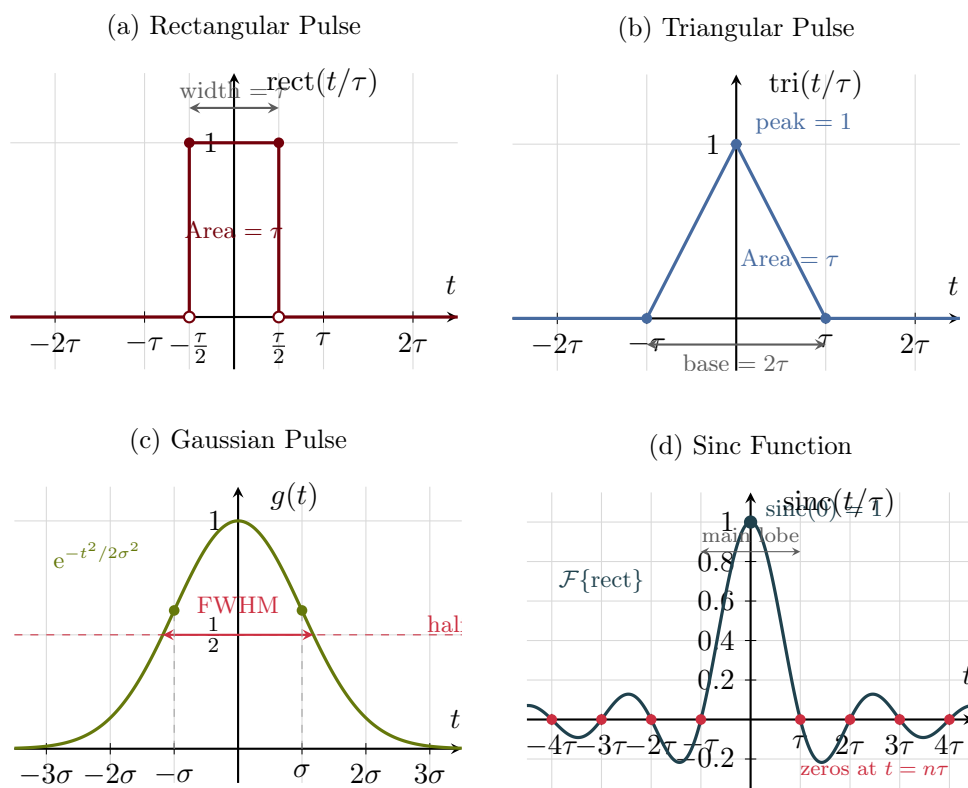


Figure 10.2: Common pulse-type signals and their characteristics. (a) **Rectangular pulse** $\text{rect}(t/\tau)$: unity height over width τ , fundamental building block with Fourier transform proportional to sinc. (b) **Triangular pulse** $\text{tri}(t/\tau)$: linear rise and fall with base 2τ , equals the convolution $\text{rect} * \text{rect}$, smoother spectrum than rectangular. (c) **Gaussian pulse** $e^{-t^2/2\sigma^2}$: infinitely smooth with no sharp edges, characterized by standard deviation σ and full-width-at-half-maximum (FWHM) $\approx 2.355\sigma$; its Fourier transform is also Gaussian. (d) **Sinc function** $\text{sinc}(t/\tau) = \sin(\pi t/\tau)/(\pi t/\tau)$: the Fourier transform of the rectangular pulse, with zeros at integer multiples of τ and oscillating side lobes; appears in ideal low-pass filter impulse response and sampling theory.

10.3 The Gamma Function

The gamma function extends the concept of factorial to non-integer and complex arguments, making it essential for generalizing Laplace transform pairs.

Definition 10.7: Gamma Function

The **gamma function** is defined for $\text{Re}(z) > 0$ by:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (10.54)$$

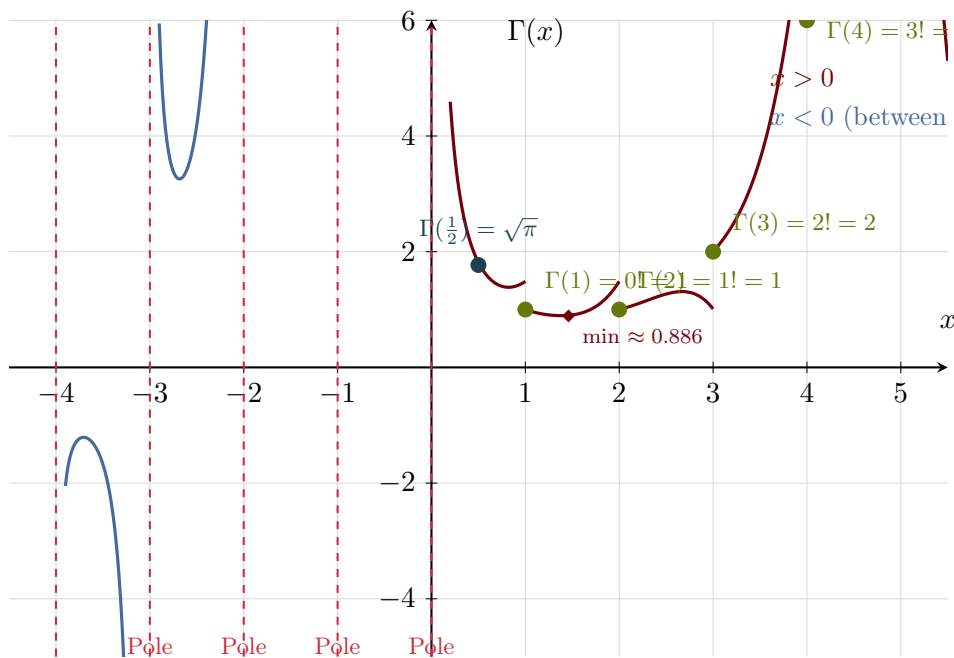


Figure 10.3: The Gamma function $\Gamma(x)$ extends the factorial to real (and complex) numbers via the integral $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ for $x > 0$. Key properties: (1) $\Gamma(n) = (n-1)!$ for positive integers (Horseshoe markers); (2) Recurrence relation $\Gamma(x+1) = x\Gamma(x)$; (3) Special value $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (Congaree marker); (4) Simple poles at non-positive integers $x = 0, -1, -2, \dots$ (Rose dashed lines). The function has a unique minimum at $x \approx 1.462$ where $\Gamma(x) \approx 0.886$.

10.3.1 Fundamental Properties

Gamma Function Recurrence

The gamma function satisfies the fundamental recurrence relation:

$$\Gamma(z+1) = z\Gamma(z) \quad (10.55)$$

Proof. Using integration by parts with $u = t^z$ and $dv = e^{-t} dt$:

$$\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt \quad (10.56)$$

$$= [-t^z e^{-t}]_0^\infty + z \int_0^\infty t^{z-1} e^{-t} dt \quad (10.57)$$

$$= 0 + z\Gamma(z) = z\Gamma(z) \quad (10.58)$$

where the boundary term vanishes for $\text{Re}(z) > 0$. $\square$

Relation to Factorial

For positive integers:

$$\boxed{\Gamma(n+1) = n!} \quad (10.59)$$

This follows from the recurrence relation and $\Gamma(1) = 1$:

$$\Gamma(1) = \int_0^\infty e^{-t} dt = 1 \quad (10.60)$$

$$\Gamma(2) = 1 \cdot \Gamma(1) = 1 = 1! \quad (10.61)$$

$$\Gamma(3) = 2 \cdot \Gamma(2) = 2 = 2! \quad (10.62)$$

$$\Gamma(n+1) = n \cdot (n-1) \cdots 2 \cdot 1 = n! \quad (10.63)$$

Special Values

$$\Gamma(1) = 1 \quad (10.64)$$

$$\Gamma(2) = 1 \quad (10.65)$$

$$\Gamma(1/2) = \sqrt{\pi} \quad (10.66)$$

$$\Gamma(3/2) = \frac{1}{2}\sqrt{\pi} \quad (10.67)$$

$$\Gamma(n+1/2) = \frac{(2n)!}{4^n n!} \sqrt{\pi} \quad (10.68)$$

Example 10.3: Computing $\Gamma(1/2)$

Show that $\Gamma(1/2) = \sqrt{\pi}$.

Solution:

$$\Gamma(1/2) = \int_0^\infty t^{-1/2} e^{-t} dt \quad (10.69)$$

Substitute $t = u^2$, so $dt = 2u du$:

$$\Gamma(1/2) = \int_0^\infty \frac{1}{u} e^{-u^2} \cdot 2u du = 2 \int_0^\infty e^{-u^2} du \quad (10.70)$$

The Gaussian integral $\int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2}$, therefore:

$$\boxed{\Gamma(1/2) = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}} \quad (10.71)$$

10.3.2 Use in Laplace Transforms

The gamma function appears in the Laplace transform of power functions:

$$\mathcal{L}\{t^\alpha\} = \frac{\Gamma(\alpha+1)}{s^{\alpha+1}} \quad \text{for } \alpha > -1 \quad (10.72)$$

For integer $\alpha = n$, this reduces to:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad (10.73)$$

Example 10.4: Laplace Transform of $t^{1/2}$

Find $\mathcal{L}\{t^{1/2}\}$.

Solution: Using the generalized formula:

$$\mathcal{L}\{t^{1/2}\} = \frac{\Gamma(1/2+1)}{s^{1/2+1}} = \frac{\Gamma(3/2)}{s^{3/2}} \quad (10.74)$$

Since $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \frac{\sqrt{\pi}}{2}$:

$$\mathcal{L}\{t^{1/2}\} = \frac{\sqrt{\pi}}{2s^{3/2}} \quad (10.75)$$

10.3.3 Other Properties

Reflection Formula

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)} \quad (10.76)$$

Duplication Formula

$$\Gamma(z)\Gamma(z+1/2) = \frac{\sqrt{\pi}}{2^{2z-1}}\Gamma(2z) \quad (10.77)$$

Stirling's Approximation

For large $|z|$:

$$\Gamma(z+1) \approx \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \quad (10.78)$$

10.4 Bessel Functions

Bessel functions arise in problems with cylindrical or spherical symmetry and appear in control systems involving wave propagation, heat conduction, and vibration analysis.

Definition 10.8: Bessel Functions of the First Kind

The **Bessel function of the first kind** of order n is defined by:

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k} \quad (10.79)$$

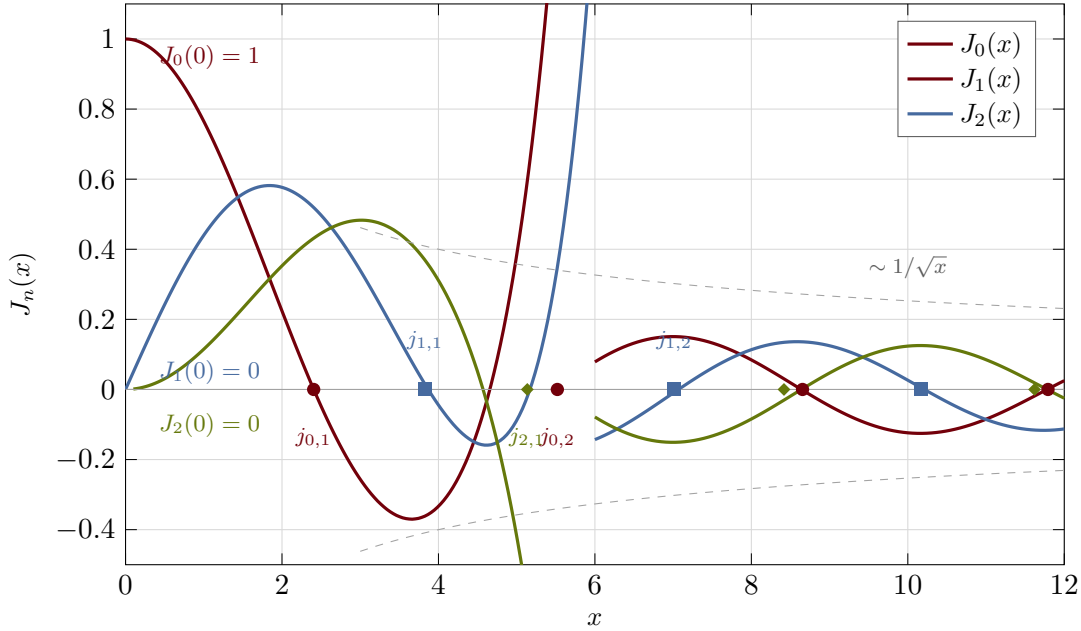


Figure 10.4: Bessel functions of the first kind $J_n(x)$ for orders $n = 0, 1, 2$. These functions arise in problems with cylindrical symmetry, including heat conduction, vibrating membranes, and electromagnetic waveguides. Key features: $J_0(0) = 1$ while $J_n(0) = 0$ for $n \geq 1$; all functions exhibit damped oscillatory behavior with amplitude decaying as $1/\sqrt{x}$. The zeros $j_{n,k}$ (marked points) determine natural frequencies in boundary value problems.

10.4.1 Properties of Bessel Functions

Recurrence Relations

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad (10.80)$$

$$J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x) \quad (10.81)$$

Derivative Relations

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x) \quad (10.82)$$

$$\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x) \quad (10.83)$$

Special Values

$$J_0(0) = 1 \quad (10.84)$$

$$J_n(0) = 0 \quad \text{for } n > 0 \quad (10.85)$$

Integral Representation

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta - n\theta) d\theta \quad (10.86)$$

10.4.2 Bessel Functions of the Second Kind

The **Bessel function of the second kind** (Weber function or Neumann function) is defined as:

$$Y_n(x) = \frac{J_n(x) \cos(n\pi) - J_{-n}(x)}{\sin(n\pi)} \quad (10.87)$$

For integer n , this is defined by a limit process. Unlike $J_n(x)$, $Y_n(x)$ is singular at $x = 0$.

10.4.3 Modified Bessel Functions

For problems with real exponential rather than oscillatory behavior:

Definition 10.9: Modified Bessel Functions

The **modified Bessel functions** are:

$$I_n(x) = j^{-n} J_n(jx) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k} \quad (10.88)$$

$$K_n(x) = \frac{\pi}{2} \frac{I_{-n}(x) - I_n(x)}{\sin(n\pi)} \quad (10.89)$$

$I_n(x)$ grows exponentially for large x , while $K_n(x)$ decays exponentially.

10.4.4 Physical Applications

Bessel functions appear in control-relevant applications:

1. **Vibrating Circular Membranes:** The radial modes of a drum are described by J_n .
2. **Heat Conduction in Cylinders:** Temperature distribution in cylindrical coordinates involves Bessel functions.
3. **Electromagnetic Waveguides:** Field distributions in circular waveguides use J_n and Y_n .
4. **FM Signal Analysis:** The spectrum of frequency-modulated signals involves J_n .
5. **Diffusion Processes:** Radial diffusion in cylindrical geometry uses modified Bessel functions.

Example 10.5: Bessel Functions in Vibration Analysis

The deflection $w(r, t)$ of a circular membrane satisfies:

$$\frac{\partial^2 w}{\partial t^2} = c^2 \left(\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} \right) \quad (10.90)$$

For a membrane clamped at radius a , the natural frequencies are determined by the zeros of J_0 :

$$\omega_n = \frac{c \cdot z_n}{a} \quad (10.91)$$

where z_n are the zeros of $J_0(x)$: $z_1 \approx 2.405$, $z_2 \approx 5.520$, $z_3 \approx 8.654$, etc.

10.4.5 Laplace Transforms Involving Bessel Functions

$$\mathcal{L}\{J_0(at)\} = \frac{1}{\sqrt{s^2 + a^2}} \quad (10.92)$$

$$\mathcal{L}\{J_1(at)\} = \frac{\sqrt{s^2 + a^2} - s}{a\sqrt{s^2 + a^2}} \quad (10.93)$$

$$\mathcal{L}\{I_0(at)\} = \frac{1}{\sqrt{s^2 - a^2}} \quad \text{for } s > a \quad (10.94)$$

10.5 Other Special Functions

Several other special functions appear in control systems and signal processing.

10.5.1 Error Function**Definition 10.10: Error Function**

The **error function** is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (10.95)$$

The **complementary error function** is:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt \quad (10.96)$$

Properties:

- $\operatorname{erf}(0) = 0$, $\operatorname{erf}(\infty) = 1$
- $\operatorname{erf}(-x) = -\operatorname{erf}(x)$ (odd function)
- $\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$

The error function appears in:

- Diffusion and heat transfer problems

- Probability calculations with normal distributions
- Communications systems (bit error rates)

Laplace transform:

$$\mathcal{L} \left\{ \operatorname{erf} \left(\frac{a}{2\sqrt{t}} \right) \right\} = \frac{1 - e^{-a\sqrt{s}}}{s} \quad (10.97)$$

10.5.2 Sinc Function

Definition 10.11: Sinc Function

The **sinc function** (normalized) is defined as:

$$\operatorname{sinc}(x) = \begin{cases} \frac{\sin(\pi x)}{\pi x} & x \neq 0 \\ 1 & x = 0 \end{cases} \quad (10.98)$$

The unnormalized version is $\operatorname{sinc}_u(x) = \frac{\sin x}{x}$.

Properties:

- $\operatorname{sinc}(n) = 0$ for all nonzero integers n
- $\int_{-\infty}^{\infty} \operatorname{sinc}(x) dx = 1$
- The Fourier transform of a rectangular pulse is a sinc function

The sinc function is fundamental in:

- Signal reconstruction (Shannon interpolation)
- Ideal low-pass filter impulse response
- Antenna pattern analysis

10.5.3 Gaussian Function

Definition 10.12: Gaussian Function

The **Gaussian function** is:

$$g(t) = e^{-at^2} \quad (10.99)$$

The normalized Gaussian (probability density) is:

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} \quad (10.100)$$

Properties:

- The Fourier transform of a Gaussian is another Gaussian
- $\int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\frac{\pi}{a}}$
- The Gaussian minimizes the time-bandwidth product

The Gaussian appears in:

- Noise modeling (thermal noise is Gaussian distributed)
- Pulse shaping in communications
- Uncertainty principles
- Smoothing filters

Laplace transform (one-sided, causal approximation):

$$\mathcal{L}\{e^{-at^2}u(t)\} = \frac{1}{2}\sqrt{\frac{\pi}{a}}e^{s^2/(4a)}\operatorname{erfc}\left(\frac{s}{2\sqrt{a}}\right) \quad (10.101)$$

10.5.4 Summary of Special Functions

Function	Definition	Primary Applications
Gamma $\Gamma(z)$	$\int_0^\infty t^{z-1}e^{-t}dt$	Generalizing factorials
Bessel $J_n(x)$	Series expansion	Cylindrical problems
Error $\operatorname{erf}(x)$	$\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$	Diffusion, statistics
Sinc $\operatorname{sinc}(x)$	$\frac{\sin(\pi x)}{\pi x}$	Signal reconstruction
Gaussian e^{-at^2}	Exponential of $-t^2$	Noise, uncertainty

Table 10.2: Summary of special functions and their applications.

Chapter 11

Numerical Methods

Numerical methods form the computational backbone of modern control systems. While analytical solutions provide elegant insights, the vast majority of real-world control problems require numerical computation—from simulating system dynamics to finding controller gains that satisfy stability constraints. This chapter develops the fundamental numerical techniques that underpin control system analysis and design.

11.1 Introduction to Numerical Methods

11.1.1 Why Numerical Methods Matter

Control systems engineering presents numerous problems that cannot be solved analytically:

- **Simulation:** Time-domain simulation of nonlinear systems requires numerical integration of differential equations.
- **Root finding:** Determining closed-loop poles involves finding roots of high-order polynomials.
- **Optimization:** Tuning controller gains to minimize a performance index requires iterative numerical algorithms.
- **Frequency response:** Computing Bode plots and Nyquist diagrams involves evaluating transfer functions at many frequencies.
- **State estimation:** Implementing Kalman filters requires numerical solution of Riccati equations.

The Role of Numerical Methods in Control

Numerical methods bridge the gap between mathematical theory and practical implementation. Every modern control system—from autopilots to industrial process controllers—relies on numerical algorithms running on digital computers to compute control actions in real time.

11.1.2 Sources of Error

All numerical computations introduce errors. Understanding these errors is essential for reliable engineering calculations.

Definition 11.1: Truncation Error

Truncation error arises from approximating a mathematical procedure. For example, approximating a derivative by a finite difference or truncating an infinite series introduces truncation error:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad (11.1)$$

The exact derivative involves the limit $h \rightarrow 0$, but numerically we must use finite h .

Definition 11.2: Round-off Error

Round-off error arises from representing real numbers with finite precision. Computers typically use IEEE 754 floating-point representation with approximately 15-16 significant decimal digits (double precision).

The Truncation-Round-off Trade-off

Reducing step size h decreases truncation error but increases round-off error (more operations accumulate errors). There exists an optimal h that minimizes total error—typically around $h \sim \sqrt{\epsilon_{\text{machine}}}$ for first-order methods, where $\epsilon_{\text{machine}} \approx 10^{-16}$ for double precision.

Definition 11.3: Order of Accuracy

A numerical method has **order of accuracy** p if the local error behaves as $O(h^{p+1})$, meaning the global error over a fixed interval is $O(h^p)$. Higher-order methods achieve the same accuracy with larger step sizes, requiring fewer computations.

11.2 Root Finding

Finding roots of nonlinear equations is fundamental to control systems analysis. Applications include:

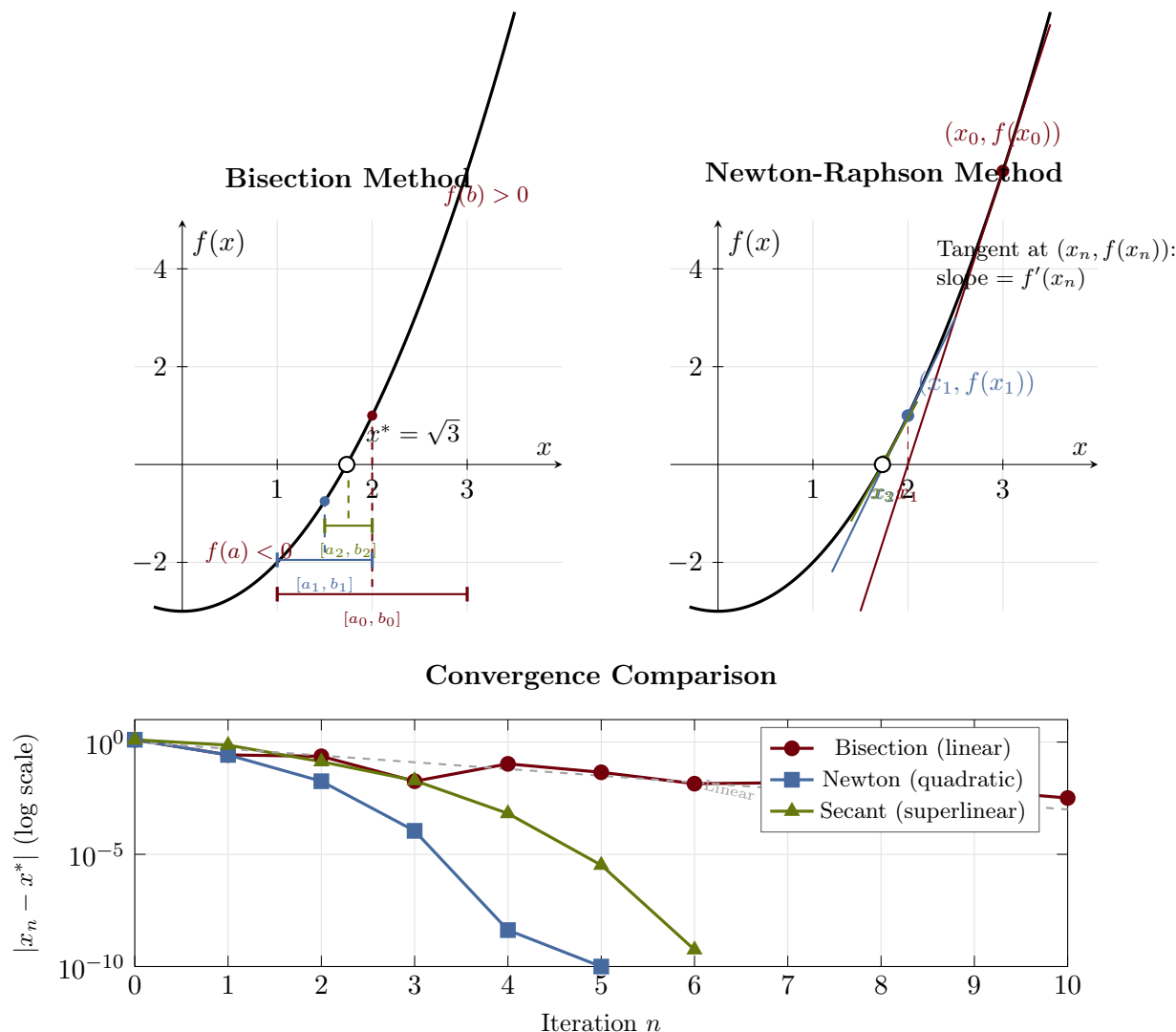
- Finding poles and zeros of transfer functions
- Solving characteristic equations for stability analysis
- Computing equilibrium points of nonlinear systems
- Determining gain and phase margins

11.2.1 Bisection Method

The bisection method is the simplest root-finding algorithm. It is guaranteed to converge for continuous functions but is relatively slow.

Definition 11.4: Bisection Method

Given a continuous function f with $f(a) \cdot f(b) < 0$ (a sign change indicating a root), the bisection method repeatedly halves the interval:



Bisection: $c = \frac{a+b}{2}$; if $f(a)f(c) < 0$: $b \leftarrow c$, else $a \leftarrow c$ (Linear: error halves each step)

Newton-Raphson: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ (Quadratic: digits of accuracy double each step)

Secant: $x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$ (Superlinear: order ≈ 1.618)

Figure 11.1: Root finding methods for $f(x) = x^2 - 3$ with root $x^* = \sqrt{3} \approx 1.732$. **Top left:** Bisection method iteratively halves the bracketing interval $[a, b]$ where $f(a)f(b) < 0$. It is guaranteed to converge but only linearly. **Top right:** Newton-Raphson method follows tangent lines from the current approximation to the x -axis, achieving quadratic convergence when sufficiently close to the root. **Bottom:** Convergence comparison showing that Newton's method (Atlantic blue) converges in about 5 iterations to machine precision, while bisection (Garnet) requires many more iterations for the same accuracy. The secant method (Horseshoe green) achieves superlinear convergence without requiring derivative evaluation.

1. Compute midpoint: $c = \frac{a+b}{2}$
2. If $f(a) \cdot f(c) < 0$, the root is in $[a, c]$; set $b = c$
3. Otherwise, the root is in $[c, b]$; set $a = c$
4. Repeat until $|b - a| < \epsilon$ (tolerance)

Bisection Convergence

The bisection method has **linear convergence** with rate $1/2$. After n iterations, the interval width is $(b - a)/2^n$. To achieve tolerance ϵ , we need:

$$n \geq \log_2 \left(\frac{b - a}{\epsilon} \right) \quad (11.2)$$

This is predictable but slow—each iteration gains only one binary digit of accuracy.

Example 11.1: Bisection Method for Finding a Closed-Loop Pole

Find the positive real root of $f(x) = x^3 - 2x - 5$ using bisection on $[2, 3]$.

Solution:

First, verify the sign change: $f(2) = 8 - 4 - 5 = -1 < 0$ and $f(3) = 27 - 6 - 5 = 16 > 0$. ✓

n	a	b	$c = (a + b)/2$	$f(c)$	New interval
1	2.0000	3.0000	2.5000	5.125	$[2, 2.5]$
2	2.0000	2.5000	2.2500	1.891	$[2, 2.25]$
3	2.0000	2.2500	2.1250	0.346	$[2, 2.125]$
4	2.0000	2.1250	2.0625	-0.351	$[2.0625, 2.125]$
5	2.0625	2.1250	2.0938	-0.009	$[2.0938, 2.125]$

After 5 iterations: $x \approx 2.094$ with error bound $|b - a|/2 = 0.016$.

The exact root is $x^* \approx 2.0946$.

11.2.2 Newton-Raphson Method

The Newton-Raphson method uses local linearization to achieve much faster convergence than bisection, but requires the derivative $f'(x)$.

Definition 11.5: Newton-Raphson Method

Given an initial guess x_0 and a differentiable function f , the Newton-Raphson iteration is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (11.3)$$

Geometrically, this finds where the tangent line to f at x_n crosses zero.

Newton-Raphson Convergence

Near a simple root (where $f'(x^*) \neq 0$), Newton-Raphson has **quadratic convergence**:

$$|x_{n+1} - x^*| \approx \frac{|f''(x^*)|}{2|f'(x^*)|} |x_n - x^*|^2 \quad (11.4)$$

The number of correct digits approximately doubles with each iteration.

Example 11.2: Newton-Raphson for the Same Problem

Find the positive real root of $f(x) = x^3 - 2x - 5$ using Newton-Raphson with $x_0 = 2$.

Solution:

We have $f(x) = x^3 - 2x - 5$ and $f'(x) = 3x^2 - 2$.

The iteration formula is:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	2.000000	-1.000	10.00	2.100000
1	2.100000	0.061	11.23	2.094569
2	2.094569	0.000086	11.16	2.094551
3	2.094551	$< 10^{-10}$	11.16	2.094551

Newton-Raphson converges to 6-digit accuracy in only 3 iterations, compared to bisection needing about 20 iterations for similar accuracy.

Observation: Note the quadratic convergence—the error decreases from ~ 0.1 to ~ 0.005 to $\sim 10^{-5}$ to $\sim 10^{-10}$, roughly squaring each time.

Newton-Raphson Failure Modes

Newton-Raphson can fail or diverge when:

- $f'(x_n) = 0$ (division by zero)
- The initial guess is too far from the root
- Multiple roots exist ($f'(x^*) = 0$), causing only linear convergence
- The function has inflection points that send iterates far away

Always verify convergence and consider bisection as a backup.

11.2.3 Secant Method

The secant method approximates the derivative using finite differences, avoiding the need to compute $f'(x)$ analytically.

Definition 11.6: Secant Method

Given two initial guesses x_0 and x_1 , the secant iteration is:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \quad (11.5)$$

This uses the secant line through $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$ instead of the tangent line.

Secant Method Convergence

The secant method has **superlinear convergence** with order $\phi = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio):

$$|x_{n+1} - x^*| \approx C|x_n - x^*|^{1.618} \quad (11.6)$$

While slower than Newton-Raphson's quadratic rate, each iteration requires only one function evaluation instead of two (function and derivative).

Example 11.3: Secant Method Iteration

Find the root of $f(x) = x^3 - 2x - 5$ using the secant method with $x_0 = 2$ and $x_1 = 3$.

Solution:

The secant iteration is:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

n	x_n	$f(x_n)$	Error $ x_n - x^* $
0	2.000000	-1.000	0.0946
1	3.000000	16.000	0.9054
2	2.058824	-0.390	0.0358
3	2.096559	0.022	0.0020
4	2.094511	-0.0004	0.00004
5	2.094551	$< 10^{-7}$	$< 10^{-6}$

The secant method converges in 5 iterations without requiring derivative computation.

11.2.4 Comparison of Root-Finding Methods

Method	Convergence Order	Evaluations/Iter	Robustness
Bisection	1 (linear)	1	Guaranteed
Newton-Raphson	2 (quadratic)	2 (f and f')	May diverge
Secant	1.618 (superlinear)	1	May diverge

Practical Root-Finding Strategy

A robust approach combines methods:

1. Use bisection to bracket the root and get a reasonable initial guess

2. Switch to Newton-Raphson or secant for rapid convergence
3. Monitor for divergence and fall back to bisection if needed

This hybrid approach is implemented in MATLAB's `fzero` function.

11.3 Numerical Integration

Numerical integration (quadrature) computes definite integrals that cannot be evaluated analytically. In control systems, this arises when:

- Computing integral performance indices (ISE, IAE, ITAE)
- Evaluating convolution integrals for system response
- Calculating areas under probability density functions

11.3.1 Rectangle Rules

The simplest integration methods approximate the integrand as piecewise constant.

Definition 11.7: Rectangle Rules

For integrating $f(x)$ over $[a, b]$ with n subintervals of width $h = (b - a)/n$:

Left Rectangle Rule:

$$\int_a^b f(x) dx \approx h \sum_{i=0}^{n-1} f(x_i), \quad x_i = a + ih \quad (11.7)$$

Right Rectangle Rule:

$$\int_a^b f(x) dx \approx h \sum_{i=1}^n f(x_i), \quad x_i = a + ih \quad (11.8)$$

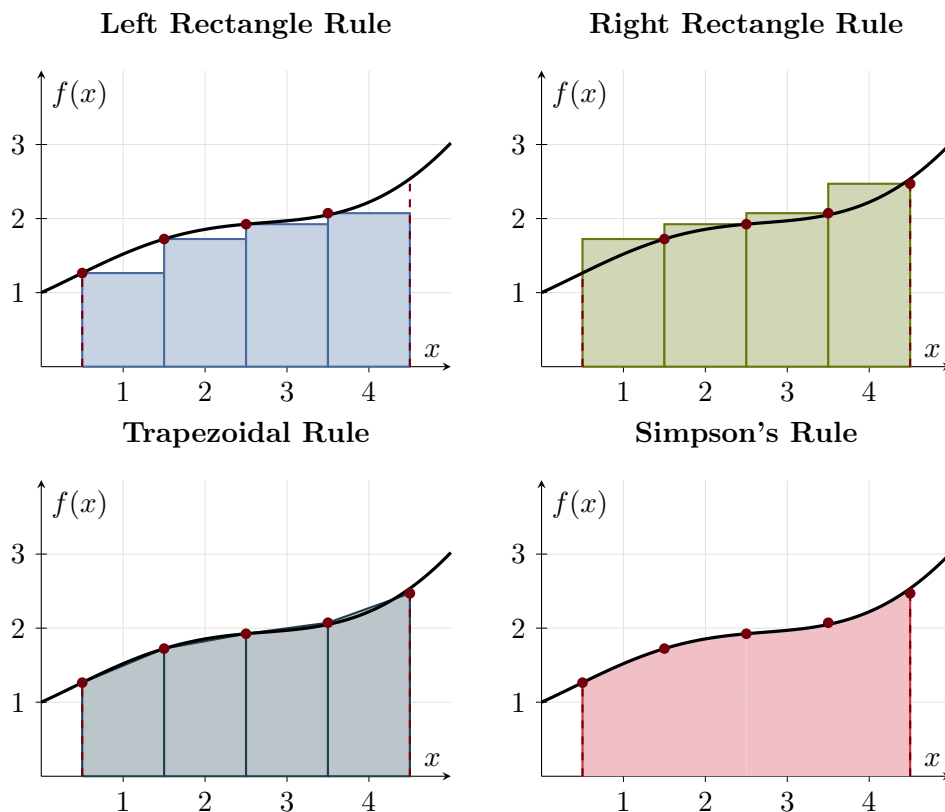
Midpoint Rule:

$$\int_a^b f(x) dx \approx h \sum_{i=0}^{n-1} f\left(x_i + \frac{h}{2}\right) \quad (11.9)$$

The left and right rules have error $O(h)$, while the midpoint rule has error $O(h^2)$ due to symmetry.

11.3.2 Trapezoidal Rule

The trapezoidal rule approximates the integrand with piecewise linear functions.



Left Rectangle: $\int_a^b f(x) dx \approx h \sum_{i=0}^{n-1} f(x_i)$
Right Rectangle: $\int_a^b f(x) dx \approx h \sum_{i=1}^n f(x_i)$

Trapezoidal: $\int_a^b f(x) dx \approx \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$

Simpson's: $\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{\text{odd } i} f(x_i) + 2 \sum_{\text{even } i} f(x_i) + f(x_n) \right]$

Figure 11.2: Comparison of numerical integration methods for approximating $\int_a^b f(x) dx$. **Top left:** Left rectangle rule uses function values at left endpoints, typically underestimating for increasing functions. **Top right:** Right rectangle rule uses function values at right endpoints. **Bottom left:** Trapezoidal rule connects adjacent function values with straight lines, achieving second-order accuracy $O(h^2)$. **Bottom right:** Simpson's rule fits parabolas through triplets of points, achieving fourth-order accuracy $O(h^4)$ when n is even. The step size is $h = (b - a)/n$ where n is the number of subintervals.

Definition 11.8: Trapezoidal Rule

$$\int_a^b f(x) dx \approx h \left[\frac{f(a)}{2} + \sum_{i=1}^{n-1} f(x_i) + \frac{f(b)}{2} \right] \quad (11.10)$$

where $x_i = a + ih$ and $h = (b - a)/n$.

The error is $O(h^2)$:

$$E_{\text{trap}} = -\frac{(b-a)h^2}{12} f''(\xi), \quad \xi \in (a, b) \quad (11.11)$$

Example 11.4: Trapezoidal Rule for Integral Performance Index

Compute $\int_0^2 t \cdot e^{-t} dt$ (related to ITAE) using the trapezoidal rule with $n = 4$ subintervals.

Solution:

We have $h = (2 - 0)/4 = 0.5$ and $f(t) = t \cdot e^{-t}$.

i	t_i	$f(t_i) = t_i e^{-t_i}$
0	0.0	0.0000
1	0.5	0.3033
2	1.0	0.3679
3	1.5	0.3347
4	2.0	0.2707

Applying the trapezoidal rule:

$$\int_0^2 t e^{-t} dt \approx 0.5 \left[\frac{0}{2} + 0.3033 + 0.3679 + 0.3347 + \frac{0.2707}{2} \right] \quad (11.12)$$

$$= 0.5[0 + 0.3033 + 0.3679 + 0.3347 + 0.1354] \quad (11.13)$$

$$= 0.5 \times 1.1413 = 0.5707 \quad (11.14)$$

The exact value is $1 - 3e^{-2} \approx 0.5940$. Our approximation has relative error of about 4%.

11.3.3 Simpson's Rule

Simpson's rule uses quadratic interpolation, achieving higher accuracy.

Definition 11.9: Simpson's Rule

For an even number of subintervals n :

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(a) + 4 \sum_{\text{odd } i} f(x_i) + 2 \sum_{\text{even } i} f(x_i) + f(b) \right] \quad (11.15)$$

The error is $O(h^4)$:

$$E_{\text{Simpson}} = -\frac{(b-a)h^4}{180} f^{(4)}(\xi), \quad \xi \in (a, b) \quad (11.16)$$

Why Simpson's Rule is $O(h^4)$

Although Simpson's rule uses quadratic polynomials (which would suggest $O(h^3)$ error), symmetry causes the cubic error term to cancel, resulting in $O(h^4)$ accuracy. Simpson's rule is exact for polynomials up to degree 3.

Example 11.5: Simpson's Rule Comparison

Recompute $\int_0^2 te^{-t} dt$ using Simpson's rule with $n = 4$.

Solution:

Using the same function values as before:

$$\int_0^2 te^{-t} dt \approx \frac{0.5}{3} [f(0) + 4f(0.5) + 2f(1.0) + 4f(1.5) + f(2.0)] \quad (11.17)$$

$$= \frac{0.5}{3} [0 + 4(0.3033) + 2(0.3679) + 4(0.3347) + 0.2707] \quad (11.18)$$

$$= \frac{0.5}{3} [0 + 1.2132 + 0.7358 + 1.3388 + 0.2707] \quad (11.19)$$

$$= \frac{0.5}{3} \times 3.5585 = 0.5931 \quad (11.20)$$

Simpson's rule gives 0.5931, compared to the exact value 0.5940—a relative error of only 0.15%, much better than the trapezoidal rule with the same number of points.

11.3.4 Error Analysis Summary

Method	Error Order	To Halve Error
Left/Right Rectangle	$O(h)$	Double n
Midpoint	$O(h^2)$	Multiply n by $\sqrt{2}$
Trapezoidal	$O(h^2)$	Multiply n by $\sqrt{2}$
Simpson's	$O(h^4)$	Multiply n by $\sqrt[4]{2} \approx 1.19$

11.4 ODE Solvers

Numerical integration of ordinary differential equations (ODEs) is perhaps the most important numerical method in control systems. Given an initial value problem:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(t, \mathbf{x}), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \quad (11.21)$$

we seek to compute $\mathbf{x}(t)$ at discrete times $t_n = t_0 + nh$.

11.4.1 Euler's Method (Forward Euler)

The simplest ODE solver uses forward difference approximation of the derivative.

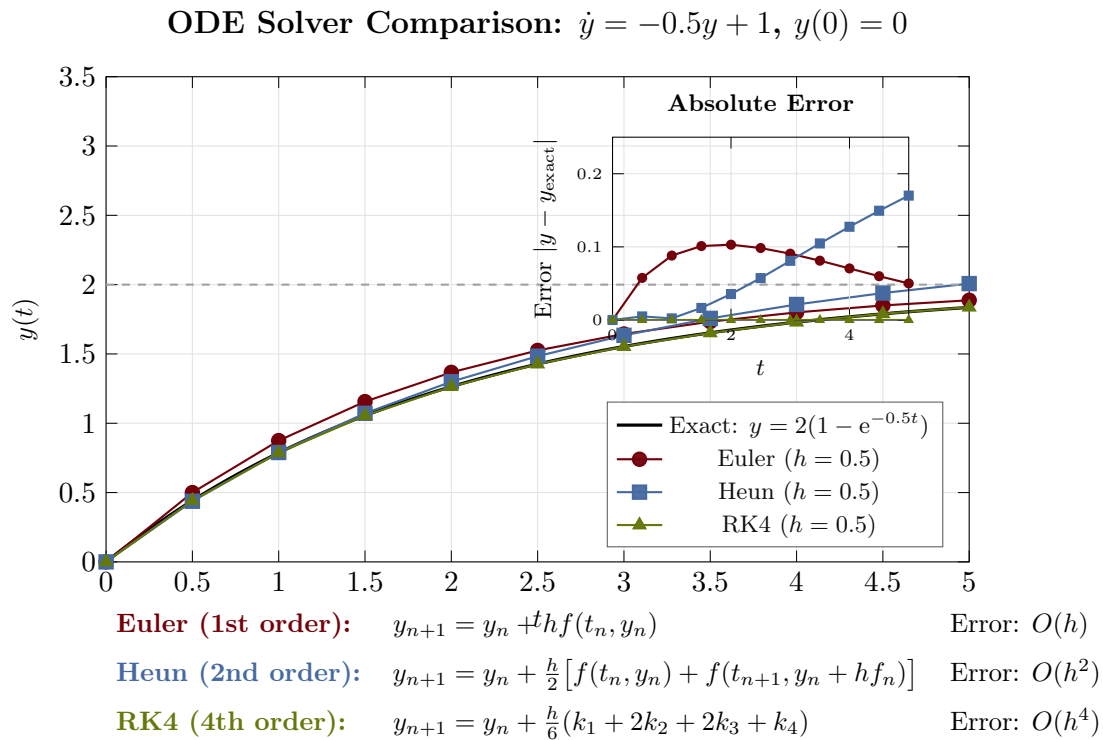


Figure 11.3: Comparison of numerical ODE solvers for the first-order linear equation $\dot{y} = -0.5y + 1$ with initial condition $y(0) = 0$. The exact solution $y(t) = 2(1 - e^{-0.5t})$ approaches the steady state $y_\infty = 2$. **Euler's method** (Garnet) exhibits first-order accuracy with visible deviation from the exact solution. **Heun's method** (Atlantic blue), also called the improved Euler or trapezoidal method, achieves second-order accuracy. **RK4** (Horseshoe green) provides fourth-order accuracy and is nearly indistinguishable from the exact solution even with a relatively large step size $h = 0.5$. The inset shows the absolute error versus time for each method.

Definition 11.10: Forward Euler Method

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{f}(t_n, \mathbf{x}_n) \quad (11.22)$$

This is an **explicit method**— $\mathbf{x}_{n+1}$ is computed directly from known quantities.
Local truncation error: $O(h^2)$. Global error: $O(h)$ (first-order method).

Example 11.6: Euler's Method for a First-Order System

Simulate $\dot{x} = -2x$ with $x(0) = 1$ from $t = 0$ to $t = 2$ using Euler's method with $h = 0.5$.

Solution:

The exact solution is $x(t) = e^{-2t}$.

Euler iteration: $x_{n+1} = x_n + h(-2x_n) = x_n(1 - 2h) = x_n(1 - 1) = 0$

Wait—this is wrong! With $h = 0.5$, we have $1 - 2h = 0$, so:

n	t_n	x_n (Euler)	$x(t_n)$ (exact)	Error
0	0.0	1.000	1.000	0.000
1	0.5	0.000	0.368	0.368
2	1.0	0.000	0.135	0.135
3	1.5	0.000	0.050	0.050
4	2.0	0.000	0.018	0.018

This example shows Euler's method with too large a step size. Let's try $h = 0.25$:

With $h = 0.25$: $x_{n+1} = x_n(1 - 0.5) = 0.5x_n$

n	t_n	x_n (Euler)	$x(t_n)$ (exact)	Error
0	0.0	1.000	1.000	0.000
2	0.5	0.250	0.368	0.118
4	1.0	0.0625	0.135	0.073
6	1.5	0.0156	0.050	0.034
8	2.0	0.0039	0.018	0.014

Better, but still significant error. Euler's method requires very small step sizes for accuracy.

11.4.2 Improved Euler (Heun's Method)

Heun's method improves accuracy by averaging slopes at both ends of the interval.

Definition 11.11: Heun's Method (Improved Euler)

$$\tilde{\mathbf{x}}_{n+1} = \mathbf{x}_n + h\mathbf{f}(t_n, \mathbf{x}_n) \quad (\text{predictor}) \quad (11.23)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{2} [\mathbf{f}(t_n, \mathbf{x}_n) + \mathbf{f}(t_{n+1}, \tilde{\mathbf{x}}_{n+1})] \quad (\text{corrector}) \quad (11.24)$$

This is a second-order method with global error $O(h^2)$.

Example 11.7: Heun's Method Comparison

Repeat Example 11.6 using Heun's method with $h = 0.5$.

Solution:

For $\dot{x} = -2x$:

$$\tilde{x}_{n+1} = x_n + 0.5(-2x_n) = 0 \cdot x_n = 0 \quad (11.25)$$

$$x_{n+1} = x_n + 0.25[(-2x_n) + (-2 \cdot 0)] = x_n - 0.5x_n = 0.5x_n \quad (11.26)$$

n	t_n	x_n (Heun)	$x(t_n)$ (exact)	Error
0	0.0	1.000	1.000	0.000
1	0.5	0.500	0.368	0.132
2	1.0	0.250	0.135	0.115
3	1.5	0.125	0.050	0.075
4	2.0	0.0625	0.018	0.044

Heun's method gives non-zero values (unlike Euler with $h = 0.5$), but still has noticeable error. For this problem, the predictor step gives zero, which limits the method's effectiveness.

11.4.3 Runge-Kutta 4th Order (RK4)

The fourth-order Runge-Kutta method is the workhorse of ODE simulation, offering an excellent balance of accuracy and computational cost.

Definition 11.12: Classical RK4 Method

$$k_1 = \mathbf{f}(t_n, \mathbf{x}_n) \quad (11.27)$$

$$k_2 = \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{x}_n + \frac{h}{2}k_1\right) \quad (11.28)$$

$$k_3 = \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{x}_n + \frac{h}{2}k_2\right) \quad (11.29)$$

$$k_4 = \mathbf{f}(t_n + h, \mathbf{x}_n + hk_3) \quad (11.30)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (11.31)$$

This fourth-order method has global error $O(h^4)$.

Why RK4 is Popular

RK4 requires 4 function evaluations per step but achieves $O(h^4)$ accuracy. This means:

- Halving the step size reduces error by factor of 16
- Often allows 10-100× larger steps than Euler for same accuracy
- Self-starting (unlike multistep methods)

- Easy to implement and understand

Example 11.8: RK4 for a Second-Order System

Simulate the damped oscillator $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2y = 0$ with $\zeta = 0.3$, $\omega_n = 2$ rad/s, $y(0) = 1$, $\dot{y}(0) = 0$ using RK4 with $h = 0.1$ s for one step.

Solution:

First, convert to state-space form. Let $x_1 = y$ and $x_2 = \dot{y}$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -1.2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Initial condition: $\mathbf{x}_0 = [1, 0]^\top$.

Computing the k values:

$$\begin{aligned} k_1 &= \mathbf{f}(0, [1, 0]^\top) = [0, -4]^\top \\ k_2 &= \mathbf{f}(0.05, [1, 0]^\top + 0.05[0, -4]^\top) = \mathbf{f}(0.05, [1, -0.2]^\top) \\ &= [-0.2, -4 + 0.24]^\top = [-0.2, -3.76]^\top \\ k_3 &= \mathbf{f}(0.05, [1, 0]^\top + 0.05[-0.2, -3.76]^\top) = \mathbf{f}(0.05, [0.99, -0.188]^\top) \\ &= [-0.188, -3.96 + 0.226]^\top = [-0.188, -3.734]^\top \\ k_4 &= \mathbf{f}(0.1, [1, 0]^\top + 0.1[-0.188, -3.734]^\top) = \mathbf{f}(0.1, [0.981, -0.373]^\top) \\ &= [-0.373, -3.924 + 0.448]^\top = [-0.373, -3.476]^\top \end{aligned}$$

Final update:

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{x}_0 + \frac{0.1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\ &= [1, 0]^\top + \frac{0.1}{6}([0, -4] + 2[-0.2, -3.76] + 2[-0.188, -3.734] + [-0.373, -3.476])^\top \\ &= [1, 0]^\top + \frac{0.1}{6}[-1.149, -22.464]^\top \\ &= [1, 0]^\top + [-0.0192, -0.374]^\top = [0.981, -0.374]^\top \end{aligned}$$

After one RK4 step: $y(0.1) \approx 0.981$, $\dot{y}(0.1) \approx -0.374$.

The exact solution gives $y(0.1) \approx 0.9805$ —RK4 is accurate to 3 decimal places with just one step!

11.4.4 Stability Regions

Not all step sizes lead to stable numerical solutions. Each method has a **stability region** in the complex $h\lambda$ plane, where λ is an eigenvalue of the system.

Definition 11.13: Absolute Stability Region

For the test equation $\dot{x} = \lambda x$ (where λ may be complex), a numerical method is **absolutely stable** if $|x_n| \rightarrow 0$ as $n \rightarrow \infty$ when $\text{Re}(\lambda) < 0$.

The **stability region** is the set of $z = h\lambda \in \mathbb{C}$ for which the method is stable.

Stability Regions of Common Methods

- **Forward Euler:** Stable for $|1 + z| < 1$, i.e., a disk of radius 1 centered at $z = -1$.
- **RK4:** A larger region extending to about $z \approx -2.8$ on the real axis.
- **Backward Euler:** Stable for $|1/(1 - z)| < 1$, i.e., the entire left half-plane (A-stable).

For a stable system ($\text{Re}(\lambda) < 0$), the step size must satisfy $h|\lambda| < C$ where C depends on the method.

Example 11.9: Stability Constraint on Step Size

A system has eigenvalues $\lambda = -10$ and $\lambda = -0.1$. What is the maximum step size for forward Euler to remain stable?

Solution:

Forward Euler is stable for $|1 + h\lambda| < 1$. For real negative λ :

$$|1 + h\lambda| < 1 \implies -1 < 1 + h\lambda < 1 \implies -2 < h\lambda < 0$$

For $\lambda = -10$: $-2 < -10h < 0 \implies h < 0.2$

For $\lambda = -0.1$: $-2 < -0.1h < 0 \implies h < 20$

The stability is limited by the fast mode: $h < 0.2$ s.

Note: The slow mode ($\lambda = -0.1$, time constant 10 s) requires simulation for many time constants to see behavior, but stability forces small steps due to the fast mode. This is characteristic of *stiff* systems.

11.4.5 Stiff Equations

Definition 11.14: Stiff ODEs

An ODE system is **stiff** if its eigenvalues span a wide range of magnitudes, typically with stiffness ratio:

$$\text{Stiffness ratio} = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|} \gg 1 \quad (11.32)$$

Stiff systems have widely separated time scales.

The Stiffness Problem

For stiff systems, explicit methods (Euler, RK4) require tiny step sizes for stability, even when accuracy would permit larger steps. This makes simulation extremely slow.

Solution: Use implicit methods (backward Euler, implicit Runge-Kutta) or specialized stiff solvers (MATLAB's `ode15s`, `ode23s`).

Stiff systems commonly arise in control when:

- Mixing fast actuator dynamics with slow plant dynamics

- Systems with parasitic capacitances or inductances
- Chemical reaction networks with fast and slow reactions
- Multiphysics simulations (electrical-mechanical-thermal)

11.5 Interpolation and Approximation

Interpolation constructs a function passing through given data points. This is essential for:

- Reconstructing continuous signals from sampled data
- Look-up tables in embedded controllers
- Generating smooth trajectories for motion control
- Approximating transfer functions from frequency response data

11.5.1 Linear Interpolation

The simplest interpolation connects data points with straight lines.

Definition 11.15: Linear Interpolation

Given points (x_0, y_0) and (x_1, y_1) , the linear interpolant at $x \in [x_0, x_1]$ is:

$$y(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) = y_0(1 - t) + y_1 t \quad (11.33)$$

where $t = (x - x_0)/(x_1 - x_0) \in [0, 1]$.

Properties of Linear Interpolation

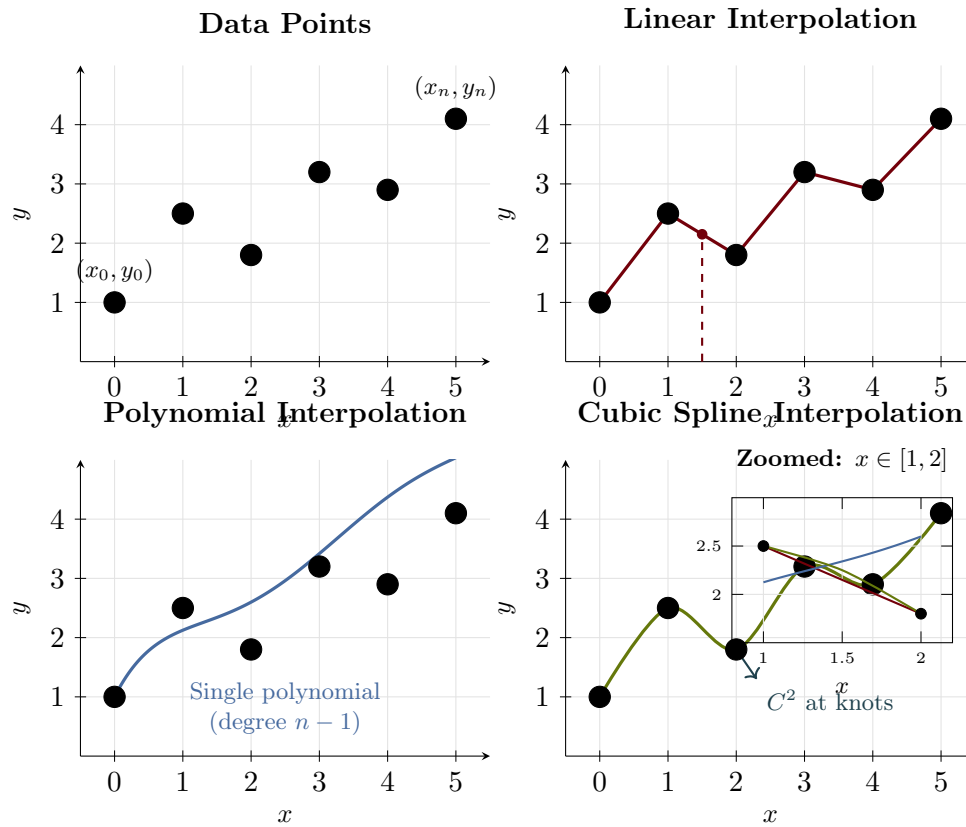
- **Simplicity:** Easy to compute and implement in real-time systems
- **Continuity:** The piecewise linear function is C^0 (continuous)
- **Not smooth:** Derivative is discontinuous at data points (C^0 but not C^1)
- **Error:** $O(h^2)$ where h is the spacing between points

Example 11.10: Linear Interpolation for Gain Scheduling

A gain-scheduled controller uses the following gain table:

Velocity v (m/s)	Gain K
0	10.0
5	8.0
10	5.0
15	3.5

Find K at $v = 7$ m/s using linear interpolation.



Linear: $p(x) = y_i + \frac{y_{i+1} - y_i}{x_{i+1} - x_i}(x - x_i)$ for $x \in [x_i, x_{i+1}]$ (C^0 continuity)

Polynomial (Lagrange): $p(x) = \sum_{i=0}^n y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$ (Unique degree $n - 1$ polynomial)

Cubic Spline: Piecewise cubic $S_i(x)$ with $S_i(x_i) = y_i$, $S_i(x_{i+1}) = y_{i+1}$, S'_i, S''_i continuous (C^2 continuity)

Figure 11.4: Comparison of interpolation methods for discrete data points (x_i, y_i) . **Top left:** The raw data points to be interpolated. **Top right:** Linear interpolation connects adjacent points with straight line segments; simple but only C^0 continuous (discontinuous derivatives at knots). **Bottom left:** Polynomial interpolation fits a single polynomial of degree $n - 1$ through n points; exact at data points but may exhibit oscillation (Runge phenomenon) for high-degree polynomials. **Bottom right:** Cubic spline interpolation uses piecewise cubic polynomials that are C^2 continuous at knot points, providing smooth interpolation without oscillation. The inset compares all three methods in the interval $[1, 2]$.

Solution:

The point $v = 7$ lies between $v = 5$ and $v = 10$.

$$t = \frac{7-5}{10-5} = \frac{2}{5} = 0.4 \quad (11.34)$$

$$K(7) = K(5)(1-t) + K(10) \cdot t = 8.0(0.6) + 5.0(0.4) = 4.8 + 2.0 = 6.8 \quad (11.35)$$

The interpolated gain is $K = 6.8$.

11.5.2 Polynomial Interpolation

Higher-order polynomials can pass exactly through more data points.

Definition 11.16: Lagrange Interpolation

Given $n+1$ points $(x_0, y_0), \dots, (x_n, y_n)$ with distinct x_i , the unique polynomial of degree $\leq n$ through these points is:

$$P(x) = \sum_{i=0}^n y_i L_i(x), \quad L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j} \quad (11.36)$$

where $L_i(x)$ are the Lagrange basis polynomials satisfying $L_i(x_j) = \delta_{ij}$.

Runge's Phenomenon

High-degree polynomial interpolation on equally-spaced points can exhibit large oscillations between data points, especially near the boundaries. This is called **Runge's phenomenon**.

Mitigation strategies:

- Use Chebyshev nodes (clustered near boundaries) instead of uniform spacing
- Use piecewise polynomials (splines) instead of a single high-degree polynomial
- Limit polynomial degree to 5-7 even with many data points

Example 11.11: Quadratic Interpolation

Fit a quadratic through $(0, 1)$, $(1, 3)$, $(2, 2)$ and evaluate at $x = 1.5$.

Solution:

Using Lagrange interpolation with $n = 2$:

$$\begin{aligned} L_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2} \\ L_1(x) &= \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1} = -x(x-2) \\ L_2(x) &= \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2} \end{aligned}$$

The interpolating polynomial:

$$P(x) = 1 \cdot L_0(x) + 3 \cdot L_1(x) + 2 \cdot L_2(x)$$

At $x = 1.5$:

$$L_0(1.5) = \frac{(0.5)(-0.5)}{2} = -0.125$$

$$L_1(1.5) = -(1.5)(-0.5) = 0.75$$

$$L_2(1.5) = \frac{(1.5)(0.5)}{2} = 0.375$$

Therefore:

$$P(1.5) = 1(-0.125) + 3(0.75) + 2(0.375) = -0.125 + 2.25 + 0.75 = 2.875$$

11.5.3 Spline Interpolation

Splines are piecewise polynomials that provide smoothness while avoiding the oscillation problems of high-degree polynomials.

Definition 11.17: Cubic Spline

A **cubic spline** $S(x)$ through points $(x_0, y_0), \dots, (x_n, y_n)$ is a piecewise cubic polynomial satisfying:

1. $S(x_i) = y_i$ for all i (interpolation)
2. $S(x)$ is C^2 (continuous second derivative)
3. On each interval $[x_i, x_{i+1}]$, $S(x)$ is a cubic polynomial

The most common boundary conditions are:

- **Natural spline:** $S''(x_0) = S''(x_n) = 0$
- **Clamped spline:** $S'(x_0) = y'_0$ and $S'(x_n) = y'_n$ (specified slopes)
- **Not-a-knot:** Third derivative continuous at x_1 and x_{n-1}

Why Splines are Preferred for Control

Cubic splines are widely used in trajectory generation and motion control because:

- **Position continuity (C^0):** No jumps in position
- **Velocity continuity (C^1):** Smooth velocity profiles
- **Acceleration continuity (C^2):** Bounded jerk, reducing mechanical stress
- **Local control:** Changing one data point only affects nearby segments

Example 11.12: Spline vs. Linear Interpolation for Trajectory

A robot arm must move through waypoints at $t = 0, 1, 2$ s with positions $\theta = 0, 90, 45$ degrees. Compare linear interpolation with cubic spline.

Linear interpolation:

- Segment 1 ($0 \leq t < 1$): $\theta(t) = 90t$ degrees
- Segment 2 ($1 \leq t \leq 2$): $\theta(t) = 90 - 45(t - 1) = 135 - 45t$ degrees

Velocity: $\dot{\theta} = 90$ deg/s for $t < 1$, then $\dot{\theta} = -45$ deg/s for $t > 1$.

Problem: At $t = 1$, velocity jumps from $+90$ to -45 deg/s—an instantaneous change of 135 deg/s requiring infinite acceleration!

Cubic spline (natural boundary conditions):

The spline ensures $\ddot{\theta}$ is continuous. The velocity smoothly transitions through $t = 1$, requiring finite, bounded acceleration. This is essential for real motors with limited torque.

11.5.4 Comparison of Interpolation Methods

Method	Continuity	Local	Oscillation	Computation
Linear	C^0	Yes	None	Trivial
Polynomial (Lagrange)	C^∞	No	High (Runge)	Moderate
Cubic Spline	C^2	Yes	Low	Solve tridiag.

11.6 Numerical Linear Algebra

Control systems rely heavily on linear algebra computations. This section summarizes key algorithms.

11.6.1 Solving Linear Systems

For $\mathbf{Ax} = \mathbf{b}$:

- **Gaussian elimination with pivoting:** $O(n^3)$ operations, numerically stable
- **LU decomposition:** Factor $\mathbf{A} = \mathbf{LU}$ once, then solve for multiple right-hand sides
- **Cholesky decomposition:** For symmetric positive definite $\mathbf{A} = \mathbf{LL}^\top$, half the work of LU
- **Iterative methods:** Conjugate gradient, GMRES for large sparse systems

11.6.2 Eigenvalue Computation

For finding eigenvalues of $\mathbf{A}$:

- **QR iteration:** The standard algorithm, $O(n^3)$ per iteration, typically $O(n^3)$ total with shifts
- **Power iteration:** Finds dominant eigenvalue only, $O(n^2)$ per iteration
- **Arnoldi/Lanczos:** For large sparse matrices, find a few eigenvalues

11.6.3 Singular Value Decomposition

The SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is computed by:

- **Golub-Kahan bidiagonalization:** Reduce to bidiagonal form
- **Implicit QR on bidiagonal:** Compute singular values and vectors

Numerical Stability in Control

When computing feedback gains, poles, or system responses, always use numerically stable algorithms:

- Use SVD-based methods for rank determination and pseudoinverse
- Avoid forming $\mathbf{A}^\top \mathbf{A}$ explicitly (use QR instead)
- Work in balanced coordinates for eigenvalue computation
- Use state-space representations rather than high-order polynomials